

Survey on Publicly Available Sinhala Natural Language Processing Tools and Research

Nisansa de Silva

Abstract—

Sinhala is the native language of the Sinhalese people who make up the largest ethnic group of Sri Lanka. The language belongs to the globe-spanning language tree, Indo-European. However, due to poverty in both linguistic and economic capital, Sinhala, in the perspective of Natural Language Processing tools and research, remains a resource-poor language which has neither the economic drive its cousin English has nor the sheer push of the law of numbers a language such as Chinese has. A number of research groups from Sri Lanka have noticed this dearth and the resultant dire need for proper tools and research for Sinhala natural language processing. However, due to various reasons, these attempts seem to lack coordination and awareness of each other. The objective of this paper is to fill that gap of a comprehensive literature survey of the publicly available Sinhala natural language tools and research so that the researchers working in this field can better utilize the contributions of their peers. As such, we shall be uploading this paper to arXiv and perpetually update it periodically as a living research article to reflect the advances made in the field. This manuscript is at version 6.0.0.

Index Terms—Sinhala, Natural Language Processing, Low-Resource Languages

1 INTRODUCTION

Sinhala¹ language, being the native language of the Sinhalese people [2–4], who make up the largest ethnic group of the island country of Sri Lanka, enjoys being reported as the mother tongue (L1) of approximately 17 million people [5, 6]. When both L1 and L2 speakers are counted, 85.2% of the total Sri Lankan population is literate in Sinhala [7]² (Fig 1 shows finer granular detail at the administrative district level). A strong correlation between the island of Sri Lanka and the usage of Sinhala has been observed [9]³. This implies two things: 1) the majority of the Sinhala linguistic sources are located in the geographical area of Sri Lanka, and 2) the majority of the available linguistic sources in Sri Lanka are in Sinhala. Sri Lanka is the only country where Sinhala is recognised as an official language [10].

To give a brief linguistic background for the purpose of aligning the Sinhala language with the baseline of English, primarily it should be noted that the Sinhala language belongs to the same Indo-European language tree [11–13]. However, unlike English, which is part of the Germanic branch, Sinhala belongs to the Indo-Aryan branch. An extremely simplified version of the Indo-European language tree focusing mainly on Sinhala and its close relatives is

shown in Fig 2. Further, Sinhala, unlike English, which borrowed the Latin alphabet, has its own writing system, which is a descendant of the Indian Brahmi script [14–20]. By extension, this makes Sinhala Script a member of the Aramaic family of scripts [21, 22]. Thus, by inheritance, the Sinhala writing system is *abugida* (*alphasyllabary*), which to say, that consonant-vowel sequences are written as single units [23]. An extremely simplified evolution tree of the Sinhala script is shown in Fig 3. In it, the Sinhala script is shown as a singular derivation from *Brahmi* directly appearing with the contemporary script. But this is a simplification. We show the actual evolutionary eras of the Sinhala script in Appendix A, where the gradual change from *Brahmi* to the contemporary script is clearer. The phonetic and graphemic view of the contemporary Sinhala script is shown in Fig 4. It should be noted that, due to various historical reasons [24], the modern Sinhala language has loanwords from languages such as Tamil [25, 26], English, Portuguese, and Dutch. An analysis of the distribution of Sinhala letters towards the beginning, middle, or end of words conducted in the style of proposed by ProoffReader [27] using the *SinMin* Sinhala corpus created by Upeksha et al. [28, 29] is given in Appendix B.

Regardless of the rich historical array of literature spanning several millennia (starting between 3rd to 2nd century BCE [38, 39]), modern natural language processing tools for the Sinhala language are scarce [40, 41]. Further, Sinhala is a *diglossic* language [42] which has two variations: the literary form and the spoken form. The *Vedda* (*Veddah*) language is considered a Sinhala creole [43–45] or a language isolate [46].

Natural Language Processing (NLP) is a broad area covering all computational processing and analysis of human languages. To achieve this end, NLP systems operate at different levels [47–49]. A graphical representation of NLP

- Nisansa de Silva is with the Department of Computer Science & Engineering, University of Moratuwa.
E-mail: nisansa@cse.mrt.ac.lk

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1. Englebreton and Genetti [1] observe that in some contexts the Sinhala language is also referred as *Sinhalese*, *Singhala*, and *Singhalese*

2. This is an increment from the percentage of 79.7% reported in 2012 [8]

3. Dunn and Edwards-Brown [9] in their work to augment the Language Identification (LID) task with geography observe that for Sinhala LID *does not* benefit from the geography model given that the models already have an alignment of 99%.

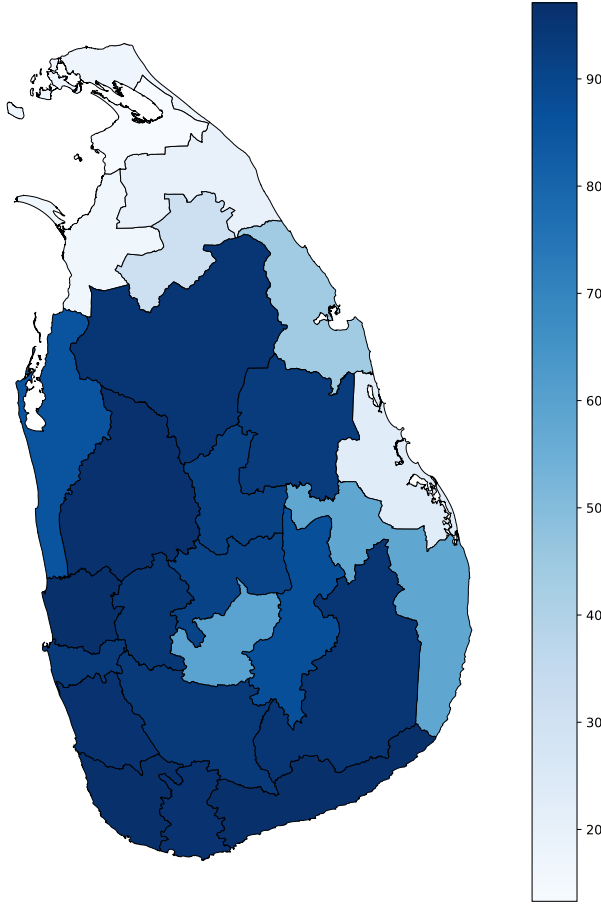


Fig. 1: Sinhala language literacy as a percentage of the population aged 10 and over in Sri Lankan administrative districts as reported by the 2024 Census [7]

layers and application domains are shown in Figure 5. On one hand, according to Liddy [48], these systems can be categorized into the following layers; *phonological*, *morphological*, *lexical*, *syntactic*, *semantic*, *discourse*, and *pragmatic*. The *phonological* layer deals with the interpretation of language sounds. As such, it consists of mainly speech-to-text and text-to-speech systems. In cases where one is working with the written text of the language rather than speech, it is possible to replace this layer with tools which handle Optical Character Recognition (OCR) and language rendering standards (such as Unicode [50]). The *morphological* layer analyses words at their smallest units of meaning. As such, analysis of word lemmas and prefix-suffix-based inflection is handled in this layer. *Lexical* layer handles individual words. Therefore, tasks such as Part of Speech (PoS) tagging happens here. The next layer, *syntactic*, takes place at the phrase and sentence level, where grammatical structures are utilised to obtain meaning. *Semantic* layer attempts to derive the meanings from the word level to the sentence level. Starting with Named Entity Recognition (NER) at the word level and working its way up by identifying the contexts they are set in until arriving at the overall meaning. The *discourse* layer handles meaning in textual units larger than a sentence. In this, the function of a particular sentence maybe contextualised within the document it is set in. Finally, the

pragmatic layer handles contexts read into contents without having to be explicitly mentioned [47, 48]. Some forms of anaphora (coreference) resolution [51–55] fall into this application.

On the other hand, Wimalasuriya and Dou [49] categorize NLP tools and research by utility. They introduce three categories with increasing complexity; *Information Retrieval* (IR), *Information Extraction* (IE), and *Natural Language Understanding* (NLU). *Information Retrieval* covers applications, which search and retrieve information which are relevant to a given query. For pure IR, tools and methods up-to and including the *syntactic* layer in the above analysis are used. *Information Extraction*, on the other hand, extracts structured information. The difference between IR and IE is the fact that IR does not change the structure of the documents in question. Be them structured, semi-structured, or unstructured, all IR does is fetch them as they are. In comparison, IE, takes semi-structured or unstructured text and puts them in a machine-readable structure. For this, IE utilizes all the layers used by IR and the *semantic* layer. *Natural Language Understanding* is purely the idea of cognition. Most NLU tasks fall under AI-hard category and remain unsolved [47]. However, with varying accuracy, some NLU tasks such as machine translation⁴ are being attempted. The *pragmatic* layer of the above analysis belongs to the NLU tasks while the *discourse layer* straddles information extraction and natural language understanding [47].

The objective of this paper is to serve as a comprehensive survey on the state of natural language processing resources for the Sinhala language. The initial structure and content of this survey are heavily influenced by the preliminary surveys carried out by de Silva [40] and Wijeratne et al. [47]. However, our hope is to host this survey at arXiv as a perpetually evolving, living research article [57, 58] which continuously gets updated as new research and tools for the Sinhala language are created and made publicly available⁵. We also discuss how the non-compliance of policies to put data and code online [60], after the research is concluded and the paper is published, has negatively impacted the growth and sustainability of Sinhala NLP. In fact, studies such as Occhini et al. [61] put the *AI readiness* of Sinhala at a low 2.9%⁶. This puts it in second place behind Tamil when considering Sri Lanka⁷ and globally at 53rd⁸. Hence, it is our hope that this work will help future researchers who are engaged in Sinhala NLP research to conduct their literature surveys efficiently and comprehensively. For the success of this survey, we shall also consider the Sri Lankan NLP tools repository, [1knlp](https://github.com/1knlp). This manuscript is at version 6.0.0. The latest version of the manuscript can be obtained from arXiv⁹ or ResearchGate¹⁰. Given that this work was initially published in 2019, when citing this work, it is recommended to cite the most recent version to avoid the reviewers questioning its continued contemporary relevance. An example

4. This is, however, not without the criticism of being nothing more than a *Chinese room* [56] rather than true NLU.

5. In contrast to subsequent similar studies [59]

6. <https://nisansa.lk/?s=45m7Kd>

7. <https://nisansa.lk/?s=ixjmZ8>

8. <https://nisansa.lk/?s=njrVq3>

9. <https://arxiv.org/abs/1906.02358>

10. <https://bit.ly/31AhvvR>

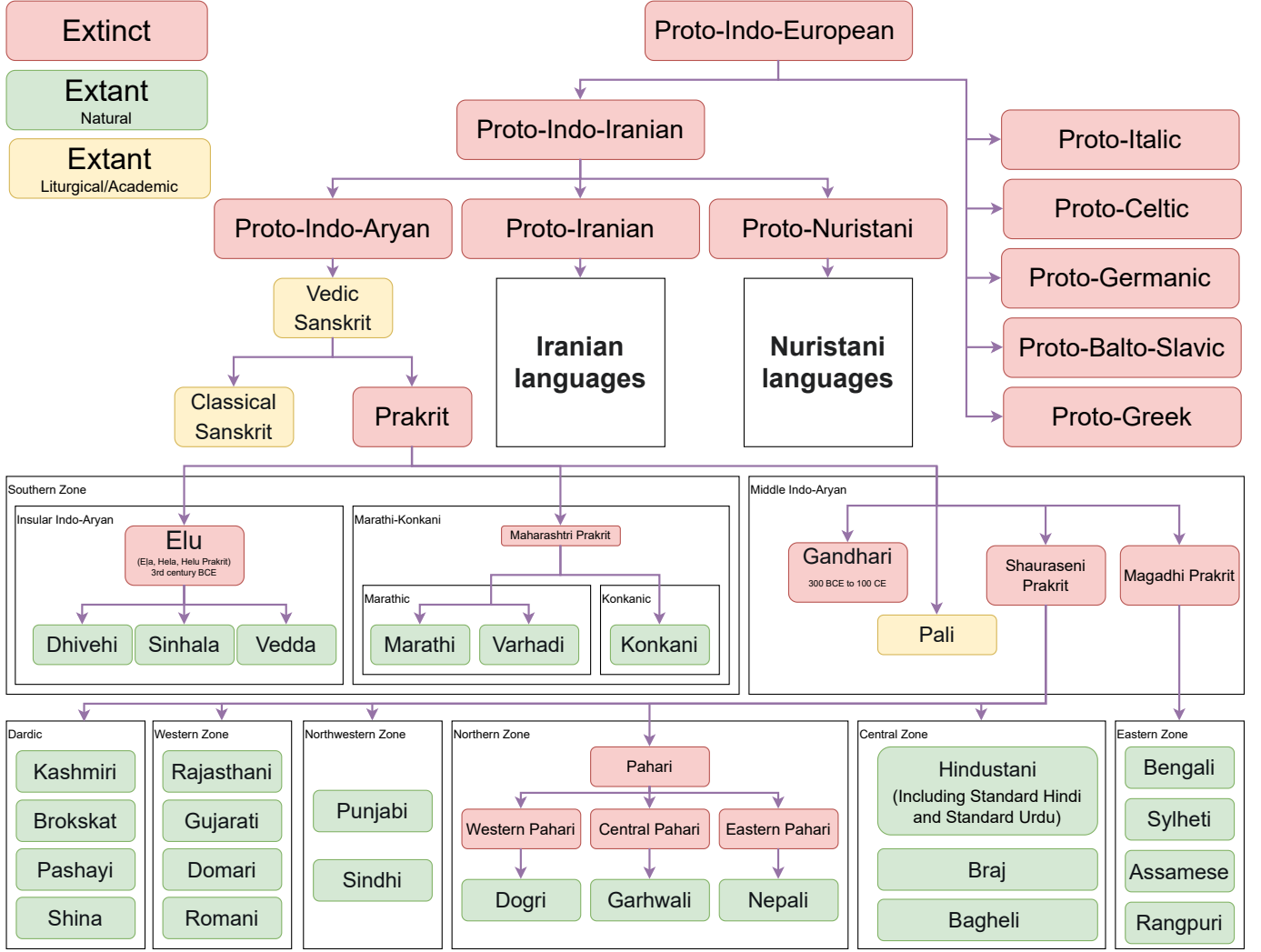


Fig. 2: A portion of the Indo-European language tree with a focus on Sinhala and its close relatives. Only the *Elu* sub-tree is shown exhaustively. All other sub-trees only list some languages for representation purposes, chosen randomly or otherwise. Some of the languages shown as *Extinct* have archaeological proof (eg; *Prakrit*) while others were retroactively reconstructed by linguists by observing the commonalities of languages that they begot (eg; *Proto-Indo-Aryan*). The status of some languages have spread beyond linguistics to be a social and political question. In this depiction, we have attempted to side with academic consensus when available and the majority when not. As an example of one contentions point on this diagram, note the positioning of *Pali*. While what we show here is the academically accepted relationship, there are other schools of thought: 1) Some argue that *Pali* is a constructed language (ConLang) of *Sinhala* and thus should be depicted as a child of *Sinhala*., 2) Others claim it to be a sibling of *Sinhala*, 3) Further, some claim it to be either be the link between *Elu* and *Sinhala* or the parent of *Elu*, 4) Some claim it needs to be merged with *Gandhari*, and 5) Some claim it to be the same as *Magadhi Prakrit* given that in some Buddhist traditions, it is noted Buddha to have spoken *Magadhi*.

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@article{de2026survey,
  title={Survey on Publicly Available Sinhala
    Natural Language Processing Tools and
    Research}},
  author={de Silva, Nisansa},
  journal={arXiv preprint arXiv:1906.02358v27},
  year={2026}
}
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Code Block 1: Example of an updated reference

of an updated reference is given in Code Block 1.

Figure 13 in Appendix C shows the most prolific researchers in the domain of Sinhala NLP. The nodes contain the name of the researcher along with the total number of Sinhala NLP papers that the researcher has authored. The edges between the two researchers are labelled with the number of Sinhala NLP papers the relevant pair of researchers have co-authored. When selecting authors, we have applied a threshold of 3 Sinhala NLP publications. Given that the objective of the visualization is to portray the cooperation between researchers, we have also added the strongest edge that connects each researcher to the rest of the researchers in the graph. The few isolated nodes are researchers who have authored at least 3 Sinhala NLP

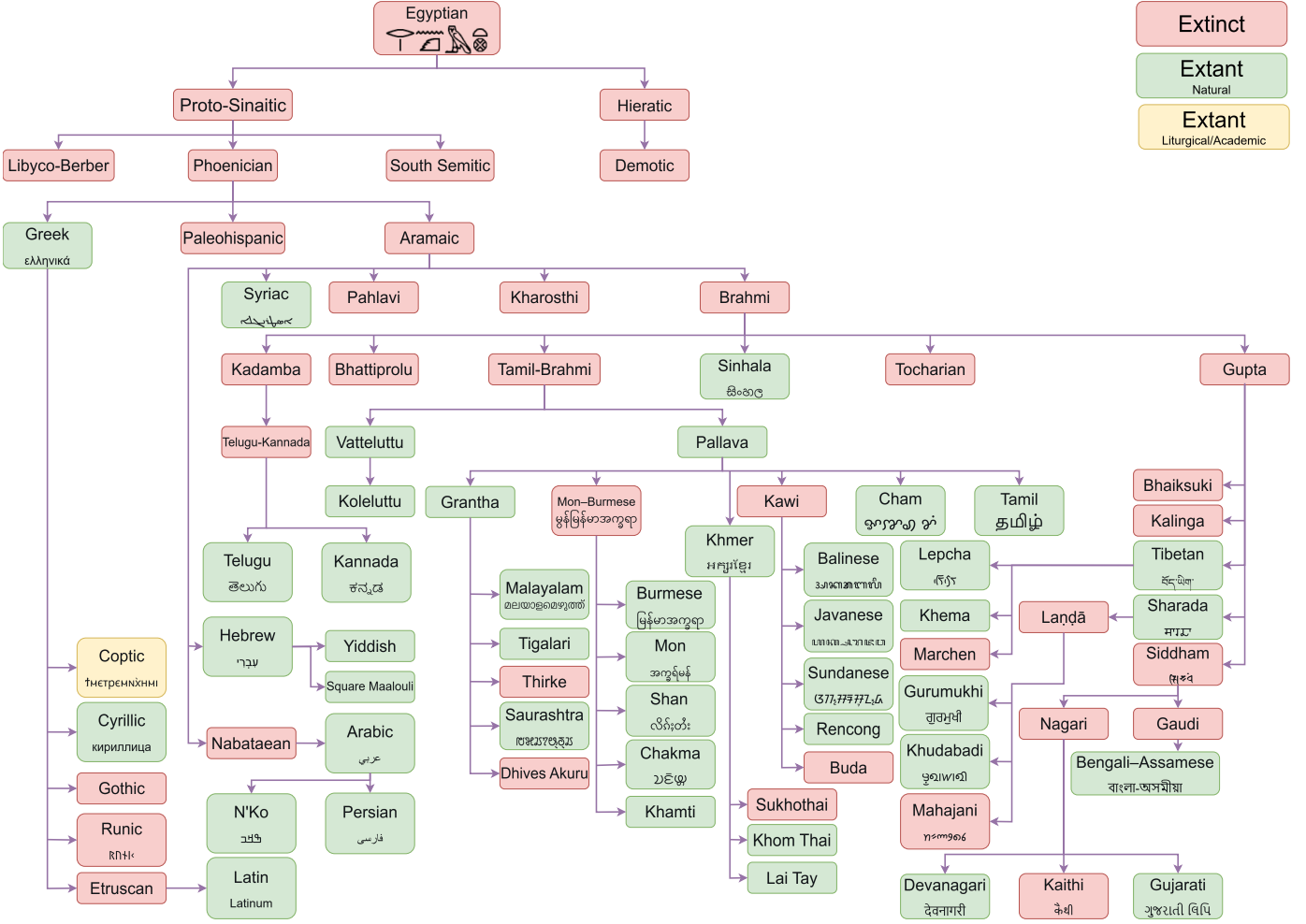


Fig. 3: A portion of the writing systems (scripts) tree with a focus on Sinhala and its close relatives. It is important to note that this is distinct from the language tree shown in Fig 2. The writing systems used by a language being related, not being related, or the degree to which they are related, is not directly correlated to the said languages being related, not being related, or the degree to which they are related. As an example, note that the writing systems of *Sinhala* and *Tamil* are reasonably closely related. But *Tamil* is not shown on Fig 2 given that it does not belong to the *Indo-European* language family. It belongs to the *Dravidian* language family. A language that is shown as a single entity in Fig 2 may have multiple entries in this diagram. An example is the language shown as *Hindustani* in Fig 2 splits into *Hindi* and *Urdu* in this diagram, where the former uses the *Devanagari* script and the latter a derivation of *Arabic* script. Conversely, languages that may be shown as distinct entities in Fig 2, may map to a singular entity in the script tree. An example is *English* which would go under *Proto-Germanic* and *Italian* which would go under *Proto-Italic* in an extended version of Fig 2, would both map to *Latin* in this diagram, given that both of them use the *Latin* script.

publications but do not have any coauthored papers with anyone else on the graph. We have also added labels to clusters in cases where all or the majority of researchers in those clusters have the same affiliation. It is observable that the cluster from the *Department of Computer Science & Engineering, University of Moratuwa* is the most prolific in Sinhala NLP research.

Figure 14 in Appendix C shows the probability of studies from the institutions to which the most prolific authors from Figure 13 are affiliated citing institutions of the same set. We observe both interesting and disturbing trends which we discuss in the figure caption. Figure 15 in Appendix C shows the mapping between authors with at least 10 papers in the Sinhala NLP domain and their research interests, denoted by the subsection titles of the Section 3 of this

paper. If two or more authors on the diagram have co-authored a paper, it gets counted for each of the authors separately without any bias on the author order listed on the publication. For example, the paper *Building a wordnet for Sinhala* [62] is counted for both *Nisansa de Silva* and *Gihan Dias*. If a single paper contributes to more than one research area, that paper is counted for all of the research areas to which it contributes. For example, the paper *Sinhala Text Classification: Observations from the Perspective of a Resource Poor Language* [40] which introduces a new Sinhala text classification data set is counted for both *Data Sets* (Section 3.2) and *Text Classification* (Section 3.15).

The remainder of this survey is organized as follows: Section 2 introduces some important properties and conventions of the Sinhala language, which are important for the

Short Vowels	 ¹ [a], [ə]	 ¹ [æ]	 ¹ [i]	 ¹ [u]	 ¹ [e]	 ¹ [o]	 ²	Key Suddha Vowel Diacritic Form Transliteration [IPA] Misra Vowel Diacritic Form Transliteration [IPA] Other Diacritic Transliteration [IPA] Suddha Consonant Transliteration [IPA] Misra Consonant Transliteration [IPA] Prenasalized Consonant Transliteration [IPA]		
	 ¹ [a:]	 ¹ [æ:]	 ¹ [i:]	 ¹ [u:]	 ¹ [e:]	 ¹ [o:]				
Long Vowels	 ¹ [ru]	 ¹ [ru:]	 ¹ [aj]	 ¹ [au]	 ¹ [li]	 ¹ [li:]	 ² [h]			

	Unvoiced		Voiced		Nasal	Approximant	Sibilant	Fricative	Other	Prenasalized
	Inaspirate	Aspirated	Inaspirate	Aspirated						
Velar	 ka [ka]	 kha [ka]	 ga [ga]	 gha [ga]	 ŋa [ŋa]			 ha [ha]		 ŋga [ŋga]
Palatal	 ca [tʃa]	 cha [tʃa]	 ja [dʒa]	 jha [dʒa]	 ɲa [ɲa]	 ya [ja]	 sa [ja]		 dʒa [dʒa]	 ɲja [ɲja]
Retroflex	 ta [ta]	 tha [ta]	 da [da]	 dha [da]	 ɳa [ɳa]	 ra [ra]	 ʂa [ja]		 la [la]	 ɳda [ɳda]
Dental	 ta [ta]	 tha [ta]	 da [da]	 dha [da]	 na [na]	 la [la]	 sa [sa]			 ɳda [ɳda]
Labial	 pa [pa]	 pha [pa]	 ba [ba]	 bha [ba]	 ma [ma]	 va/wa [va]		 fa [f]		 mba [mba]

Fig. 4: The phonetic and graphemic view of the contemporary Sinhala script. **Notes:** (1) Each unmarked consonant in Sinhala carries an inherent vowel. Thus, there is no diacritic form here. Typically, [a] is sounded when an unmarked consonant appears at the beginning of a word, and [ə] is sounded when an unmarked consonant appears anywhere else in a word. However, there are exceptions to this rule. These exceptions are the primary reason Sinhala cannot be said to have *regular (perfect) orthography* and stops at *near regular (perfect) orthography* [30]. (2) The *hal kirīma* mark is used when the aforementioned inherent vowel is removed from a consonant and thus has no sound by itself but rather the negation of one [31]. (3) There is controversy in including the voiceless affricate in the *suddha* set because some purists object to the fact, pointing out that it does not appear in the main text of *Sidatsangarāwa*. Nevertheless, *Sidatsangarāwa* uses it in some examples; so we can be sure that the character did exist in *Elu* [32]. (4) The Retroflex pronunciations, which could be found in historic speech, are no longer phonemic in modern Sinhala [33, 34]. This change was resisted for a time by forcing children to recite *Sakaskadaya* intended to train them to use the *correct* pronunciations. But with that practice falling into disuse and with the popular usage suppressing the academic, it is rare to encounter them with distinct pronunciations [35, 36]. (5) This letter is not used in ancient or contemporary use. Its usefulness is questionable, but it is included in the standard alphabet [37].

development and understanding of Sinhala NLP. Section 3 discusses the various tools and research available for Sinhala NLP. In this section, we will discuss both pure Sinhala NLP tools and research, as well as hybrid Sinhala-English work. We will also discuss research and tools that contribute to Sinhala NLP, either along with or with the help of Tamil, the other official language of Sri Lanka. Section 4 gives a brief introduction to the primary language sources used by the studies discussed in this work. Finally, Section 5, concludes the survey.

2 PROPERTIES OF THE SINHALA LANGUAGE

Before moving on to discussing Sinhala NLP resources, we shall give a brief introduction to some of the important properties of Sinhala language, which impact the development of Sinhala NLP resources. Sinhala grammar has two forms: written (literary) and spoken. These forms differ from each-other in their core grammatical structures [1, 42, 63, 64].

The written form strictly adheres to the SOV (Subject, Object, and Verb) configuration [65, 66]. Further, in the written form, *subject-verb agreement* is enforced [67] such that, in order to be grammatically correct, the subject and the verb must agree in terms of: gender (male/female), number (singular/plural) and person (1st/2nd/3rd). However, in spoken Sinhala, the SOV order can be neglected [68] and *male singular 3rd person verb* can be used for all nouns [67]. Sinhala is also a head-final language, where the complements and modifiers would appear before their heads [69] this is similar to that of English and dissimilar to that of French. In total, according to Abhayasinghe [70], there are 25 types of simple sentence structures in Sinhala and according to Madushani [71] there are 8 types of Verb Phrases. Similar to many Indo Aryan languages, *animacy* plays a major role in Sinhala grammar in syntactic and semantic roles [72–74]. Comparative studies done by Noguchi [75] and by Miyagishi [64, 76] have found that *animacy* extends

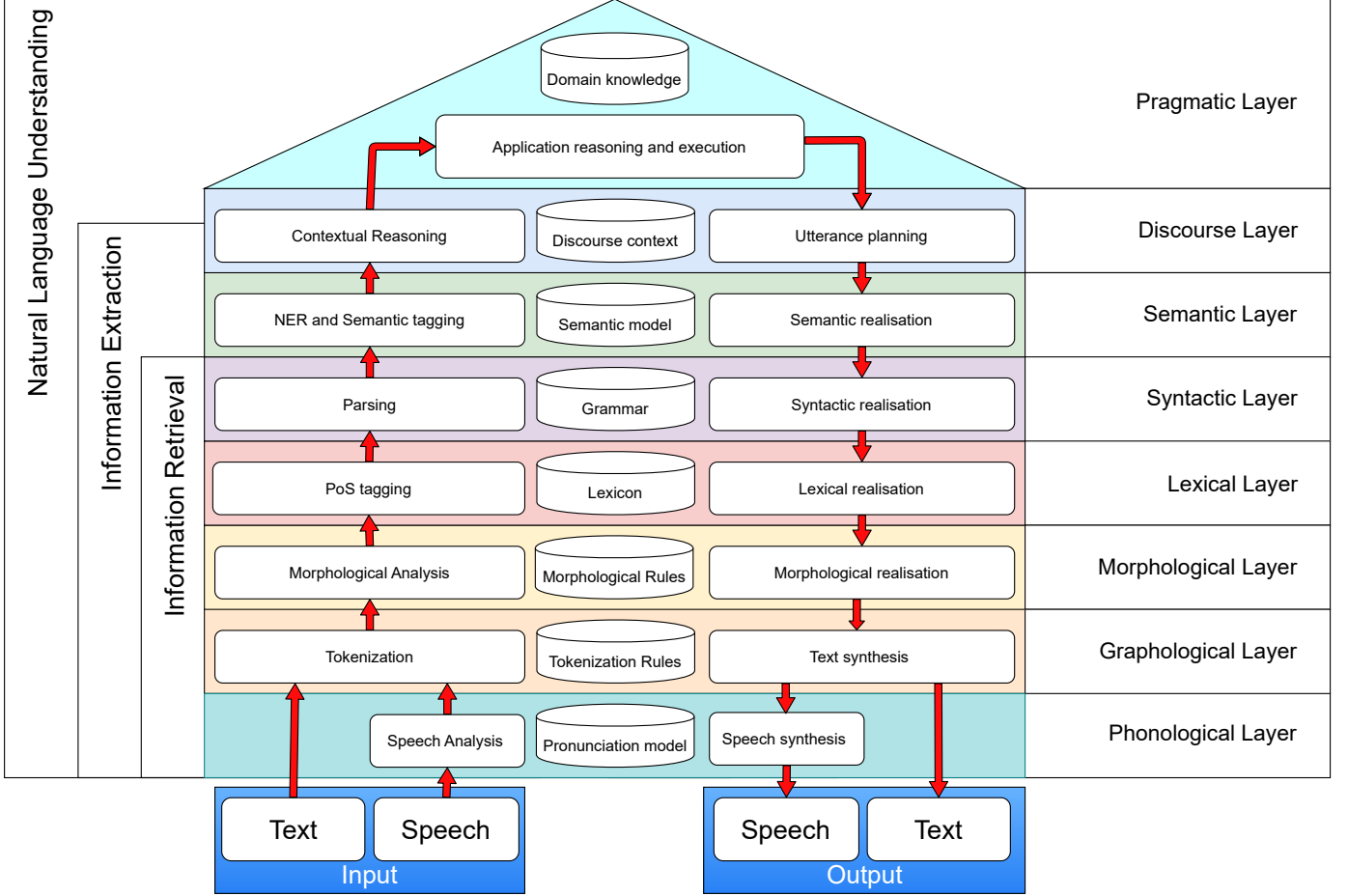


Fig. 5: Traditional NLP layers and tasks (Derived from the work of Wijeratne et al. [47] and extended to encapsulate a wider range by this work.)

its influence from phrase level to sentence level in Sinhala (e.g., Usage of post-positions [30, 65]). On this matter, Table 1 explains grammatical cases and inflections of animate common nouns while Table 2 explains grammatical cases and inflections of inanimate common nouns. We provide a comparative analysis of parsing the very simple English sentence “I eat a red apple” and its Sinhala, Hindi, and French translations in Fig 6. English and French parsing was done using the *Stanford Parser*¹¹. Hindi parsing was done using the *IIT-Hyderabad Parser*¹² and the study by Singh et al. [77].

Herath et al. [78, 79] argue that pure Sinhala words did not have suffixes and that adding suffixes was incorporated to Sinhala after 12th century BC with the influx of Sanskrit words. With this, they declare Sinhala to have the following types of words:

- 1) Suffixes
- 2) Nouns
- 3) Cases
- 4) Verbs
- 5) Conjunctions and articles
- 6) Adjectives
- 7) Demonstratives, Interrogatives, and negatives
- 8) Particles and prefixes

They further divide nouns into five groups: material, agentive, common, abstract, and proper. In addition to these, they also introduce compound nouns. We show the noun categorization proposed by Herath et al. [78] in Table 3.

Herath et al. [79] categorize Sinhala suffixes along the attributes of: gender, number, definiteness, case, and conjunctive. They further claim that there are 3 types of suffixes: *Suf1* adds gender, number, and definiteness; *Suf2* adds case; and *Suf3* adds conjunctive. Conjunctive is claimed to be equivalent to *too* and *and* in English. We show an extension of the suffix structure proposed by Herath et al. [79] in Table 4. In their analysis on register variation (vocabulary) of 60 languages, Li et al. [80] observes that Sinhala exhibits homogeneity between 0.5 and 1.0 in the three corpora considered: CC (macro-web register - Common Crawl), TW (social media register - Twitter), and WK (Wikipedia register - March 2020). Dissanayake and Perera [81] discusses a few Sinhala discourse markers that have seeped into Sri Lankan English.

3 SINHALA NLP RESOURCES

This section surveys the resources, tools, and applications that constitute the Sinhala NLP ecosystem. We broadly follow the hierarchical organisation illustrated in Figure 5, while also covering language resources that do not naturally

11. <http://nlp.stanford.edu:8080/parser/>

12. <http://ltrc.iiit.ac.in/analyzer/>

TABLE 1: Examples for grammatical cases and inflection of animate common nouns

Form	Case	Singular				Plural	
		Masculine		Feminine		Masculine	Feminine
		Definite	Indefinite	Definite	Indefinite		
1	Nominative	මිනිසා	මිනිසෙක්	ගැහැණිය	ගැහැණියක් ගැහැණියෙක්	මිනිස්සු	ගැහැණු
2	Accusative	මිනිසා	මිනිසෙකු	ගැහැණිය	ගැහැණියක	මිනිසුන්	ගැහැණුන්
3	Auxiliary						
4	Dative	මිනිසාට	මිනිසෙකුට	ගැහැණියට	ගැහැණියකුට ගැහැණියකට	මිනිසුන්ට	ගැහැණුන්ට
5	Genitive	මිනිසාගේ	මිනිසෙකුගේ	ගැහැණියගේ	ගැහැණියකගේ	මිනිසුන්ගේ	ගැහැණුන්ගේ
6	Locative						
7	Instrumental	මිනිසාගෙන්	මිනිසෙකුගෙන්	ගැහැණියගෙන්	ගැහැණියකගෙන්	මිනිසුන්ගෙන්	ගැහැණුන්ගෙන්
8	Ablative						
9	Vocative	මිනිස	මිනිස	ගැහැණිය	ගැහැණිය	මිනිසුනි	ගැහැණුනි

TABLE 2: Examples for grammatical cases and inflection of inanimate common nouns

Note that the grammatical cases of *Auxiliary* and *Vocative* do not exist for inanimate nouns in Sinhala.

Form	Case	Singular		Plural
		Definite	Indefinite	
1	Nominative	පොත	පොතක්	පොත්
2	Accusative	පොතට	පොතකට	පොත්වලට
4	Dative			
5	Genitive	පොතේ	පොතක	පොත්වල
6	Locative	පොතෙහි		
7	Instrumental	පොතෙන්	පොතකින්	පොත්වලින්
8	Ablative	පොතින්		

TABLE 3: Noun categorization by Herath et al. [78]

Type	Examples
Material	පුටු, ගෙය
Agentive	දුවන්නා, බීම
Common	ගොවියා, මිනිසා
Abstract	සුදු, උස
Proper	කොළඹ, අමර
Compound	කිරිබත් (කිරි+බත්), සුදුමල් (සුදු+මල්)

fit within that hierarchy, including corpora, datasets, dictionaries, and WordNets. Throughout this survey, we focus on NLP resources and computational methods rather than the underlying mechanics of Sinhala script handling, which have been comprehensively discussed elsewhere [82–87].

Research on Sinhala NLP spans more than three decades. One of the earliest documented efforts was the pioneering work of Herath et al. [88]. Although ambitious for its time, the project was constrained by the limited computational resources and linguistic data available during the early 1990s, resulting in relatively limited subsequent development. Later, Nandasara [89] reviewed the state of Sinhala language computing. Given that it was a decade old by the time the first edition of this survey was compiled, we observe the existence of many new discoveries in Sinhala

NLP which have not been taken into account by it. Similarly, Nandasara and Mikami [90] discussed the challenges and opportunities associated with Sinhala in computer science, providing valuable context for subsequent developments in the field.

Since those surveys were published, Sinhala NLP has experienced substantial growth across nearly every layer of the NLP pipeline. Nevertheless, research activity has been unevenly distributed. Optical character recognition (OCR) has attracted the greatest volume of research. In contrast, comparatively fewer studies have addressed higher-level language understanding tasks such as semantic analysis, discourse processing, and reasoning. Another notable direction is the development of end-to-end machine translation systems. Among these, the long-running Sinhala–English translation project [23, 91–108] represents one of the most sustained and comprehensive research efforts in Sinhala NLP. The increasing maturity of the Sinhala NLP ecosystem is also reflected in the availability of reusable software. For example,  **Sinling** provides an open-source collection of NLP utilities, including a tokenizer, stemmer, part-of-speech tagger, morphological joiner, and morphological splitter. The toolkit is distributed through *PyPI*.

Although this survey focuses on Sinhala, it is important to recognise its close relationship with Tamil, the other official language of Sri Lanka. Tamil is likewise considered a low-resource language; however, its substantially larger global speaker population has fostered a richer ecosystem of datasets, tools, and research [47]. Consequently, many studies have explored multilingual approaches that jointly leverage Sinhala and Tamil resources. The bilingual nature of Sri Lankan government publications, legislation, and news media naturally produces valuable parallel corpora that have supported numerous multilingual NLP studies discussed later in this survey. Beyond Tamil, several studies have also investigated Sinhala in combination with other languages, including Japanese [12, 39, 109–111].

Finally, Sinhala has increasingly been incorporated into broader multilingual resources. For example, it is included in the metadata collection of more than 2,800 languages compiled by van Esch et al. [112], enabling comparative analyses across diverse linguistic typologies. In addition,

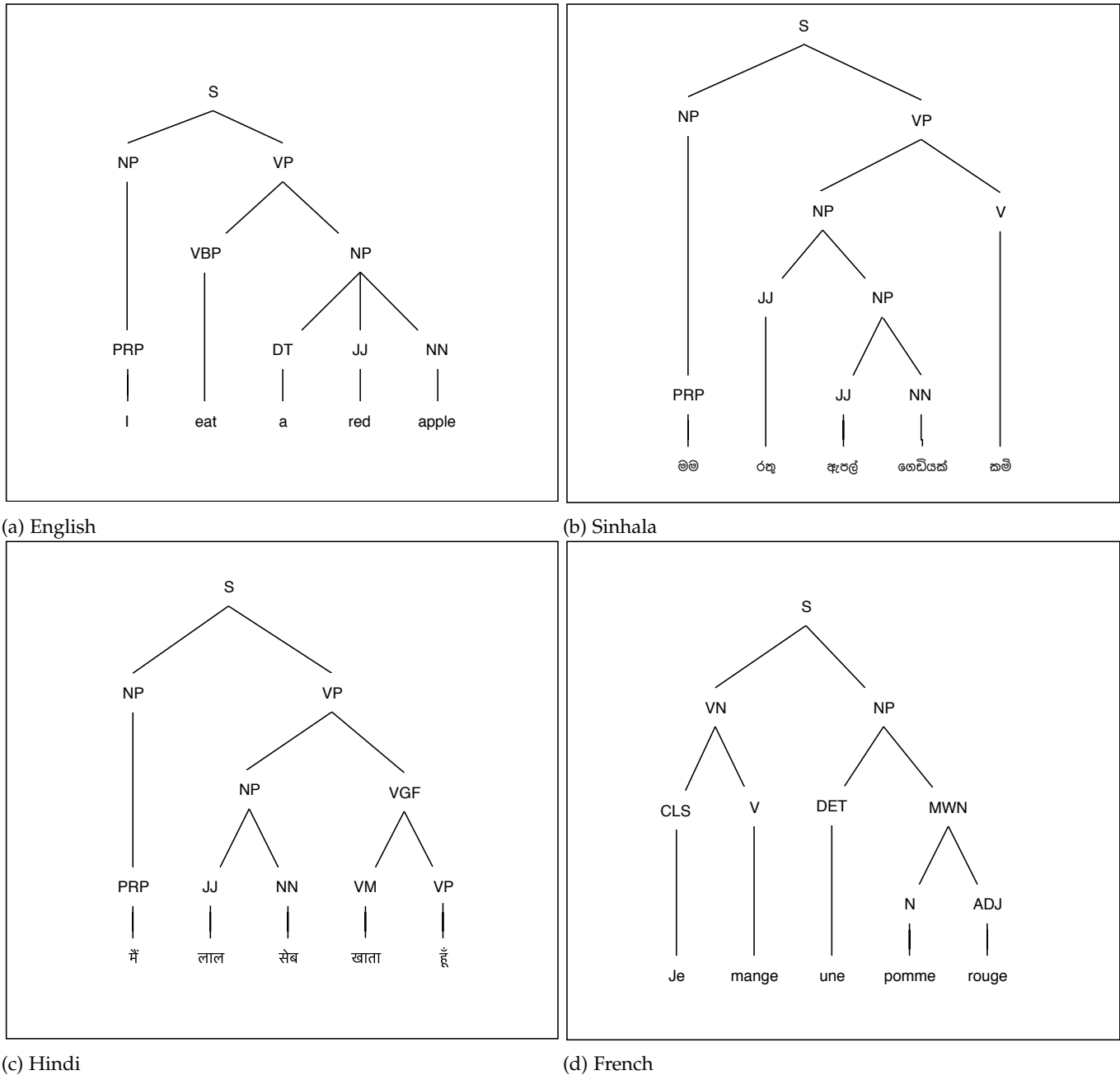


Fig. 6: Parse trees for the sentence “I eat a red apple” in four languages.

Herath et al. [113], Herath and Medagoda [114] compiled a comprehensive report on the Sinhala lexicon, providing an important linguistic foundation for subsequent NLP research.

3.1 Non-Parallel Corpora

The availability of large, diverse, and publicly accessible text corpora is one of the primary factors determining the progress of NLP research for a language. Non-parallel corpora are particularly valuable because they support a wide range of tasks, including language modelling, pre-training, information retrieval, text classification, and the construction of lexical resources. This subsection surveys

publicly reported Sinhala non-parallel corpora. Parallel corpora, which are primarily intended for machine translation and multilingual learning, are discussed separately in Section 3.27.1.

Early publicly available Sinhala corpora were relatively modest in size and were typically created through web crawling. One of the earliest substantial efforts was the [SinMin¹³](#) corpus introduced by Upeksha et al. [28, 29]. The corpus was constructed by crawling Sinhala web pages and was subsequently extended with additional resources, including [Jathaka Stories](#) and a collection of [web-crawled news articles](#). Although modest

13. 😊 [Third-party mirror](#)

TABLE 4: Extension of the suffix structure proposed by Herath et al. [79]

Number = Singular (S) / Plural (P)

Definite = Definite (D) / Indefinite (I) / Undecided (U)

Case = Nominative (N) / Accusative (A) / Dative (Da) / Genitive (G) / Instrumentive (In) / Auxiliary (Au) / Locative (L) / Ablative (Ab)

Conjun = with suffix *th* (Y) / without suffix *th* (N)

Suf1	Suf2	Suf3	Noun	-tree-	Attributes			
					Number	Definite	Case	Conjun
-	-	-	ගස්	(-s)	P	U	N, A, V	N
අ	-	-	ගස	(the-)	S	D	N, A, V	N
අක්	-	-	ගසක්	(a-)	S	I	N, A	N
අක	-	-	ගසක	(of a-)	S	I	G	N
අ	ට	-	ගසට	(to the-)	S	D	Da	N
-	ඒ	-	ගසේ	(in the-)	S	D	Au	N
-	වල	-	ගස්වල	(on -s)	P	U	L	N
-	වලට	-	ගස්වලට	(to -s)	P	U	Da	N
-	එහි	-	ගසෙහි	(of the-)	S	D	G	N
අක්	ඉන්	-	ගසකින්	(from a-)	S	I	Au	N
-	-	උත්	ගසුත්	(-s and)	P	U	N, A	Y
අ	-	ත්	ගසත්	(-and)	S	D	N, A	Y
අක්	-	උත්	ගසකුත්	(a- too)	S	I	N, A	Y
අක	-	ත්	ගසකත්	(of a -, too)	S	U	N, A	Y
අ	ට	ත්	ගසටත්	(to the- too)	S	D	Da	Y
-	ඒ	ත්	ගසේත්	(in -s too)	S	D	Au	Y
-	වල	ත්	ගස්වලත්	(on -s too)	P	U	L	Y
-	වලට	ත්	ගස්වලටත්	(to -s too)	P	U	Da	Y
-	එහි	ත්	ගසෙහිත්	(of the- too)	S	D	G	Y
අක්	ඉන්	උත්	ගසකිනුත්	(from a- too)	S	I	Au	Y

by the standards of contemporary multilingual corpora, SinMin represented an important milestone by providing one of the first publicly reusable corpora for Sinhala NLP research. Later, a smaller  [Sinhala news corpus](#) was created by de Silva [40].

The emergence of large multilingual language models substantially increased the demand for web-scale corpora. Consequently, Sinhala became part of several multilingual corpus construction efforts. Nguyen et al. [115] introduced  [CulturaX](#), a cleaned multilingual corpus intended for pre-training large language models. Their release contains 753,655 Sinhala documents comprising approximately 880 million tokens. Nevertheless, Sinhala remains severely under-represented within the overall collection, accounting for only 0.01% of the multilingual corpus. By comparison, Tamil contributes 0.07%, Hindi 0.27%, and English 45.13%.

A similar trend is evident in Kudugunta et al. [116], who introduced the  [MADLAD-400](#) multilingual corpus spanning 419 languages. For  [Sinhala](#), the authors report approximately 349k cleaned documents (out of 788k noisy), 16M clean sentences (out of 22.1M noisy), 1.9B clean tokens (out of 3.4B noisy). In addition to monolingual text, the project reports more than 7.3 million sentence pairs involving Sinhala. They also released the corresponding

multilingual  [trained checkpoints](#)

Several other multilingual corpora also include substantial Sinhala text. The OSCAR¹⁴ corpus [117, 118] contains 301,066 Sinhala documents comprising approximately 173 million words obtained through large-scale web mining. More recently,

Likewise, Imani et al. [119] introduced the  [Glot500](#) corpus covering 500 languages, including a Sinhala component comprising over 7.2 million sentences. Beyond releasing the corpus, the authors evaluated pretrained Sinhala language models on tasks such as sentence retrieval, named entity recognition, part-of-speech tagging, text classification, round-trip alignment, and perplexity estimation.

Alongside these multilingual resources, several corpora have been developed specifically for Sinhala. Dhananjaya et al. [120] introduced the  [sin-cc-15M](#) corpus, which the authors reported to be the largest monolingual Sinhala corpus available at the time of publication. A large-scale  [Sinhala NEWS corpus](#) containing more than 500,000 articles collected from multiple online news outlets was later released by Hettiarachchi et al. [121]. Similarly, a corpus collected from the Sinhala blog *Kalaya*¹⁵, together with the

14. <https://oscar-project.org/>

15. <http://www1.kalaya.org/>

associated processing code, has been made publicly available on [GitHub](#).

Social media has also emerged as an important source of Sinhala language data. Wijeratne and de Silva [122] publicly released a large corpus and accompanying stop-word list derived from a [decade of Sinhala Facebook posts](#). Although the original repository is no longer maintained, a third-party [mirror](#) has preserved public access to the complete collection.

Several additional corpora have contributed to the broader Sinhala NLP ecosystem. Weerasinghe et al. [123] developed a word corpus containing approximately 35,000 entries, although it no longer appears to be publicly accessible. Furthermore, Guzmán et al. [124] released two monolingual Sinhala corpora as part of the FLORES project, consisting of more than 155,000 filtered [Sinhala Wikipedia sentences](#) and over 5.1 million [Sinhala common crawl sentences](#). Although originally intended to support multilingual machine translation, these corpora have subsequently been adopted in a variety of Sinhala NLP tasks beyond translation.

Overall, the development of Sinhala non-parallel corpora has progressed from relatively small web-crawled collections to web-scale multilingual resources containing billions of tokens. Nevertheless, the quantitative comparison with higher-resource languages demonstrates that Sinhala remains substantially underrepresented in publicly available corpora. Addressing this resource gap remains one of the most important prerequisites for further advances in Sinhala NLP, particularly for data-intensive approaches such as large language models and foundation models.

3.2 Data Sets

This subsection surveys publicly available Sinhala datasets that are either task-agnostic or intended to support multiple NLP tasks. Datasets developed specifically for individual applications, such as sentiment analysis, named entity recognition, or machine translation, are discussed in their respective sections.

The growing popularity of multilingual benchmark suites has led to increasing efforts to incorporate Sinhala into standardised evaluation datasets. One such initiative is [Global-MMLU](#), introduced by Singh et al. [125] as a multilingual extension of the original Massive Multitask Language Understanding (MMLU) benchmark [126]. The authors report improvements over the earlier multilingual extension proposed by Üstün et al. [127]. Nevertheless, Sinhala remains underrepresented within the benchmark, and the project actively solicits community volunteers to extend coverage for low-resource languages such as Sinhala¹⁶. At present, contributors are acknowledged on the project website but are not formally included as co-authors of the associated research publication.

Several datasets have been developed to capture naturally occurring Sinhala text from online platforms. For example, De Mel and de Silva [128] released a [corpus of 63,471 Sinhala YouTube comments](#)

collected from Sinhala music videos. Of these, 35,428 comments are written directly in Sinhala script, whereas the remaining 28,043 comments were transliterated from Singlish, making the collection useful for studying both native Sinhala text and transliterated user-generated content.

In addition to individual datasets, several curated repositories aggregate Sinhala resources from multiple domains. Senaratna [129] introduced and continuously maintains the Sri Lanka Document Datasets, a collection of publicly available datasets containing Sinhala text from a variety of sources. These include [news articles](#), [educational publications](#) issued by the Educational Publications Department, and government press releases from organizations such as the [Treasury](#) and the [Presidential Media Division](#). The legal and parliamentary resources contained within this collection are discussed separately in Section 3.35.

General-purpose knowledge resources also provide valuable supplementary data for Sinhala NLP. A wiki version of the Sinhala Encyclopedia maintained by the Government of Sri Lanka is publicly available¹⁷. Although the resource appears to have received relatively few recent updates, several of its articles provide more comprehensive coverage than their counterparts in Sinhala Wikipedia¹⁸.

A number of lightweight lexical resources have likewise been released to facilitate downstream NLP research. While Wijeratne and de Silva [122] extracted a stop-word list algorithmically from a large Sinhala corpus, Lakmal et al. [130] published a manually curated [Sinhala stop-word list](#).

Reusable software packages often bundle both linguistic resources and baseline processing tools. For example, the open-source package [sinling](#) provides a rudimentary Sinhala tokeniser, stemmer, and part-of-speech tagger, together with supporting lexical resources. Although these components are discussed individually in later sections, the package itself represents one of the earliest attempts to provide an integrated toolkit for Sinhala NLP.

Overall, the range of publicly available Sinhala datasets has expanded considerably in recent years, evolving from isolated resources to curated repositories and multilingual benchmark collections. Nevertheless, compared with higher-resource languages, the ecosystem remains fragmented, with relatively few standardised benchmark datasets that are widely adopted across the research community. Continued efforts toward dataset curation, long-term maintenance, and standardised evaluation benchmarks will therefore play an important role in improving the reproducibility and comparability of future Sinhala NLP research.

3.3 Dictionaries

Bilingual and monolingual dictionaries form one of the most fundamental lexical resources in NLP. They support a wide range of applications, including machine translation, lexical alignment, information retrieval, word sense disambiguation, and language learning. Despite the long history of

16. <https://nisansa.lk/?s=Ww20rp>

17. <https://t.co/k8gFT9puM0>

18. <https://si.wikipedia.org/>

Sinhala lexicography, however, relatively few dictionaries are openly available in formats suitable for computational research.

The earliest and most comprehensive English–Sinhala dictionary remains that of Malalasekera [131]. But this resource, together with several other widely used printed dictionaries [132–137], is protected by copyright and is therefore unavailable for unrestricted research and software development. Although the dictionary developed by Kulatunga [138] is publicly accessible through an online web interface¹⁹, it neither provides an application programming interface (API) nor allows direct access to the underlying lexical database.

Among publicly accessible lexical resources, one of the largest English–Sinhala dictionary datasets originates from the discontinued Firefox plug-in *EnSiTip* [139]. The released lexicon bears a strong resemblance to the dictionary of Kulatunga [138], although the relationship between the two resources has not been formally documented. In addition, Hettige and Karunananda [94] reported the construction of an English–Sinhala lexicon to support their English-to-Sinhala machine translation system, while Samarawickrama and Hettige [140] investigated the design requirements for intelligent English–Sinhala bilingual dictionaries.

More recently, Wickramasinghe and De Silva [141] introduced one of the largest openly available Sinhala–English dictionaries. Their resource contains 1,368,416 unfiltered and 195,255 filtered *En-Si* word pairs, and both the dictionary and the accompanying code have been publicly released on  [GitHub](#). Dictionary development has also extended beyond bilingual lexical resources. Iniyage et al. [142] introduced *Wahara*²⁰, a publicly maintained Sinhala–Sinhala dictionary developed at the University of Moratuwa.

For multilingual applications, the Sri Lankan government has released a trilingual Sinhala–English–Tamil dictionary [143]. Although this resource is accessible through a web interface, it currently lacks an efficient API or bulk data access mechanism, limiting its practical use for computational research. Similarly, Weerasinghe and Dias [144] developed a multilingual gazetteer of Sri Lankan place names that serves both as a lexical resource and as supporting data for named entity recognition tasks.

In addition to conventional dictionaries, pronunciation lexicons have become increasingly important for speech technologies. Google has publicly released a comprehensive IPA pronunciation dictionary covering a  [large Sinhala vocabulary](#).

Overall, Sinhala lexical resources have evolved considerably from traditional printed dictionaries to openly available bilingual lexicons, multilingual dictionaries, gazetteers, and pronunciation dictionaries. Nevertheless, the majority of comprehensive lexical resources remain either proprietary or accessible only through web interfaces, creating significant barriers to reproducible research and large-scale NLP system development. The continued release of openly licensed, machine-readable dictionaries therefore remains an important priority for advancing Sinhala NLP.

3.4 WordNets

WordNets [145] are among the most influential lexical resources in NLP, representing words as interconnected semantic concepts rather than isolated lexical entries. They encode a variety of semantic relations, including *synonymy*, *hyponymy*, *hypernymy*, and *meronymy*, making them valuable for numerous downstream applications. These range from relatively simple ontology and gazetteer construction [49] to more advanced semantic processing tasks, including semantic similarity computation [146, 147] and semantic oppositeness detection [148].

Despite their importance, the development of a comprehensive Sinhala WordNet has remained an unsolved problem. One of the earliest large-scale efforts was undertaken by Wijesiri et al. [62], who developed a crowd-sourced Sinhala WordNet that was temporarily hosted online [149]. Although the project demonstrated the feasibility of collaboratively constructing a lexical semantic resource for Sinhala, the website has since become defunct, resulting in the loss of the original data and supporting software. To the best of our knowledge, the only publicly accessible remnant of the project is the subset of the resource preserved by Arukgoda et al. [150], who  [cloned](#) the available data for use in their word sense disambiguation system. Even during its active development, however, the resource remained incomplete because the crowd-sourced methodology failed to attract a sufficiently large volunteer community.

An independent effort to construct a Sinhala WordNet was also initiated by Welgama et al. [151]. However, this project appears to have ceased development at an even earlier stage and did not reach the level of lexical coverage reported by Wijesiri et al. [62].

Overall, unlike many higher-resource languages, Sinhala still lacks a comprehensive, actively maintained, and publicly accessible WordNet. This absence continues to limit research on lexical semantics and constrains the development of knowledge-based NLP methods that rely on structured semantic relations. The construction of an open, sustainable, and community-maintained Sinhala WordNet therefore remains an important open challenge for the language.

3.5 Tokenising

Tokenisation is one of the most fundamental preprocessing steps in NLP, as it determines how text is segmented into units that are subsequently processed by language models. While tokenisation has traditionally been regarded as a relatively straightforward preprocessing task, recent work has shown that its design has a substantial impact on the performance and efficiency of multilingual large language models, particularly for morphologically rich and low-resource languages such as Sinhala.

A comprehensive evaluation of multilingual tokenisers by Petrov et al. [152] compared 17 off-the-shelf tokenisation schemes across a large number of languages, including Sinhala. The authors observed that higher-resource languages such as English generally exhibit tokenisation lengths close to the ideal value of 1.0, whereas some languages require substantially longer token sequences, with the worst-performing language reaching a value of approximately 15.0. For Sinhala, most tokenisers produced

19. <https://www.maduraonline.com/>

20. <http://crawler.nlp.uom.lk/>

relatively poor tokenisation lengths ranging from 8.83 to 12.86, highlighting the difficulty of efficiently representing Sinhala text using tokenisers designed primarily for higher-resource languages. Notable exceptions were MBart50 [153], mT5 [153, 154], and CANINE [155], which achieved tokenization lengths of 1.35, 1.66, and the ideal value of 1.00, respectively.

Subsequent work has investigated the causes of these discrepancies. Rather than focusing solely on the tokenisation algorithm itself, Velayuthan and Sarveswaran [156] demonstrated that the pre-tokenisation stage has an even greater influence on tokenisation efficiency for languages such as Sinhala in the context of large language models. Their findings suggest that language-specific preprocessing strategies are at least as important as the choice of the underlying subword tokenisation algorithm. Later Nisfer et al. [157] introduced a Python toolkit named `grapheme-kit` building on this work.

Task-specific tokenisers have also been proposed for Sinhala. Ranasinghe [158] introduced the `SLTK` tokeniser, which attempts to improve tokenisation by explicitly handling Sinhala compound words. However, the proposed approach relies heavily on a manually curated `dictionary` that currently contains only seven entries. Although this may be sufficient for the accompanying `news title dataset`, the limited lexical coverage raises concerns regarding its applicability to general-domain Sinhala text.

Building upon these observations, Darshana [159] proposed Syllable-aware Grapheme Pair Encoding (SGPE), an extension of Grapheme Pair Encoding (GPE) [156] that explicitly incorporates the syllabic structure of Sinhala during tokenisation. The proposed method was evaluated using Llama 4 Scout and DeepSeek V3.

Overall, recent studies indicate that tokenisation remains a significant challenge for Sinhala despite being largely considered a solved problem for higher-resource languages. The evolution from conventional subword tokenisers to language-aware approaches that exploit graphemic and syllabic structure reflects a broader trend towards designing tokenisation methods that better accommodate the linguistic characteristics of low-resource languages.

3.6 Morphological Analysers

As shown in Fig. 5, morphological analysis is one of the foundational components of an NLP pipeline. Its importance is particularly pronounced for Sinhala, which exhibits rich inflectional morphology and complex word formation processes [40, 67, 68]. Accurate morphological analysis is therefore essential for numerous downstream tasks, including part-of-speech tagging, parsing, machine translation, information retrieval, and language modelling.

The earliest studies of Sinhala morphology were conducted by Herath et al. [78, 79]. These works focused primarily on formalising the morphological structure of Sinhala rather than developing computational tools. Consequently, their linguistic observations are discussed in Section 2. It is also worth noting that these studies predate the adoption of the Unicode standard for Sinhala and therefore represent Sinhala words using a Latin-script transliteration.

Building upon these linguistic foundations, Herath et al. [164] proposed a modular architecture for Sinhala morphological analysis. Subsequently, as part of their broader effort to develop an English-to-Sinhala machine translation system, Hettige and Karunananda [91] reported the development of a Sinhala morphological analyser that complemented their work on a Sinhala parser [92] and computational grammar [23]. Although these studies collectively describe a relatively comprehensive framework for Sinhala language processing, neither the implementation nor the underlying lexical resources were made publicly available. The same research group later proposed a multi-agent architecture for morphological analysis [103], further extending this line of work.

Several independent efforts subsequently explored alternative approaches to Sinhala morphological analysis. Welgama et al. [161] investigated the applicability of machine learning techniques, representing one of the earliest attempts to move beyond purely rule-based approaches. Independently, Fernando and Weerasinghe [162] developed a morphological parser specifically for Sinhala verbs. A similar verb-centric study was later presented by Dilshani and Dias [163]. However, neither work was extended to cover the full Sinhala lexicon, and no publicly accessible software or datasets were released beyond the associated publications.

Research has also addressed individual components of the morphological analysis pipeline. For example, Nandathilaka et al. [160] proposed a rule-based approach for Sinhala lemmatisation, while Welgama et al. [165] reported the development of a set of gold-standard definitions for Sinhala morphology. However, because the latter resource has not been made publicly available, its broader adoption and independent evaluation have not been possible. The table 5 provides a comparative summary of the discussion above.

Subsequent work focused on developing more practical and reusable morphological analysis tools. A combined study introduced both a rule-based Sinhala stemmer [166] and an enhanced tokenizer [167], addressing two closely related components of the morphological processing pipeline. Building upon these efforts, Kumarasinghe et al. [168] introduced *SinMorphy*²¹, a rule-based morphological analyser that leverages a comprehensive Sinhala vocabulary to improve lexical coverage and analysis accuracy. More recently, researchers have explored neural approaches to Sinhala morphological analysis. Ekanayaka et al. [169] compared recurrent neural network (RNN), Long Short-Term Memory (LSTM), and Gated Recurrent Unit (GRU) architectures using both Sinhala and Singlish datasets. Their experiments showed that a bidirectional GRU (BiGRU) model achieved the best performance, reporting an accuracy of 87.96%.

Beyond the development of analysers themselves, comparative linguistic studies have also sought to better understand Sinhala morphology in relation to other languages. For example, [170] compared the morphological characteristics of Sinhala with those of several other languages to facilitate the development of multilingual machine translation systems. Such studies provide valuable linguistic

21. <https://nlp-tools.uom.lk/sin-morphy/>

TABLE 5: **Morphological Analyzers comparison**

Base: Rule-based (RB) / Machine Learning (ML)

Able to Handle Part of Speech (Handles): Yes (Y) / No (N)

Outputs: Yes (Y) / No (N) / No Information (O)

Abbreviations: Nouns (Nu), Verbs (Ve), Adjectives (Aj), Adverbs (Av), Function Words (Fn), Root (R), Person (P), Number (Nb), Gender (G), Article (A), Case (C)

	Base	Modus Operandi	Handles					Output					
			Nu	Ve	Aj	Av	Fn	R	P	Nb	G	A	C
Hettige and Karunananda [91]	RB	Finite State Automata	Y	Y	N	N	N	Y	Y	Y	Y	Y	Y
Hettige et al. [103]	RB	Agent-based	Y	Y	Y	Y	Y	N	N	N	N	N	N
Nandathilaka et al. [160]	RB	N/A	Y	N	N	N	N	Y	N	N	N	N	N
Welgama et al. [161]	ML	Morfessor algorithm	Y	Y	Y	Y	Y	Y	N	N	N	N	N
Fernando and Weerasinghe [162]	RB	Finite State Transducer	N	Y	N	N	N	Y	Y	Y	Y	N	Y
Dilshani and Dias [163]	RB	N/A	N	Y	N	N	N	O	O	O	O	O	O

insights that can inform the design of future multilingual morphological analysers.

We can conclude that recent research demonstrates an increasing diversification of approaches to Sinhala morphological analysis, ranging from traditional rule-based systems to vocabulary-driven analysers and neural sequence models. Nevertheless, the scarcity of standardised evaluation benchmarks and the limited public availability of implementations continue to hinder direct comparison between competing methods. Consequently, despite more than three decades of research, there remains no widely adopted, openly maintained morphological analyser that has become the de facto standard for Sinhala NLP.

3.7 Part of Speech Taggers

Part-of-Speech (PoS) tagging is one of the fundamental stages of syntactic analysis, assigning each word in a sentence to its corresponding grammatical category. As with other morphologically rich languages, the design of an appropriate tag set is a prerequisite for developing accurate PoS taggers for Sinhala. Comprehensive Sinhala PoS tag sets have been proposed independently by Fernando et al. [171] and Dilshani et al. [172], providing a common foundation for subsequent research.

The availability of annotated datasets has enabled the development and evaluation of Sinhala PoS tagging systems. A PoS-tagged corpus containing more than 250,000 Sinhala words is publicly available through Google Groups²². In addition, another publicly available  [Sinhala PoS tagged dataset](https://github.com/fernando-silva/sinhala-pos-tagged-dataset) has been released [171–173].

Early research focused primarily on statistical sequence models. Among the earliest studies, Herath and Weerasinghe [174], Jayaweera and Dias [175], and Jayasuriya and Weerasinghe [176] investigated stochastic approaches for Sinhala PoS tagging. Building upon this line of research, Jayaweera and Dias [177] established the initial framework that was subsequently extended through a Hidden Markov Model (HMM)-based tagger [178]. Recognising that unseen words are a persistent challenge for morphologically rich languages, the same research group also investigated methods for handling unknown words [179, 180]. They later compared several Sinhala PoS tagging approaches [181] and released a RESTful PoS tagging web

service [182], which remains accessible²³ through standard HTTP POST and GET requests.

In parallel, other researchers explored alternative learning paradigms. Expanding upon the standardised tag sets, Fernando et al. [171] proposed an SVM-based Sinhala PoS tagger, while Fernando and Ranathunga [173] compared several machine learning classifiers for the same task. Although these studies represent one of the more sustained lines of research on Sinhala PoS tagging, much of the work appears to have remained within a single research group, with limited public release of the corresponding implementations and trained models. A hybrid PoS tagger combining multiple approaches was subsequently proposed by Gunasekara et al. [183]. More recently, Wijerathna [184] also employed Support Vector Machines (SVMs) using the 30-tag inventory proposed by Fernando et al. [171].

More recent studies have continued to investigate both data availability and modern learning techniques. The study by Kothalawala et al. [185] highlighted the broader problem of limited annotated resources for Sinhala NLP through a PoS tagging case study. To address data scarcity in informal text, Withanage and Silva [186] proposed a stochastic PoS tagger trained on a comparatively small corpus of 10,000 words collected from Facebook and Twitter. Sathsarani et al. [187] compared traditional rule-based and stochastic methods against deep learning models, reflecting the broader transition within NLP toward neural approaches for sequence labelling.

Overall, research on Sinhala PoS tagging has evolved from the establishment of standardised tag sets and annotated corpora to stochastic, hybrid, machine learning, and deep learning approaches. Nevertheless, compared with higher-resource languages, the field still suffers from a shortage of large, standardised benchmark corpora and widely adopted open-source implementations. Greater emphasis on publicly available datasets, reproducible evaluation protocols, and community-maintained tagging systems would substantially strengthen future research in Sinhala PoS tagging.

3.8 Parsers

The PoS tagged data then needs to be handed over to a parser. This is an area which is not completely solved even in English due to various inherent ambiguities in

22. <https://t.co/jpQz74YEdQ>

23. <http://bit.ly/2F0jKid>

natural languages. However, in the case of English, there are systems which provide adequate results [188] even if not perfect yet. The Sinhala state of affairs is that the first parser for the Sinhala language was proposed by Hettige and Karunananda [92] with a model for grammar [23]. The study by Liyanage et al. [68] is concentrated on the same, given that they have worked on formalising a computational grammar for Sinhala. While they do report reasonable results, yet again, do not provide any means for the public to access the data or the tools that they have developed. Kanduboda [67] have worked on Sinhala differential object markers relevant for parsing. A meta-study was conducted by Stephen and Zeman [189] on the Sinhala data from *Universal Dependencies (UD)* for identifying light verb constructions in Sinhala.

The first attempt at a Sinhala parser, as mentioned above, was by Hettige and Karunananda [92] where they created a prototype Sinhala morphological analyser and a parser as part of their larger project to build an end-to-end translator system. The function of the parser is based on three dictionaries: *Base Dictionary*, *Rule Dictionary*, and *Concept Dictionary*. They are built as follows:

- **The Base Dictionary:** *prakurthi* (base words), *nipatha* (prepositions), *upasarga* (prefixes), and *vibakthi* (Irregular Verbs).
- **The Rule Dictionary:** inflection rules used to generate various forms of verbs and nouns from the base words.
- **The Concept Dictionary:** synonyms and antonyms for the words found in the base dictionary.

Parsers are, in essence, a computational representation of the grammar of a natural language. As such, in building Sinhala parsers, it is crucial to create a computational model for Sinhala grammar. The first such attempt was taken by Hettige and Karunananda [23] with special consideration given to Morphology and the Syntax of the Sinhala language as an extension to their earlier work [92]. Here, it is worthy to note that, unlike in their earlier attempt [92], where they explicitly mentioned that they are building a parser, in this study [23], they use the much conservative claim of building a computational grammar. Under Morphology, they again handled Sinhala inflection. Their system is based on a Finite State Transducer (FST) and Context-Free Grammar (CFG) where they modeled 85 rules for nouns and 18 rules for verbs. The specific implementation is more partial to a rule-based composer rather than parser. It is also worthy to note that this system could only handle simple sentences which only contained the following 8 constituents: *Attributive Adjunct of Subject*, *Subject*, *Attributive Adjunct of Object*, *Object*, *Attributive Adjunct of Predicate*, *Attributive Adjunct of the Complement of Predicate*, *Complement of Predicate*, and *Predicate*. With these, they propose the following grammar rules for Sinhala:

```
S = Subject Akkyanaya
Subject = SimpleSubject | ComplexSubject
ComplexSubject = SimpleSubject ConSub
SimpleSubject = Noun | Adjective Noun
ConSub = Conjunction SimpleSubject
Akkyanaya = VerbP | Object VerbP
Object = SimpleObject | ComplexObject
ComplexObject = Conjunction SimpleObject
```

```
SimpleObject = Noun | Adjective Noun
VerbP = Verb | Adverb Verb
```

The later work by Liyanage et al. [68] also involves formalizing a computational grammar for Sinhala. They claim that Sinhala can have any order of words in practice. However, they do not note that this is happening because practices of the spoken language, which does not share the strong SOV conventions of the written language, are slowly seeping into written text. However, they do make note of how Sinhala grammar is modeled as a head-final language [69]. They propose the Sinhala Noun Phrase (*NP*) to be defined as shown in equation 1 where *NN* is a noun which can be of types: common noun (*N*), pronoun (*PrN*) or proper noun (*PropN*). The adjectival phrase (*ADJP*) is then defined as as shown in equation 2 where: *Det* is a Determiner, *Adj* is the adjective, and *Deg* is an optional operator *Degrees* which can be used to intensify the meaning of the adjective in cases where the adjective is qualitative. While they note that according to Gunasekara [32], there has to be three classes of adjectives (*qualitative*, *quantitative*, and *demonstrative*), they do not implement this distinction in their system. Similarly, they propose Sinhala Verb Phrase (*VP*) to be defined as shown in equation 3 where *V* is a *single verb*. They here note that they are ignoring *compound verbs* and *auxiliary verbs* in their grammar. The adverbial phrases (*ADVP*) are then recursively defined as as shown in equation 4.

$$NP = [ADJP][NN] \quad (1)$$

$$ADJP = \left[[Det] \left[[Deg][Adj] \right] \right] \quad (2)$$

$$VP = [ADVP][V] \quad (3)$$

$$ADVP = \left[[NP] \left[[ADVP] \left[[Deg][ADV] \right] \right] \right] \quad (4)$$

Similar to Hettige and Karunananda [23], the work by Liyanage et al. [68] also builds a CFG for Sinhala covering 10 out of the 25 types of simple sentence structures in Sinhala reported by Abhayasinghe [70]. This parser is unable to parse sentences where inanimate subjects do not consider the number. Further, sentences which contain, *compound verbs*, *auxiliary verbs*, *present participles*, or *past participles* cannot be handled by this parser. If the verbs have *imperative mood* or *negation* those too cannot be handled by this. Non-verbal sentences which end with *adjectives*, *oblique nominals*, *locative predicates*, *adverbials*, or any other language entity which is not a verb cannot be handled by this parser.

The study by Kanduboda [67] covers not the whole of Sinhala parsing but analyzes a very specific property of Sinhala observed by Aissen [190] which states that it is possible to notice Differential Object Marking (DOM) in Sinhala active sentences. Kanduboda [67] define this as the choice of */wa/* and */ta/* object markers. They further observe three unique aspects of DOM in Sinhala: (a) it is only observed in active sentences which contain transitive verbs, (b) it can occur with accusative marked nouns but not with any other

cases, (c) it exists only if the sentence has placed an animate noun in the accusative position. They do a statistical analysis and provide a number of short gazetteer lists as appendixes. However, they observe that further work has to be done for this particular language rule in Sinhala given that they found some examples which proved to be exceptions to the general model which they proposed.

3.9 Spelling and Grammar

Reliable benchmark datasets are essential for evaluating and comparing spelling correction systems. To this end, Sonnadara et al. [191] introduced a  [benchmark dataset](#) for Sinhala spelling correction together with a neural baseline model. This resource was subsequently  [extended](#) by Gunathilake et al. [192]. Although the newer dataset is hosted on Hugging Face, access currently requires submitting a request that must be approved by the dataset authors.

3.9.1 Sinhala Spelling

Early research on Sinhala spelling correction focused primarily on data-driven and rule-based approaches. The open-source systems proposed by Wasala et al. [34, 193] introduced one of the earliest publicly described data-driven spelling correction frameworks for Sinhala. Subsequently, Jayalatharachchi et al. [194] proposed combining two complementary correction algorithms to improve spelling correction performance, and this line of research [34, 194] was further extended by Subhagya et al. [195].

Rule-based spell checking has also received considerable attention. Liyanapathirana et al. [196] introduced *SinSpell*, a Sinhala spell checker built on top of *Hunspell*²⁴. In addition to describing the system, the authors provide an online demonstration²⁵, making it one of the few publicly accessible Sinhala spell-checking services. Similarly, Samarawickrama et al. [197] proposed the multi-agent spelling correction system *LaSi Spell*, while Sithamparanathan and Uthayasanker [198] extended the *Generic Environment for Context-Aware Spell Correction* to support both Sinhala and Tamil.

Several studies have addressed specialized spelling correction problems. For example, Udagedara et al. [199] focused specifically on correcting Sri Lankan names and postal addresses, while Sudesh et al. [200] introduced *Erroff*, a system designed to detect and correct real-word spelling errors in Sinhala text. More recently, Praveen et al. [201] reported a Random Forest-based spell checker achieving an accuracy of 82%.

The recent success of large language models has also influenced Sinhala spelling correction. Gunathilake et al. [192] introduced  [LMSpell](#), an LLM-based toolkit for Sinhala spelling correction. In addition to releasing the toolkit, the authors provide public access to the corresponding  [trained models](#) and an online demonstration²⁶. This work represents the latest transition from traditional rule-based and statistical approaches toward foundation-model-based spelling correction.

24. <http://hunspell.github.io/>

25. <http://nlp-tools.uom.lk/sinspell/>

26. <https://nisansa.lk/?s=eDbXYk>

3.9.2 Sinhala Grammar

Compared with spelling correction, grammatical error detection and correction has received considerably less attention in Sinhala NLP. One of the earliest efforts was presented by Pabasara and Jayalal [202], who developed a model for detecting grammatical errors. This work was subsequently extended into a grammatical error detection and correction system [203]. Around the same period, Fernando and Arudchelvam [204] proposed a Hidden Markov Model (HMM)-based Sinhala grammar checker.

Alternative approaches have also been explored. Rather than treating grammar correction as a classification task, Widyaratna [205] formulated it as a sequence-to-sequence translation problem, using an attention-based encoder-decoder model to transform grammatically incorrect sentences into their corrected forms. Meanwhile, Jayasuriya et al. [206] combined rule-based methods with Google Translate to perform grammatical correction, illustrating a hybrid approach that leverages existing machine translation systems.

Research has also addressed more specialized grammatical transformations. For example, Ilukkumbura and Rupasinghe [207] proposed a rule-based system for converting Sinhala sentences from active voice to passive voice while simultaneously correcting grammatical errors. Although designed for a specific transformation, this work demonstrates how grammar correction techniques can be integrated with broader text rewriting tasks.

While not directly addressing grammar correction, Gunasekara et al. [208] investigated annotation projection for semantic role labeling in Sinhala. Although this work falls outside the traditional scope of grammar checking, it contributes to the broader understanding of Sinhala syntax and grammatical structure.

3.9.3 Sinhala Spelling and Grammar Both

Several systems attempt to jointly address spelling and grammatical errors within a unified framework. Goonawardena et al. [209] proposed a rule-based system capable of both spell checking and grammatical error detection and correction. Similarly, Navoda et al. [210] reported the development of an automated tool that simultaneously performs Sinhala spelling and grammar checking.

3.9.4 Other Relevant Applications

Research on error correction has also influenced several related text restoration tasks. Disara and Ranathunga [211] fine-tuned a RoBERTa model using a physics-domain Sinhala corpus derived from G.C.E. Advanced Level educational materials and evaluated its ability to restore missing or corrupted words in Sinhala sentences. More recently, Fernando and Ganegoda [212] employed ByT5 [213] to reconstruct modern Sinhala sentences from direct Brahmi-Sinhala translations. Their approach used one fine-tuned ByT5 model to recover the correct word order and a second model to perform morphological correction.

3.10 Named Entity Recognition Tools

The development of Sinhala Named Entity Recognition (NER) has been supported by a small but growing collection of annotated datasets. The  [multiNER](#) dataset

introduced by Ranathunga et al. [214] provides parallel NER annotations for Sinhala, Tamil, and English and contains approximately 99,000 tagged Sinhala words. It follows a conventional annotation scheme comprising the entity classes PER, ORG, LOC, and MISC, together with the non-entity label O. The  *Ananya* dataset, created as part of the work by Manamini et al. [215], is also publicly available.

As shown in Fig. 5, NER forms an important stage in the extraction of structured information from processed text. The earliest identified attempt to develop an NER system for Sinhala was presented by Dahanayaka and Weerasinghe [216]. As no established Sinhala NER framework was available at the time, the study adapted methods previously developed for other languages, with particular attention to research on Indic languages. An important observation made in this work was that NER can be more difficult for Sinhala and other Indic languages than for English because Sinhala does not use capitalization as a cue for identifying proper nouns and named entities. Following the dominant sequence-labelling approaches of the period, they employed Conditional Random Fields (CRFs) and compared their performance against a Maximum Entropy (ME) baseline. The feature representation was relatively limited, consisting primarily of the *candidate word*, its surrounding *context words*, and basic information derived from Sinhala *suffixes*.

The follow-up study by Senevirathne et al. [217] retained the CRF architecture and the original feature set, although it did not report a corresponding comparison with an ME model. Its principal contribution was the introduction of a richer collection of language-sensitive features. These included the *length of the word*, represented using a threshold-based feature, and a *first word* feature motivated by regularities in Sinhala sentence structure. The study also refined the use of contextual information by introducing *clue words*, a selected subset of context words expected to provide stronger evidence for entity identification. In addition, the *previous map* feature incorporated the predicted named-entity label of the preceding word. Several of these features were supported by a rule-based post-processing component that relied on context-sensitive word lists.

The third major Sinhala NER system, *Ananya*, was introduced by Manamini et al. [215]. This work retained both the CRF model and the ME baseline used by Dahanayaka and Weerasinghe [216], while substantially extending the feature inventory developed by Senevirathne et al. [217]. A *frequency of the word* feature was introduced based on the assumption that highly frequent words are less likely to be named entities. Word frequency was therefore converted into a Boolean feature through the application of a predefined threshold. The positional feature introduced in the earlier work [217] was also expanded from *first word* to *first word/last word of a sentence*, with the latter motivated by the predominantly subject-object-verb (SOV) structure of Sinhala. In line with NER research for other languages, *Ananya* additionally incorporated a *Part-of-Speech (PoS) tag* feature and features derived from *gazetteer lists*. The *clue words* feature, which had previously been treated as part of the broader contextual representation by Dahanayaka and Weerasinghe [216], was formalized as an independent feature. Because *Ananya* retained the ME component unlike Dahanayaka and Weerasinghe [216], Manamini et al.

[215] were also able to introduce an *outcome prior* feature complementary to the *previous map* feature. This feature exploited the underlying probability distribution produced by the ME model. Finally, a *cutoff value* feature was incorporated to reduce overfitting by excluding features whose frequencies fell below a specified threshold.

Table 6 provides a comparative summary of the three early Sinhala NER systems discussed above. All three systems were restricted to three principal entity categories: *person names*, *location names*, and *organisation names*. The dataset associated with the  *Ananya* system by Manamini et al. [215] is publicly available. In contrast, the data and implementations associated with Dahanayaka and Weerasinghe [216] and Senevirathne et al. [217] are not publicly accessible.

Subsequent research expanded Sinhala NER beyond the relatively coarse entity inventories and feature-engineered models used in the earlier systems. Building upon their previous work on Sinhala NER [215] and PoS tagging [173], Azeez and Ranathunga [218] proposed a fine-grained NER model for Sinhala. This work represents an important shift from recognizing only broad categories such as persons, locations, and organizations toward distinguishing more specific entity types.

Alternative learning paradigms were also explored. Anuruddha [219] proposed a reinforcement-learning-based method for Sinhala NER, extending the task beyond conventional supervised sequence-labelling approaches. In parallel, Mallikarachchi et al. [220] employed Support Vector Machines to identify Sinhala named entities belonging to the *person*, *location*, and *organisation* categories. Although the entity inventory remained similar to that of the earlier systems, this work provided an additional classical machine learning formulation of the task.

Domain-specific NER has received comparatively limited attention. One notable exception is the sports-domain Sinhala NER system introduced by Wijesinghe and Tissera [221], who evaluated classical machine learning models for identifying entities in sports-related text. The restricted domain can reduce some of the ambiguity associated with general-domain NER, but it also limits the direct applicability of the resulting models to other genres.

Sinhala NER has additionally been incorporated as an intermediate component in broader NLP applications. For example, although the principal task investigated by [222] was the clustering of Sinhala news articles, named-entity information was used to support the clustering process.

More recent work has examined the use of pretrained and large language models for Sinhala NER. Gurgurov et al. [223] compared several adaptation strategies for applying language models to the task. Their strongest results were obtained using a masked language modelling objective for Sinhala, with combinations of ConceptNet (CN) and Wikipedia (*wiki*) language adapters together with task adapters (TA).

In contrast, Rathnayake and Asanka [224] applied LLaMA directly to Sinhala NER. Although the effectiveness of such approaches depends on factors such as prompt design, model coverage of Sinhala, and the availability of suitable evaluation data.

The development of Sinhala NER should also be viewed within the wider context of NER research for South Asian

TABLE 6: NER system comparison

* Denotes a baseline.

 F_1 to F_{11} denotes Context Words, Word Prefixes and Suffixes, Length of the Word, Frequency of the Word, First Word/ Last Word of a Sentence, (POS) Tags, Gazetteer Lists, Clue Words, Outcome Prior, Previous Map, and Cutoff Value

	CRF	ME	Features										
			F_1	F_2	F_3	F_4	F_5	F_6	F_7	F_8	F_9	F_{10}	F_{11}
Dahanayaka and Weerasinghe [216]	Yes	Yes*	Yes	Yes	No	No	No	No	No	No	No	No	No
Senevirathne et al. [217]	Yes	No	Yes	Yes	Yes	No	Yes	No	No	Yes	No	Yes	No
Manamini et al. [215]	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

languages. Sinhala is discussed alongside other Indic languages in the survey by Deshmukh et al. [225], which situates the challenges faced by Sinhala, including limited annotated data, weak capitalization cues, and restricted tool availability, within a broader multilingual perspective.

3.11 Word Sense Disambiguation

Word Sense Disambiguation (WSD) is the task of determining the intended meaning of an ambiguous word from its surrounding context [226–230]. As one of the core semantic analysis tasks in NLP, WSD underpins numerous downstream applications, including information retrieval, question answering, machine translation, and semantic search. Despite its importance, relatively few studies have addressed Sinhala WSD compared with higher-resource languages.

One of the earliest dedicated Sinhala WSD systems was *Aruth*, proposed by Arukgoda et al. [150]. The system adopts the Lesk algorithm [231], a knowledge-based approach that identifies the most appropriate word sense by measuring the overlap between dictionary definitions and the surrounding context. Unlike many early Sinhala NLP systems, *Aruth* was accompanied by several publicly accessible resources, including an online demonstration²⁷, a publicly available  API, and both the source code and lexical resources hosted on  GitHub. These releases have made *Aruth* one of the most reproducible Sinhala WSD systems reported in the literature.

Alternative approaches have also been investigated. Marasinghe et al. [232] proposed a probabilistic model for Sinhala WSD, representing an early attempt to move beyond purely knowledge-based methods toward statistical inference. Although the published literature provides relatively limited details regarding the long-term development of this approach, it illustrates the exploration of probabilistic techniques for lexical ambiguity resolution. Research related to lexical ambiguity has also appeared within broader semantic processing tasks. For example, Palihakkara et al. [233] developed a dialogue act recognition system that employed comparatively simple classification algorithms. While dialogue act recognition extends beyond conventional WSD, both tasks rely on contextual semantic interpretation and therefore share many underlying challenges.

More recently, Subasingha [234] introduced the WSD system *SinSense*. Rather than performing disambiguation directly in Sinhala, the proposed method adopts a cross-lingual strategy in which English word sense disambiguation is first performed and the resulting sense assignments

are then transferred to Sinhala. This approach enables Sinhala WSD to benefit from the substantially richer lexical resources available for English. However, neither the implementation nor the associated datasets have been made publicly available.

Overall, Sinhala WSD research has evolved from knowledge-based algorithms to probabilistic and cross-lingual approaches. Nevertheless, progress has been constrained by the absence of a comprehensive Sinhala WordNet, the scarcity of large annotated sense corpora, and the limited availability of openly released implementations. Consequently, reproducible evaluation remains challenging, and WSD continues to be one of the least explored areas of Sinhala semantic processing despite its importance to many downstream NLP applications.

3.12 Semantic Similarity

Measuring semantic similarity between words, sentences, or documents is a fundamental task in NLP, underpinning applications such as information retrieval, duplicate detection, question answering, automatic grading, and semantic search. Compared with other core NLP tasks, research on semantic similarity for Sinhala remains relatively limited, although the field has gradually evolved from rule-based sentence similarity measures to neural representation learning and multilingual evaluation benchmarks.

One of the earliest studies on Sinhala semantic similarity was conducted by Kadupitiya et al. [235], who proposed a semantic similarity measure for short Sinhala sentences. This work was subsequently extended by Kadupitiya et al. [236] to support the practical application of automatic *short-answer grading*. Although these studies established an important foundation for semantic similarity research in Sinhala, neither the corresponding datasets nor the software implementations have been made publicly available.

Later research shifted towards neural representation learning. Nilaxan and Ranathunga [237] developed a sentence similarity model based on Siamese neural networks and demonstrated its applicability to both Sinhala and Tamil. Building upon this direction, Deepal et al. [238] proposed a hybrid Siamese architecture combining Long Short-Term Memory (LSTM) and Convolutional Neural Networks (CNNs) for Sinhala sentence similarity estimation. Their model achieved an F1-score of 0.9041 using cosine similarity as the evaluation metric. Although the authors describe how several publicly available datasets were refined to construct their training corpus, the resulting processed dataset has not been released.

Semantic similarity has also been investigated in multilingual settings. Isuranga et al. [239] proposed a cross-

27. <http://aruth.herokuapp.com/>

lingual document similarity framework for Sinhala and English, extending semantic similarity beyond monolingual sentence comparison to multilingual document retrieval and comparison. Such approaches are particularly valuable for low-resource languages, where multilingual transfer can compensate for the limited availability of large annotated corpora.

More recently, emphasis has shifted toward standardized multilingual evaluation. Ranasinghe et al. [240] introduced the  **MUSTS** benchmark, a multilingual semantic textual similarity dataset that includes Sinhala. The Sinhala portion of the benchmark is derived from the short-sentence similarity dataset originally developed by Kadupitiya et al. [236], providing one of the first standardized resources for evaluating multilingual semantic similarity systems involving Sinhala.

Overall, research on semantic similarity for Sinhala has progressed from handcrafted sentence-level similarity measures to neural representation learning and multilingual benchmarking. Nevertheless, the field continues to be constrained by the limited public availability of annotated datasets and reproducible implementations. Larger and more diverse semantic similarity datasets remain necessary to support the development and comparison of modern embedding-based and large language model approaches.

3.13 Embeddings and Small Language Models

Distributed representations have become the dominant paradigm for representing linguistic information in modern NLP. Unlike manually engineered lexical features, embeddings encode semantic and syntactic properties in continuous vector spaces, enabling models to generalize beyond individual words. The evolution of Sinhala embeddings mirrors the broader development of NLP, progressing from static word embeddings to contextual transformer models and, more recently, task-specific Small Language Models (SLMs).

The earliest publicly available pretrained embeddings for Sinhala were released by Facebook as part of the FastText project [241–243]. These models, trained on the Sinhala Wikipedia corpus, are available both as text models²⁸ and binary models²⁹. Building upon these embeddings, Lakmal et al. [130] trained an extended FastText model using a substantially more diverse corpus consisting of Wikipedia articles, news publications, and official government documents. The resulting binary model³⁰ has also been released publicly.

The availability of pretrained embeddings naturally led to comparative studies of their effectiveness. Lakmal et al. [130] conducted one of the earliest evaluations of Sinhala word embeddings, while Silva [244] revisited the problem several years later to assess the progress achieved by subsequent developments in Sinhala NLP. Cross-lingual representation learning has likewise received attention. For example, Liyanage et al. [245] analysed Sinhala FastText and Word2Vec embeddings within bilingual embedding spaces,

investigating their suitability for multilingual representation learning.

Research has also extended beyond word-level representations to sentence embeddings. Weeraprameshwara et al. [246] proposed a two-tier architecture for generating Sinhala sentence embeddings and reported that a *Seq2Seq* Gated Recurrent Unit (GRU) model with attention operating on FastText embeddings achieved the strongest performance for Sinhala sentence embedding. More recently, Ranaweera and Dayaratna [247] introduced *AshenBERTo*, a RoBERTa model specifically pretrained for Sinhala sentence similarity estimation.

The emergence of transformer-based language models significantly changed the landscape of Sinhala representation learning. Dhananjaya et al. [120] released two pretrained Sinhala BERT models,  *SinBERT-large* and  *SinBERT-small*, providing dedicated transformer encoders for Sinhala. A further independently developed model,  *SinhalaBERTo*, was trained by adapting the RoBERTa architecture [248] using the OSCAR corpus [117, 118]. Although this model has no accompanying research publication, it demonstrates the growing community interest in developing publicly available pretrained Sinhala language models.

Several studies have subsequently explored multilingual embedding resources and parameter-efficient adaptation techniques. Demotte et al. [249] publicly released both the  *word embeddings* and the accompanying  *implementation* used in their embedding-based Sinhala sentiment analysis system. These embeddings were later incorporated into the multilingual *LowREm* repository [250], which provides embedding resources for 87 low-resource languages and explicitly credits Demotte et al. [249] as the Sinhala source. Building upon multilingual pretrained models, Gurgurov et al. [251] argued that multilingual language models (mLMs) such as mBERT and XLM-R are better suited than contemporary large language models for many low-resource languages, including Sinhala, and proposed parameter-efficient adapter-based methods for their adaptation. Their implementation has been made publicly available on  *GitHub*.

As pretrained representations became increasingly widespread, standardized evaluation benchmarks also began to emerge. Enevoldsen et al. [252] introduced  *MMTEB*, together with a public  *leaderboard* for multilingual embedding evaluation. The benchmark includes Sinhala through the news classification dataset developed by de Silva [40]. More recently, Ranasinghe et al. [253] introduced  *SINHALA-GLUE*, a comprehensive language understanding benchmark accompanied by three new Sinhala encoder models: *Sinhala-BERT* (Raja), *Sinhala-RoBERTa* (Koliya), and *Sinhala-Electra* (Mahasen). These models are not publicly available.

The recent emergence of Small Language Models (SLMs) has further expanded the scope of Sinhala representation learning. Devine et al. [254] introduced *Kakugo*, a general-purpose SLM training pipeline evaluated across multiple languages, including Sinhala. While the absolute performance obtained for Sinhala is moderate, the authors report consistently favourable pairwise win rates for the *Kakugo*

28. <http://bit.ly/2JXAYL8>

29. <http://bit.ly/2JY5J9c>

30. <http://bit.ly/2WowH0h>

models, suggesting that carefully trained SLMs can provide competitive performance even for relatively low-resource languages.

Overall, Sinhala representation learning has evolved rapidly from static FastText embeddings to contextual transformer encoders, multilingual adaptation methods, benchmark suites, and dedicated Small Language Models. Compared with many other areas of Sinhala NLP, this is one of the fastest-moving research directions. Nevertheless, challenges remain in terms of publicly releasing pretrained models, establishing standardized evaluation protocols, and developing sufficiently large pretraining corpora. Continued progress in these areas is likely to benefit virtually every downstream Sinhala NLP task discussed throughout this survey.

3.14 Embedding Alignment and Bilingual Lexicon Induction

Embedding alignment and Bilingual Lexicon Induction (BLI) seek to map the vector spaces of different languages into a common semantic space, thereby enabling cross-lingual transfer, multilingual information retrieval, and machine translation without requiring large parallel corpora. For low-resource languages such as Sinhala, these techniques are particularly valuable because they leverage the abundant lexical resources available for higher-resource languages while minimizing the need for expensive manual annotation.

One of the first comprehensive studies of Sinhala–English embedding alignment was conducted by Wickramasinghe and de Silva [255]. Using the Sinhala–English dictionary introduced by Wickramasinghe and De Silva [141], the authors established one of the first benchmark datasets for Sinhala–English embedding alignment and compared its performance with several widely studied language pairs, including English–Spanish, English–French, English–German, English–Russian, and English–Chinese. To facilitate reproducible research, both the benchmark data and the corresponding implementation have been released on  [GitHub](#).

Subsequent work shifted attention toward improving alignment algorithms. Rathnayake et al. [256] investigated unsupervised BLI for the Sinhala–English language pair using Unsupervised VecMap (UVecMap) [257] as the baseline alignment method. Their experiments demonstrated that combining UVecMap with linear transformation achieved the strongest performance for English–Sinhala Word2Vec embeddings, reporting a score of 33.18. For English–Sinhala FastText embeddings, the highest score of 29.49 was obtained by combining UVecMap, linear transformation, and CSCBLI [258].

More recently, Wickramasinghe and de Silva [259] conducted a broader comparative study of multiple BLI techniques across several language pairs. Their experiments showed that Language-agnostic BERT Sentence Embeddings (LaBSE) [260] combined with Cross-domain Similarity Local Scaling (CSLS) [261] consistently achieved the strongest overall performance. In addition to comparing existing methods, the authors proposed vocabulary pruning and stemming as effective preprocessing techniques when

one or both languages exhibit rich inflectional morphology, as is the case for Sinhala.

Embedding alignment has also been explored in broader multilingual semantic representation settings. Yu [262] employed the FLORES-200 [263] dataset to evaluate a cross-model semantic alignment framework and demonstrated improvements over both CLIP [264] and ALIGN [265]. Notably, Sinhala was identified as one of the languages exhibiting the largest improvements in Recall@1 and Mean Rank.

Overall, research on Sinhala embedding alignment has progressed from establishing benchmark datasets to developing increasingly sophisticated alignment algorithms and multilingual evaluation frameworks. A recurring theme across these studies is that the linguistic characteristics of Sinhala, particularly its rich morphology, necessitate language-aware preprocessing strategies in addition to generic multilingual alignment methods. As multilingual language models continue to improve, embedding alignment and bilingual lexicon induction are likely to remain key enabling technologies for cross-lingual transfer involving Sinhala.

3.15 Text Classification

Text classification is one of the most widely studied applications at the semantic layer of the NLP stack. Sinhala research in this area spans news categorization, source and writing-style identification, authorship attribution, fake-news detection, code-mixed text analysis, textual entailment, and topic discovery. The field has progressed from rule-based and conventional machine learning methods to neural architectures, multilingual transformers, parameter-efficient adaptation, and active learning. However, the availability and quality of datasets remain uneven across these applications.

Several publicly accessible datasets support Sinhala text classification research. Although the primary objective of Ranasinghe [158] was to develop a Sinhala tokenizer rather than a classification benchmark, the associated repository contains a  [dataset](#) of approximately 26,000 labelled Sinhala news titles. An earlier and considerably smaller implementation of Sinhala news classification was presented by de Silva [40]. As discussed in Section 3.2, the corresponding  [news corpus](#) is publicly available. A further set of classification resources was released by Dhananjaya et al. [120]. The authors derived three datasets from previously published corpora: a  [Sinhala news source classification](#) dataset based on the corpus introduced by Sachintha et al. [266]; a second  [Sinhala news source classification](#) dataset derived from the corpus of de Silva [40]; and a  [Sinhala writing style classification](#) dataset constructed from the corpus introduced by Upeksha et al. [28]. For authorship analysis, Faumi et al. [267] released  [SimpleAA](#), a writing-style identification dataset that also includes Romanised (transliterated) Sinhala text. Although the research paper associated with Ekanayaka et al. [268] appears to be unavailable now, the corresponding news corpus has been released through  [GitHub](#). It contains 1,000 labelled and 50,000 unlabelled Sinhala news reports

collected from multiple sources, while the wider repository also provides some of the associated  [code](#).

Early Sinhala text classification research predominantly relied on conventional statistical and rule-based methods. Gallege [269] investigated basic classification using Naïve Bayes, Support Vector Machines (SVMs), and features motivated by Zipf’s law. Lakmali and Haddela [270] subsequently compared six popular rule-based algorithms and reported the construction of a corpus named *SinNG5*, although no mechanism was provided for obtaining the corpus. Kumari and Haddela [271] later used *SinNG5* to investigate the application of LIME [272] to interpretable Sinhala document classification, but likewise did not release the data. In a related direction, Nanayakkara and Ranathunga [273] developed a classification system based on corpus-derived similarity measures.

Feature selection and document representation were also central themes in this early research. Gunasekara and Haddela [274, 275] proposed a context-aware stop-word extraction method for Sinhala text classification based on TF-IDF, while Koralage [276] presented a simpler TF-IDF-based classification system. Haddela et al. [277] adopted a different approach by applying a Genetic Algorithm to Lucene³¹ search queries in order to obtain interpretable classification models for Sinhala documents. Later, Weerasiri et al. [278] compared Word2Vec, FastText, and Doc2vec [279] representations against TF-IDF for Sinhala news classification.

Topic-based methods have similarly been used for organizing Sinhala news. Kirindage and Godewithana [280] employed Latent Dirichlet Allocation (LDA) [281] to construct Sinhala news topic hierarchies and classify documents according to the resulting structure. Sameemdeen and Selvanthan [282] considered Naïve Bayes, SVM, and KNN while also reviewing earlier work by Nanayakkara and Ranathunga [273], Gunasekara and Haddela [274], Lakmali and Haddela [270], and Buddhika et al. [283]. The authors proposed active learning [284, 285] as an alternative for reducing annotation requirements, although no experiments were provided to demonstrate its effectiveness for Sinhala text classification. More recent work by Kariyakarane and Athuraliya [286] used Sinhala text classification to evaluate CAL-Log, a logarithmic model of annotation cost for active learning, thereby providing a more explicit investigation of annotation efficiency.

Several studies proposed classification frameworks intended to generalize across domains. Kodithuwakku and Hettiarachchi [287] introduced *AdaptText*, a domain-independent and domain-adaptive framework for Sinhala text classification. Bandara et al. [288] proposed an ontology-based approach to Sinhala fake-news detection. However, its literature review omitted foundational work in ontology-based information extraction, including Wimalasuriya and Dou [49], which weakened the methodological grounding of the proposed system.

The adoption of neural methods broadened both the models and tasks explored for Sinhala. Jayasinghe and Sirts [289] proposed an LSTM-based system for Sinhala textual entailment. Demotte and Ranathunga [290] intro-

duced a dual-state capsule network architecture and evaluated it using the Sinhala dataset established by Senevirathne et al. [291]. Capsule-based methods were also used by Chathuranga and Ranathunga [292] to classify Sinhala-English code-mixed text, following the approach recommended by Senevirathne et al. [291]. For social-media content, Caldera et al. [293] employed stacked LSTMs to classify Sinhala and Singlish discussions of COVID-19 collected from YouTube and Twitter.

Transformer-based models subsequently became an important focus. Dhananjaya et al. [120] conducted a comparative evaluation of BERT-based architectures for Sinhala text classification and publicly released the corresponding  [code](#). For code-mixed and code-switched classification, Rathnayake et al. [294] applied adapter-based fine-tuning [295–300] to XLM-R [301], demonstrating how parameter-efficient adaptation can be used when fully fine-tuning large multilingual models is impractical. More recently, Ilangeshwaran et al. [302] introduced Saliency-based token Swap (SSwap), a data augmentation method designed to improve Sinhala text classification, while Wickramasinghe et al. [303] proposed a SinBERT-based model for Sinhala news personalization.

Text classification has also been applied to several specialized tasks. Hettigoda [304] collected English-Sinhala code-mixed comments from Facebook pages operated by online businesses in Sri Lanka’s clothing industry and classified them as *Inquiries*, *Maybe Inquiries*, or *Not Inquiries*. Wijayarathna and Jayalal [305] used classical machine learning methods, including Random Forest, to classify Sinhala fake news on Twitter. Ranathunga et al. [306] investigated writing-style identification specifically for distinguishing AI-generated from human-written Sinhala answers and reported an accuracy of 86% using an LSTM operating on TF-IDF features. In literary authorship attribution, Fernando and Pitigala [307] applied a CNN to TF-IDF representations to identify the authors of Sinhala poems.

Other work has extended classification beyond assigning conventional document labels. Gunawardana and Thamarasee [308] attempted to predict the future popularity of Sinhala-language YouTube videos using multiple associated factors. Rathnayake and Asanka [224] used LLaMA to classify named entities in Sinhala documents before applying Fuzzy C-Means clustering and subsequently using LLaMA again to generate names for the resulting clusters.

Overall, Sinhala text classification has developed from small-scale rule-based and conventional machine learning experiments into a broad research area encompassing neural models, multilingual transformers, data augmentation, parameter-efficient adaptation, and active learning. News remains the dominant domain, while writing-style analysis, fake-news detection, code-mixed classification, and social-media applications have received increasing attention. Despite the growing number of publicly released corpora, several influential datasets remain unavailable. Progress would therefore benefit from consolidated benchmark suites, clearer dataset provenance, standardized train-test divisions, and evaluations that compare classical, transformer-based, and generative approaches under consistent experimental conditions.

31. <https://lucene.apache.org/>

3.16 Sentiment Analysis

Sentiment analysis is among the most extensively studied semantic processing tasks in Sinhala NLP. Research has covered binary and multiclass sentiment classification, emotion recognition, reaction prediction, aspect-based sentiment analysis, code-mixed social-media content, and multimodal emotion analysis. The methodological progression broadly follows developments in NLP more generally, moving from lexicon-based and conventional machine learning approaches to recurrent neural networks, multilingual transformers, language adapters, explainable models, and fine-tuned large language models.

Several datasets have supported this research. Je-nanthanan et al. [309] introduced the  ACTSEA dataset, which contains 318,308 Sinhala tweets annotated with emotion labels. More recently, Galwatta et al. [310] released SalAngaBhava, an aspect-tagged Sinhala sentiment analysis dataset available through both  GitHub and  HuggingFace. The dataset contains 10,989 customer-feedback records from online marketplaces, of which 1,858 were manually annotated. Using InstructABSA [311], the authors established baseline Macro F1-scores of 0.72 for Aspect Term Extraction (ATE) and 0.28 for Aspect-Level Sentiment Classification (ALSC). Most Sinhala sentiment datasets employ discrete sentiment or emotion categories. In contrast, De Mel and de Silva [312] introduced  GeeSanBhava by manually annotating the raw Sinhala YouTube comment dataset released by De Mel and de Silva [128]. This is the first reported Sinhala sentiment resource annotated according to Russell's Valence-Arousal model [313], providing a continuous dimensional representation of emotion rather than relying solely on predefined sentiment classes.

Early computational work on Sinhala sentiment analysis employed relatively simple neural and lexical methods. Building upon their earlier study [314], Medagoda [315] proposed a multilayer perceptron-based method for classifying sentiment in Sinhala text. A publicly available  Word2Vec-based tool was also developed for analysing Sinhala news comments. To address the scarcity of manually constructed lexical resources, Chathuranga et al. [316] proposed a semi-automated methodology for deriving a Sinhala sentiment lexicon from a given corpus.

Subsequent research increasingly adopted recurrent neural architectures and distributed representations. Demotte et al. [249] introduced a  sentiment analysis system based on sentence-state LSTM networks for Sinhala news comments. In follow-up work, Karunanayake et al. [317] used word similarity to construct a Sinhala semantic lexicon, while Senevirathne et al. [291] evaluated several deep learning architectures, including recurrent neural networks and bidirectional LSTMs, for Sinhala sentiment analysis. Ranathunga and Liyanage [318] further argued that word embeddings can serve as effective semantic features and partially compensate for the limited availability of Sinhala-specific linguistic resources when analysing sentiment in news comments.

Several studies compared alternative lexical representations and classification strategies. Jayasuriya et al. [319] compared word-level and character-level n -grams for the

sentiment classification of Sinhala social-media content and subsequently extended the work using an ensemble approach [320]. Liyanaarachchi et al. [321] investigated character-level embeddings for binary sentiment classification of Sinhala news comments and reported that the 1-of- m representation outperformed the Log- m representation. More recently, the short comparative study by Mohamed et al. [322] found that XLM-R and SinBERT achieved the strongest overall performance on Sinhala comment sentiment analysis, although BiLSTM and Logistic Regression remained competitive for certain tasks.

Social-media sentiment has been studied across several platforms and domains. Karunarathne [323] used word embeddings to analyse manually annotated Sinhala tweets, while Jayasuriya et al. [324] classified Sinhala sports-related posts into positive and negative sentiment categories. Jayawickrama et al. [325, 326] used the Facebook corpus released by Wijeratne and de Silva [122] to predict the reactions elicited by Sinhala posts. This work was subsequently extended by Weeraprameshwara et al. [327], who compared results obtained from the Facebook reaction dataset against those obtained using the sentiment dataset of Senevirathne et al. [291]. Sentiment-related classification has also been applied to music: Abeyratne and Jayaratne [328] conducted a multimodel analysis of emotion classification for Sinhala songs.

Code-mixed Sinhala-English content has become another prominent research direction. Aththanayaka and Naleer [329] applied Random Forest, Support Vector Machines, and Multinomial Naïve Bayes to sentiment analysis of Sinhala-English code-mixed social-media text. Uthpala and Thirukumaran [330] created a sentiment-annotated dataset of Sinhala-English code-mixed YouTube comments. However, both the publication and the dataset are behind access restrictions, and the resource is therefore not freely available for external research.

More recent work has applied multilingual pretrained language models to compensate for the scarcity of Sinhala-labelled data. Dhananjaya et al. [331] used the sentiment lexicon of a high-resource language to fine-tune pretrained multilingual language models, including mBERT and XLM-R, on an intermediate task before adapting them to Sinhala sentiment classification. Gurgurov et al. [223] compared language-model adaptation strategies for Sinhala sentiment analysis and obtained the strongest results using a masked language modelling objective. Their configurations combined ConceptNet (CN) and/or Wikipedia (*wiki*) language adapters with task adapters (TA).

Transformer models have also been evaluated directly on established Sinhala sentiment datasets. Bandaranayake and Usoof [332] compared BERT, DistilBERT, RoBERTa, and XLM-R using the Sinhala news-comment dataset introduced by Senevirathne et al. [291], reporting that XLM-R-large achieved the strongest performance. In their work on improving the cross-lingual representations of multilingual language models through linguistic entity masking, Fernando and Ranathunga [333] used Sinhala code-mixed sentiment analysis as one of the evaluation tasks. More recently, SinLlama [334] was fine-tuned by Abeynayake et al. [335] for sentiment analysis of Romanised Sinhala-English code-mixed social-media text.

Aspect-based sentiment analysis and explainability have received increasing attention. Rizvi et al. [336] used XLM-RoBERTa to extract keywords and classify aspects in Sinhala code-mixed data. In subsequent work [337], the authors again employed XLM-RoBERTa for sentiment classification and incorporated SHAP [338] and LIME [272] to explain the resulting predictions. Similarly, Senevirathna et al. [339] reported the creation of a dataset containing 13,000 aspect-tagged banking reviews in Sinhala, English, and code-mixed language. Their hybrid XLM-RoBERTa model incorporated domain-specific lexicon correction and achieved a reported accuracy of 88.4% and an F1-score of 0.84 for the Sinhala and code-mixed data. SHAP [338] and LIME [272] were again used to provide model-level explanations.

Sentiment and emotion analysis have also been extended beyond text-only classification. Peiris and Sumanathilaka [340] analysed emotion in Sinhala songs using both lyrical and acoustic information. For the textual component, the authors used RoBERTa to represent Sinhala words before combining the resulting features with audio-based evidence.

Several domain-specific sentiment datasets and models have additionally been reported. Dissanayake [341] collected a dataset of *Sinhala patient feedback in a hospital setting* and reported that a BiLSTM model achieved the highest accuracy of 98.08%. Perera and Caldera [342] reported the construction of a dataset containing 13,000 manually annotated YouTube comments for Sinhala code-mixed sentiment analysis. Their evaluation covered conventional machine learning methods—including Logistic Regression, Support Vector Machines, Multinomial Naïve Bayes, K-Nearest Neighbour, Decision Trees, and Random Forests—as well as deep learning and transformer models, including BERT, DistilBERT, ALBERT, and XLM-R.

Overall, Sinhala sentiment analysis has developed from small-scale binary classification and manually constructed lexicons into a diverse research area encompassing large social-media datasets, recurrent neural networks, multilingual transformers, adapter-based transfer, aspect-level analysis, explainable AI, multimodal emotion recognition, and fine-tuned LLMs. News comments and social-media text remain the dominant domains, although recent work has expanded into online marketplaces, banking, healthcare, music, and Romanised code-mixed communication. The field would benefit from clearer dataset licensing, standardised train-validation-test partitions, consistent sentiment and emotion taxonomies, and broader comparisons across classical, encoder-based, and generative approaches.

3.17 Hate Speech Detection

In the Sinhala NLP literature, the terms *hate speech detection*, *offensive speech detection*, and *inappropriate speech detection* are frequently used interchangeably. This contrasts with research on higher-resource languages, where these tasks are typically defined as related but distinct phenomena [343]. Hate speech generally targets individuals or groups on the basis of protected characteristics, whereas offensive or inappropriate language may include insults, profanity, abuse, or other harmful content without necessarily expressing identity-directed hatred. Because the existing Sinhala literature rarely maintains this distinction consistently, this

subsection jointly reviews work described under one or more of these task labels.

A growing collection of annotated resources supports research in this area. NLLB Team et al. [263] introduced a large  [Sinhala toxicity detection dataset](#) by extending material from FLORES [124]. The resource can support direct toxicity detection as well as translation-based approaches to hate-speech analysis. Gammanpila and Perera [344] reported the creation of SOCOLD, an obfuscated and code-mixed Sinhala offensive-language dataset designed to capture deliberately altered harmful expressions.

Platform-specific datasets have also been released. De Saa and Ranathunga [345] published a  [YouTube comment dataset](#) together with a feature model for Sinhala hate-speech detection. A separate collection of Sinhala hate-speech comments obtained from Facebook is available through Kaggle³², although it has no accompanying research publication. Perera et al. [346] released a  [hate-annotated dataset](#) containing more than 1,600 Sinhala tweets. In addition, a lexical resource of Sinhala swear and obscene words is publicly available in both  [Unicode](#) and  [Singlish](#) formats.

More recently, Chavinda and Thayasivam [347] introduced  [Facebook](#) and  [Twitter](#) datasets for Sinhala hate-speech detection. They evaluated multilingual large language models using Dual Contrastive Learning (DCL) and reported that TwHIN-BERT-base [348] achieved the strongest performance.

The most influential benchmark in this area is the  [Sinhala Offensive Language Dataset \(SOLD\)](#) introduced by Ranasinghe et al. [349]. SOLD contains 10,000 Sinhala Twitter posts annotated at both sentence and token levels using the binary classes *offensive* and *not offensive*. The same study introduced  [SemiSOLD](#), a substantially larger collection of 145,000 Sinhala tweets annotated using a semi-supervised procedure. The accompanying code is publicly available through  [GitHub](#). SOLD subsequently became the principal shared benchmark for comparing Sinhala offensive-language detection systems.

Early Sinhala hate-speech detection research predominantly relied on conventional machine learning and manually engineered features. De Silva [350] proposed a machine learning approach for Sinhala hate-speech detection, while Kariyawasam [351] investigated the identification of toxic Sinhala content on social media. Sandaruwan et al. [352] collected a corpus of 3,000 Sinhala comments and applied Multinomial Naïve Bayes, whereas Sheran [353] reported a machine learning system for detecting hate speech written in either Sinhala or Singlish.

Further studies expanded the range of harmful-content categories and platforms considered. Sandaruwan et al. [354] used text mining and machine learning methods to identify abusive Sinhala comments on social media. Amali and Jayalal [355] examined Sinhala cyberbullying classification using classical machine learning algorithms, while Het-tiarachchi et al. [356] applied similar methods to Romanised Sinhala social-media posts. Although the latter work did not explicitly describe the text as *transliterated* Sinhala, its treat-

32. <https://bit.ly/3j9r3Za>

ment of Romanised content corresponds to a transliteration-based setting.

The transition toward neural methods began with studies such as Samarasinghe et al. [357], who proposed a Convolutional Neural Network for Sinhala hate-speech detection. Munasinghe and Thayasivam [358] later explored deep learning for detecting hate speech in Sinhala tweets, while Fernando et al. [359] reported the use of both machine learning and deep learning methods for the task. Gamage et al. [360] compared several embedding representations with classical frequency-based features, providing a broader evaluation of alternative text representations for Sinhala hate-speech detection.

Classical models nevertheless remained competitive in several comparative studies. Guruge et al. [361] evaluated an ensemble of Naïve Bayes, Support Vector Machines, XG-Boost, multilayer perceptrons, and AdaBoost using 49,019 tweets collected between February and April 2021. Shalinda and Munasinghe [362] compared a Linear Support Vector Classifier, Random Forest, SGD, Logistic Regression, XG-Boost, and Multinomial Naïve Bayes on Sinhala and Singlish content to identify *hate words*. Their strongest result was obtained using an SGD classifier with TF-IDF representations constructed from unigrams and bigrams.

Beyond classifying individual messages, researchers have also investigated the users and temporal dynamics associated with harmful content. Perera et al. [363] attempted to predict Sinhala hate speech using Twitter user behaviour and ensemble machine learning, with users represented according to the Big Five personality traits. Their follow-up study [346] analysed hate-speech propagation on Twitter and released the accompanying  dataset of more than 1,600 annotated Sinhala tweets. Rajapaksha et al. [364] used deep learning to identify periods during which particular hate-related topics became prominent on Sinhala Twitter.

Several studies have addressed inappropriate or harmful content through indirect or specialised formulations. Arachchi et al. [365] used a web-based Sinhala-English translation service as part of a system for detecting inappropriate Sinhala word usage. Ehelepola [366] analysed hate speech in Sinhala text obtained from both social-media and e-commerce platforms. Hewage and Thanthriwatta [367] concentrated specifically on *body shaming*, constructing a corpus containing both Sinhala-script and Romanised Sinhala instances.

Obfuscated profanity represents another important challenge because users frequently alter spellings to evade automated moderation. Premachandra et al. [368] reported a manually annotated dataset of 4,000 Facebook comments containing deliberately modified profanity. Using SinBERT Dhananjaya et al. [120], they reported F1-scores of 0.671 for profanity detection and 0.387 for altered-speech detection. The comparatively lower performance on altered expressions attests to the difficulty of detecting adversarial orthographic variation.

Code-mixed and Romanised Sinhala have likewise received substantial attention. Muthuthanthri and Smith [369] evaluated TF-IDF and *fastText* representations with SVM, CNN, RNN, LSTM, BERT, and GPT-2 models using an annotated Facebook dataset. They concluded that BERT achieved the strongest performance for detecting hate speech in

English-Sinhala code-mixed text. Tanjim [370] studied cyberbullying detection across several low-resource languages and, when treating Sinhala as the target language, recommended Assamese, Bengali, Nepali, Urdu, Hindi, or Bodo as transfer-learning source languages.

More recent systems increasingly rely on pretrained multilingual transformers. Jahnavi and Chaturvedi [371] combined BERT with CNN and LSTM components for Sinhala hate-speech detection. Perera et al. [372] used XLM-RoBERTa to assign Sinhala YouTube titles and comments to different hate-severity levels, thereby formulating the task as multi-class rather than binary classification.

Hate-speech analysis has also expanded beyond text-only inputs. Dikwatta et al. [373] applied supervised algorithms to Sinhala text extracted from Internet memes. In subsequent work [374], the authors introduced a Sinhala meme dataset annotated using both textual and visual components and reported an accuracy of 95% for their SVM-based approach. Although the reported model is not inherently multimodal, the dataset enables future work that jointly models the two modalities.

A related multimodal formulation was explored by Wickramaarachchi et al. [375], who used an LSTM over BART-derived representations to compare the titles and descriptions of Sinhala YouTube videos with their audio content in order to determine whether the videos contained hate speech. This extends the task from message-level classification to consistency analysis across textual and acoustic evidence.

The release of SOLD [349] led to a more standardised phase of Sinhala offensive-language research. Fernando and Deng [376] evaluated traditional machine learning algorithms on the benchmark. Ranasinghe and Zampieri [377] compared pretrained mT5 performance on Sinhala SOLD tweets with corresponding results for German, Spanish, Hindi, Korean, and Marathi. Bestgen [378] compared classical machine learning models using simple n -gram features on SOLD against equivalent experiments for Assamese and Bengali. The benchmark has also supported research on translation, transliteration, and cross-lingual transfer. Dmonte et al. [379] used SOLD to investigate whether machine translation affects offensive-language detection performance. Chamikara and Sumanathilaka [380] first transliterated SOLD into Romanised Sinhala and then compared XLM-RoBERTa, IndicBERT, and distil-mBERT on the transformed dataset. These studies demonstrate that script representation and translation quality can materially affect downstream classification. SOLD was adopted as the Sinhala dataset for the Hate Speech and Offensive Content (HASOC) identification task introduced by Ranasinghe et al. [381]³³. For this shared task, Narayan et al. [382] compared several pretrained deep learning models and reported that XLM-RoBERTa-base achieved a Macro F1-score of 83.49%, compared with 75.30% for an LSTM baseline. Ojo et al. [383] used mBERT for the same Sinhala hate- and offensive-content identification task. Generative models have also been evaluated on SOLD. Rostamkhani and Eetemadi [384] compared the zero-shot performance of ChatGPT against several Sinhala models and concluded that zero-shot Chat-

33. <https://hasocfire.github.io/hasoc/2023/task1.html>

GPT outperformed systems that had not themselves been fine-tuned on SOLD. More recently, Haturusinghe et al. [385] introduced the *Subasa* family of models and evaluated them on SOLD. Their trained  *PMLs* and  *LLMs* are publicly available in separate repositories. The authors report that *Subasa-XLM-R* outperforms even state-of-the-art commercial LLMs on the SOLD benchmark. Machine-translated offensive-language resources have provided another route for expanding Sinhala data. Dmonte et al. [386] used SOLD to evaluate the  *MT-Offense* dataset, which was created by translating the English Offensive Language Identification Dataset (OLID) [387] using neural machine translation. In their detailed evaluation, Dmonte et al. [386] reported a best F1-score of 0.550 for translations generated by nllb-200-3.3B [263]. This comparatively modest result illustrates the limitations of constructing offensive-language datasets through automatic translation alone, particularly when culturally and linguistically specific expressions do not transfer directly.

Overall, Sinhala hate- and offensive-speech detection has developed from small, independently collected social-media corpora and conventional classifiers into a diverse field encompassing code-mixed and Romanised text, obfuscated profanity, user behaviour, temporal propagation, multimodal memes, multilingual transfer, adapter and transformer models, and generative LLMs. SOLD has played a central role in enabling more reproducible comparison and shared-task evaluation. Nevertheless, the field continues to face inconsistent task definitions, overlapping annotation schemes, limited coverage of targeted identities and hate categories, and insufficient evaluation of model robustness to transliteration, spelling variation, cultural context, and domain shift. Future benchmarks should clearly distinguish hate speech, offensiveness, profanity, cyberbullying, and other forms of harmful language while supporting multi-class, target-aware, token-level, and multimodal evaluation.

3.18 Fake News Detection

Fake news and misinformation detection has received growing attention in Sinhala NLP, driven largely by the widespread use of social media and online news platforms. Compared with sentiment analysis or hate-speech detection, however, research in this area remains relatively limited. Existing work spans misinformation corpora, hybrid credibility-assessment frameworks, semantic similarity approaches, multimodal verification, and, more recently, evaluations using large language models.

The principal publicly available benchmark for Sinhala misinformation detection was introduced by Jayawickrama et al. [388]. The authors released a  *dataset* containing 3,576 Sinhala news articles collected from Sri Lankan news websites and annotated with four credibility labels: *CREDIBLE*, *FALSE*, *PARTIAL*, and *UNCERTAIN*. Alongside the dataset, they also established baseline misinformation classification models, providing one of the first standardised benchmarks for Sinhala fake-news detection.

Early research primarily relied on hybrid systems that combined textual analysis with external credibility indicators. Wijayarathna and Jayalal [389] proposed a framework that verified the textual content of social-media posts against

trusted news sources while simultaneously evaluating the credibility of the user account through a rule-based scoring scheme.

Several studies have instead focused on semantic representations of the textual content itself. Wijayarathna and Jayalal [390] collected a dataset containing 120 fake-news tweets and 250 genuine tweets. Each tweet was represented by averaging the FastText embeddings of its constituent words, after which vector similarity was used to predict whether the associated news item was genuine or fabricated.

Subsequent work explored increasingly sophisticated hybrid scoring frameworks. Udurawana et al. [391] proposed combining an accuracy score derived from analysing the news content with a credibility score obtained through a rule-based scoring mechanism. Their framework additionally incorporated passive-aggressiveness classification. The adoption of deep learning methods has been comparatively limited but continues to increase. Adihetti and Jayalal [392] investigated the use of autoencoders for detecting fake news in Sinhala social-media posts. Researchers have also explored multimodal formulations of the problem. Wickramarachchi et al. [375] detected fake content in Sinhala YouTube videos by comparing the titles and descriptions of videos against automatically derived representations of the spoken audio using an LSTM operating on BART representations.

Recent work has shifted toward benchmarking and evaluating large language models. Atthanayake and Ruzaik [393] conducted a comparative study between the English *Fakeddit* benchmark [394] and the Sinhala misinformation corpus introduced by Jayawickrama et al. [388], highlighting similarities and differences between misinformation datasets in high-resource and low-resource languages. Building upon the rapid progress of generative language models, Premadasa and Thanthriwatta [395] evaluated GPT-4o, DeepSeek-Chat, and Gemini-2.5-Flash-Lite for misinformation detection in Sinhala online news articles under zero-shot, few-shot, Chain-of-Thought (CoT), and role-based prompting settings. This represents a shift away from task-specific supervised classifiers toward prompt-based reasoning with foundation models.

Overall, Sinhala fake-news detection has evolved from small-scale embedding-based classifiers and hybrid credibility-scoring systems to multimodal verification frameworks and prompt-driven large language model approaches. Compared with related areas such as sentiment analysis and hate-speech detection, however, the availability of publicly accessible benchmark datasets remains relatively limited, with the LIRNEasia misinformation corpus serving as the primary evaluation resource. Future research would benefit from larger multi-domain datasets, multimodal benchmarks that jointly model text, images, videos, and user metadata, and standardised evaluation protocols that enable robust comparisons between conventional machine learning, neural architectures, and modern generative language models.

3.19 Language Identification (LID) Task

Language Identification (LID), also referred to as language identification (LangID), is the task of automatically de-

termining the language in which a given piece of text is written. For Sinhala, this task is generally regarded as comparatively straightforward because the language is predominantly written using the unique Sinhala script. Consequently, most existing benchmarks report near-perfect performance for Sinhala. However, this apparent success masks several practical challenges that are largely absent from current evaluations, including Romanised (Singlish) text, code-mixed content, closely related languages such as Pali and Sanskrit written in the Sinhala script, and deliberately obfuscated spelling.

One of the largest publicly available multilingual resources for this task was introduced by Burchell et al. [396], who released a  dataset together with multilingual language identification models covering Sinhala and many other languages. Their models generally outperform the language identification component of NLLB [263] across most languages. For Sinhala, however, they reported performance comparable to that of NLLB, ostensibly suggesting that conventional Sinhala language identification is already approaching saturation under existing benchmark conditions.

Several studies have investigated factors beyond purely textual representations. Dunn and Edwards-Brown [9] explored whether incorporating geographical priors could improve language identification. While geographic information benefited certain languages, the authors found virtually no improvement for Sinhala because the language identification model and the geographical model already agreed on approximately 99% of the instances.

Research has also focused on developing lightweight multilingual language identification tools. Kargaran et al. [397] introduced *GlottScript*, a written language identification system that achieved over 80% accuracy for Sinhala. More recently, Van Der Goot [398] surveyed the remaining challenges of language identification across more than 2,000 languages. Their experiments reported F1-scores exceeding 99% for Sinhala, with both *Textcat* [399]³⁴ and *GLOT500* [119] achieving 99.9%. However, the authors' evaluation considered only Sinhala text written in the Sinhala script. Consequently, the reported performance reflects script identification as much as language identification. This is further illustrated by the original  *GLOT500* study, where Imani et al. [119] reported only 45.0% accuracy for Sinhala under substantially more challenging multilingual evaluation settings.

These studies imply that conventional Sinhala language identification is largely a solved problem when the language is written using the Sinhala script. Nevertheless, this conclusion should be interpreted with caution because current benchmarks do not adequately capture many real-world scenarios encountered in Sri Lankan digital communication. Existing evaluations largely ignore Romanised Sinhala, Sinhala-English code-mixed text, closely related languages such as Pali and Sanskrit written using the Sinhala script, dialectal variation, and adversarial spellings. Future research should therefore move beyond script-based discrimination and develop benchmarks that evaluate language identification under these more challenging and practically relevant conditions.

3.20 Plagiarism Detection

An extremely simple plagiarism detection tool which only uses n-grams of simply tokenized text was proposed by Basnayake et al. [400]. Another simple plagiarism detection tool that uses synonymy and Hyponymy-Hypernymy (which they call *Generalization* in the paper) was attempted by Rajamanthri and Thelijjagoda [401]. They later extended this work [402] to propose a more advanced plagiarism detection tool which uses Internet resources. KasthuriArachchi and Charles [403] proposed using Word2Vec vector cosign similarity to detect plagiarism. A multi-document Sinhala similarity detection tool to detect plagiarism was proposed by Piyaarathna [404]. Punchihewa et al. [405] developed a character-level model which can identify the author for Sinhala text in student answers. The work by Jayasundara and Tissera [406] used cosine similarity on TF-IDF vectors to identify plagiarism in Sinhala English mixed student essays and reports. Primal et al. [407] have proposed a plagiarism checking system which uses LaBSE embeddings.

3.21 Sinhala-English Code-Mixing

The problem of recognizing Sinhala and English code-mixed data where the Sinhala text is written in Singlish was explored by Smith and Thayasivam [408] and later by Smith [409] using an XGB classifier and a CRF model building on their previous work [410], which analysed such data. Shanmugalingam and Sumathipala [411] also attempted to identify the language in Sinhala-English code-mixed text using Support Vector Machines (SVM), Naive Bayes, Logistic Regression, Random Forest, and Decision Trees. A dictionary based approach to standardize Sinhala Code-Mixed text was proposed by Kugathasan and Sumathipala [412]. They later used it for NMT in Sinhala-English Code-Mixed text [413]. Shakir and Deuber [414] claims to have developed a corpus of South Asian languages in the context of code-mixing which includes Sinhala. However, this is not publicly available. It should further be noted that they report Sinhala data using the heading *Sri Lanka* rather than *Sinhala*. In their follow-up work Shakir and Deuber [415] claim to extend this code-mixed data set to include meme data.

As discussed in Section 3.15, Rathnayake et al. [294] used adapter-based [295–300] fine-tuning on XLM-R [301], for classifying code-mixed and code-switched Sinhala text. Hettigoda [304] classified English-Sinhala code-mixed comments from Facebook. As discussed in Section 3.16, Aththanayaka and Naleer [329] utilized traditional machine learning methods for sentiment analysis on Sinhala-English code-mixed text from social media. Chathuranga and Ranathunga [292] used capsule-based methods to classify Sinhala-English code-mixed data. Dissanayake et al. [416] claim that the web-based code-less chatbot development platform for Sinhala proposed by them is capable of handling Sinhala-English code-switching. The feature set derived by Walawage and Ranathunga [417] for text on social media images (memes) attempts to separately identify Sinhala and English text. The study by Fazal and Farook [418] used classical machine learning algorithms on TF-IDF features to predict depression in Sinhala-English code-mixed data from Twitter and Facebook. The study

34. <https://www.let.rug.nl/vannoord/TextCat/>

by Udawatta et al. [419] shows that prompt-based learning of pre-trained language models (PLMs) outperforms full fine-tuning of PLMs on Code-mixing and code-switching (CMCS) English-Sinhala data across various NLP tasks such as sentiment classification, hate-speech detection, and humour detection. Uthpala and Thirukumaran [330] claims to have created a sentiment-annotated Sinhala-English code mixed dataset using comments from YouTube videos. However, given that even their paper is behind a paywall, there is no free and public way to access this dataset. Senanayaka et al. [420] proposes *SingRAG*, a RAG and translation framework built on LLaMA that converts code-mixed Sinhala to English for processing and then converts whatever the English results is of the attached backend to code-mixed Sinhala before sending it back to the user.

3.22 Question Answering Systems (Q&A)

Romero et al. [421] introduced CVQA, a multilingual visual Question Answering benchmark which includes Sinhala. They have publicly released both their  [data](#) and leader board³⁵. Vayani et al. [422] created a dataset  [ALM-Bench](#) by machine translating English QA pairs from *LLaVA-Bench* (In-the-Wild) dataset [423] and extended it by searching for images under *country name*, *language name*, and *cultural category* on the web so that they can include culturally relevant questions and answer for each language. They then reported that for Sinhala Q&A (especially visual Q&A), GPT-4o [424] is superior to other tested LLMs.

Dissanayake and Hettige [425] implemented a question-and-answer generator for Sinhala with the limited PoS: pronouns, adjectives, verbs, and adverbs. Jayakody et al. [426] uses simple KNN and SVM methods on a PoS tagged Sinhala corpus to create a question-answering system which they name *Mahoshadha*. Shah and Kartik [427] introduced a  [small benchmark dataset](#) of 90 items in the public-service question domain where systems are evaluated on whether they should answer the question, ask for further clarifications, or escalate the user to an official channel.

Wansekara and Jayasekara [428] developed a visual Sinhala question and answer system using YOLO for object identification and a BERT model trained on a Sinhala corpus for Sinhala text. Dhanawardhana et al. [429] frame their work just as a multi-model deep learning approach. But in actuality, their lost-and-found application is utilising techniques that are relevant to Sinhala VQA. The system created by Dasanayaka et al. [430] is claimed to be capable of first, generating radiology reports for orthopantomography (related to dentistry) using Llama on captions generated by a Blip-2-based caption generator on dental panoramic tomography (DPT) images and then serve a Sinhala chatbot where the patients may ask questions on the said report and get answers. Shafique et al. [431] introduced ViMUL-Bench a multimodal video benchmark for LLMs that includes Sinhala. With 49.6, they show that GPT4o has the best results for Sinhala; however, the Sinhala accuracy is the lowest of all the tested languages across almost all the LLMs.

3.23 Text Summarizing

Hasan et al. [432] introduced the  [XL-sum](#) dataset, which includes 3,414 Sinhala documents and their summaries collected from the Sinhala BBC website³⁶. The  [M3LS](#) multi-modal dataset introduced by Verma et al. [433] contains 10,148 Sinhala documents, relevant images (making this a multi-modal dataset), and their summaries collected from the Sinhala BBC website³⁷. Further, they claim that MT5 [154] obtains the best ROUGE-1 and ROUGE-L f scores for their Sinhala dataset. The first Sinhala multi-document summarizing dataset,  [M2DS](#), was created by Hewapathirana et al. [434]. The *M2DS* dataset consists of 23.5k Sinhala documents in 5.5k clusters, with each cluster having a golden summary. Their code and pre-trained models are also available on  [GitHub](#).

A deterministic process flow for automatic Sinhala text summarizing was proposed by Welgama [435]. The study by Wimalasuriya [436], which has the same name as the above work by Welgama [435], uses the graph-based TextRank algorithm for automatic Sinhala text summarizing. The use case of Sinhala Text summarization for government gazettes was explored by Jayawardane [437]. The study by Rathnayake et al. [438] compared the results of extractive and abstractive summarization on Sinhala textbooks. The study by Jahan and Wijesekara [439] compared the abstractive summarizing methods of TF-IDF and Text-Rank for Sinhala using ROUGE as the evaluation score. They concluded TF-IDF to be the superior choice. Patabandi et al. [440] used BART and DistilBART to summarise Romanised Sinhala (Singlish) messages collected from WhatsApp group chats. Seneviratne and Weerasinghe [441] claims to have created a dataset of human-validated 2,493 Sinhala article-summary pairs and established a benchmark of ROUGE-1 0.286 with a fine-tuned Gemma-3-4B model.

3.24 Text Simplification

The  [MultiLS](#) data set introduced by Shardlow et al. [442] for  [MLSP 2024 Shared Task](#) includes a Sinhala text simplification data set with data from News and Religious domains. Ranathunga et al. [443] introduced  [SiTSE](#), a Sinhala text simplification dataset. It has 1000 complex Sinhala texts along with two separate simplifications for each of them. Further, they provide the evaluation metrics that can be used to evaluate any subsequent research conducted using the same data. Liu et al. [444] created  [OasisSimp](#), a text simplification dataset that includes Sinhala entries that were sourced from the SiTa trilingual parallel corpus [445].

The text simplification shared task of Building Educational Applications (BEA) at MLSP 2024 [446] included the *NSINA* [121] Sinhala corpus. The aggregated data from the shared task is available online as  [Train-Test data](#) and  [cleaned gold data](#). Enomoto et al. [447] experimented with GPT-4 and noted that re-ranking systems outperformed the base system for Sinhala while the opposite was true for all the other nine languages. Goswami et al. [448] also tested GPT-4 (Terbo) for this task against the top

35. <https://nisansa.lk/?s=sOrx7r>

36. <https://www.bbc.com/sinhala>

37. <https://www.bbc.com/sinhala>

10 suggestion model and reported better results for GPT-4 for Sinhala than the top 10 suggestion model. Seneviratne and Suominen [449] compared multiple models for the English-Sinhala language pair simplification for the lexical complexity prediction task and lexical substitution task. In the submission by Cristea and Nisioi [450] word frequency and other meta-knowledge are not used for Sinhala text simplification while they are used for the other languages in the task. Their Sinhala models are trained only using Sinhala text Google translated to English. As such the results they report for Sinhala for this task is quite weak compared to their results for the other languages. Even though the main focus on the work by Nohejl et al. [451] is introducing a Japanese text simplification data set and a model, to demonstrate the generalizability of the system, they use the MultiLS data set [442] which includes Sinhala.

3.25 Text Generation

The study by Kao and Ilmini [452] used LSTM to generate Sinhala lyrics. Bandara et al. [453] introduced *Sibil AI*, a GTP-2 based Children’s story generator for Sinhala. However, they did not train a GTP-2 model in Sinhala. Instead, they trained an English GTP-2 model and generated stories in English, which were then translated to Sinhala using Google Translate API. Amarasinghe and Ranathunga [454] used an ontology-based approach to generate Sinhala essay questions. Warnasooriya et al. [455] used a Sinhala-adapted LLaMA model to generate Sinhala stories and quizzes. Ranasinghe et al. [456] trained a Sinhala image caption generator model on Google translated MS COCO [457] dataset and tested it on Google translated Flickr8K [458] and MS COCO datasets. Chen et al. [459] introduced 🤖 *MUNICHus*, a multilingual news image captioning benchmark dataset which includes Sinhala. However, with 2,418 training examples and 998 testing examples, Sinhala is the least represented language in this dataset, and the authors specifically observed that Sinhala has the worst results among the included languages. Nevertheless, they also release two models fine-tuned on the dataset: 🤖 *Aya-vision-8b* and 🤖 *Llama-3.2-11B*. The work by Liyanage and Ranathunga [460, 461] attempt to use LSTMs for mathematical word problem generation in Sinhala and other languages. The follow-up work by Niyarepola et al. [462] used pre-trained mBART [463] and mT5 [153, 154] models for the same task and showed better results. Gamage et al. [464] created a dataset named 🤖 *MultiMWP* which consists of Math Word Problems (MWPs) in nine languages, including Sinhala.

A Sinhala programming language named *Helaa* based on Java was proposed by Yasasri and Karunarathna [465]. Senarathne [466] introduced a model to *translate* Sinhala pseudocode to computer code written in C#. The works by De Silva and Athukorala [467, 468] claim to be able to provide Sinhala language explanations to JAVA code as well as generate JAVA code snippets from Sinhala descriptions. They later extend this work to create a JAVA programming assistance tool [469–471] for Sinhala native speakers.

3.26 Chatbots

A simple Sinhala chatbot which utilizes a small knowledge base has been proposed by Hettige and Karunananda [93].

A study on the effect of word embeddings on a Sinhala chatbot was conducted by Gamage et al. [472] where they used, the fasttext model trained by Facebook [241–243], on a RASA³⁸ [473] chat bot. A Sinhala chat bot for train information was proposed by Harshani [474]. Similarly, the tool proposed by Chandrasena et al. [475] serves as a chat bot-based recommendation system for Sri Lankan traditional dancers. The chat bot discussed by Kumanayake [476] has the very specific purpose of answering user inquiries about the degree programs at University of Ruhuna. Avishka et al. [477, 478] used off-the-shelf RASA NLU Engine [473] and Microsoft Bot Network [479] to set up a generic Sinhala chat bot architecture. They demonstrated the effectiveness of their architecture by creating a food ordering chat bot. A web-based code-less chat bot development platform for Sinhala was proposed by Dissanayake et al. [416]. Further, they claimed that their system can handle OOV tokens as well as Sinhala-English code-switching. The work by Dasanayaka and Warnajith [480] used a deep learning Intent Mapping (IM) model to map consumer responses in their Sinhala chat bot framework. Rajapakshe et al. [481] proposed a Sinhala conversational interface for the purpose of scheduling medical appointments and giving medical advice. The chat bot component was in the back end while the front end contained speech recognition and voice synthesizer components. The end product of work by Dasanayaka et al. [430] is a Sinhala chatbot that can answer patient questions on orthopantomography radiology reports which were previously generated by the system itself by analysing dental panoramic tomography (DPT) images. Kandamby et al. [482] introduced *Cloud Era*, a multilingual chatbot utilising *LightRAG* [483] for agentic behaviour in multi-cloud setups. They claim to have achieved 81% recall for Sinhala. The work by Weerakoon et al. [484] set out to build a Sinhala speech-to-speech chatbot. They select *Wav2Vec2-BERT* [485] for ASR and utilise a RAG approach for the chatbot. The TTS component is handled with the 🤖 *VITS* [486] model.

3.27 Translators

A meta-study on the viability of machine translators replacing English to Sinhala human translators was conducted by Dilshani and Senevirathna [487]. However, this study only involves 100 combined and complex English sentences translated to Sinhala by human translators as well as MT software. Given that reason and the fact that they seem to only used Google translate and *Akura* Sinhala dictionary app for comparison, the conclusions of this study may not be generalized. Another meta-study on the impact of pre-trained multilingual sequence-to-sequence models on low-resource language translation has been conducted by Lee et al. [488]; while they consider Sinhala as one of the examples, they do not go much into the specific impact due to the general nature of the paper. The meta-analysis by Ramadasa et al. [489] attempts to evaluate the goodness of the Google Sinhala-to-English translation by using the Google cloud API to translate Sinhala to English and then analysing the accuracy of the *Sentiment Analysis* task and

38. <https://rasa.com/>

the *Named Entity Recognition* task on the translated text. The meta-analysis by Das et al. [490] compared the results of translating English to 15 Indic languages including Sinhala using statistical translation methods. They used datasets from OPUS [491] for model building and utilized Flores-200 for fine-tuning. The NMT for Indic languages study conducted by Sheshadri et al. [492] discusses the Sinhala language translation in the abstract and conclusion but the paper itself focuses more on the languages spoken in India. Nevertheless, it puts Sinhala into a regional linguistic perspective. The meta-study by Bapna et al. [493] discusses the task of building clean, web-mined datasets for a number of languages including Sinhala for the task of machine translation. This discussion was continued by Jones et al. [494] who discussed the bilingual lexica (BLEXs) in the context of a number of languages including Sinhala. Yet another meta-analysis by Halpege [495] compared the commercially available English-Sinhala translation systems provided by Google and Bing through a comprehensive error analysis. This study used academic articles as its domain and ultimately concluded that both translation systems need significant improvements before they can be recommended for widespread usage. The survey by Tafa et al. [496] compared the performance of a number of MT models on Low-Resource languages, including Sinhala. De Silva and Hansadi [497] have also conducted another shallow study on rule-based, statistical, neural, and hybrid MT models for Low-Resource languages. The short survey by Dias and Sumanathilaka [498] discusses translation and transliteration of few code-mixed Indo-Aryan languages which includes a major focus on Sinhala.

3.27.1 Parallel Corpora

The open parallel corpus (OPUS)³⁹ by Tiedemann [491] curates the largest collection of parallel data between Sinhala and other languages. Zhang et al. [499] sampled data from OPUS for 100 languages (including Sinhala) and created the  OPUS100 data set as a benchmark for neural machine translation (NMT). Their  code is also available. Further, Tiedemann [500] created an NMT benchmark named  Tatoeba MT Challenge using OPUS data based on Tatoeba⁴⁰. The English-Sinhala data set can also be directly downloaded from a link on  GitHub. A benchmark Sinhala-English translation data set named FLORES⁴¹ was created by Guzmán et al. [124]. This includes a 600k+ Sinhala-English  subtitle pairs initially collected by [501], 45k+ Sinhala-English sentence pairs from  GNOME,  KDE, and  Ubuntu. They further provided two monolingual corpora for Sinhala. Those were a 155k+ sentences of filtered  Sinhala Wikipedia and 5178k+ sentences of Sinhala  common crawl. In addition to the data set, they have also made their  code publicly available. This work was then extended by NLLB Team et al. [263], of which the data and code are available under the  No Language Left Behind (NLLB) project. Sofalas et al. [502] introduced  SinFoS, a parallel data set of 2,344 Sinhala figures of speech including proverbs,

idioms, adages, idiosyncratics, and other sayings. Pushpananda et al. [503] claims to have created a manually annotated parallel corpus named TamSiPara with 25k parallel sentences in the Si-Ta direction and 22k parallel sentences in the Ta-Si direction. But it is not freely and openly accessible. Burchell et al. [504] introduced  HPLT v2 a multilingual monolingual and parallel corpora that includes 273k Sinhala Sentences. These sentences were extracted from the Internet Archive(IA) and the Common Crawl (CC). They have also publicly released their  translation models. Ranathunga et al. [505] have  publicly released their data, code, and models for Sinhala-English translation. Very importantly, this includes a human-cleaned portion of the NLLB dataset [263] in  En-Si. Fernando et al. [506] has created aligned corpora for Sinhala-English, Tamil-English, and Sinhala-Tamil language pairs. The  corpora as well as  code for their document and sentence alignment task, are available. A parallel corpus of Sinhala and English was collected by Bañón et al. [507] containing 217,407 sentences and available to download from their website⁴². However, the later audit by Caswell et al. [508] raised issues on the quality of that data set. A  parallel corpus of aligned Sinhala-English documents and sentences obtained from crawling the web was released by Sachintha et al. [266]. As for Sinhala-Tamil corpora, Hameed et al. [509] claim to have built a sentence-aligned Sinhala-Tamil parallel corpus and Mohamed et al. [510] claim to have built a word-aligned Sinhala-Tamil parallel corpus. However, at the time of writing this paper, neither of them was publicly available. A very small  Sinhala-Tamil aligned parallel corpus created by Farhath et al. [511] using order papers of government of Sri Lanka is available to download. Vasantharajan and Thayasivam [512] used Printed Character Recognition (PCR) to create a  Tamil-Sinhala-English parallel corpus. Their follow-up work [513] on adapting the Tesseract engine to handle non-Unicode (legacy fonts) in pdf documents resulted in an improved version of the corpus, which also can be found on  GitHub.

3.27.2 Translation Quality studies

As mentioned in Section 3.1, the study by [508] raised questions on the quality of the existing Sinhala-English corpora. In a follow-up study by Ranathunga et al. [505], it was pointed out that using an automated ranking based on a similarity measure on web-mined corpora and using the resultant top samples can yield better translation models for English-Sinhala and Sinhala-Tamil. These models sometimes were better performing than a model trained on the full dataset. They further showed that using human labour to clean web-mined parallel corpora only gives marginal benefits over automated ranking and filtering. Thus they concluded that using expensive human labour for this task might not be efficient. The study by Team et al. [514] conducted a study on the quality of the translator models trained on their NLLB data set [263]. Naturally, this includes Sinhala. Sen et al. [515] on the other hand

39. <https://opus.nlpl.eu/>

40. <https://tatoeba.org/>

41. <https://tinyurl.com/flores200dataset>

42. <https://www.paracrawl.eu/>

attempted to improve the quality of the Sinhala-English parallel corpora using fuzzy string matching where they tried to match the English translation of the given Sinhala sentence to the English sentence in the dataset pair. Omnilingual MT Team et al. [516] introduce 🍌 BOUQuET, a translation benchmark. However, Sinhala is not included in the 8 handcrafted languages; it is only included under the 260+ machine-translated languages. They do welcome annotators to donate their labour⁴³ to change this. Abubacar et al. [517] claimed to have created a 🍌 quality estimation dataset that includes 7k Sinhala-English pairs and 🍌 associated code, but neither of the links currently work.

A study on the viability of using Google Translate for the legal domain English-Sinhala and Sinhala-English translation was conducted by [518]. They used human experts to extensively analyse the end result of the translation with many concrete examples of legal phrases that needed to be translated. The multilingual lexicon for low-resource machine translation dataset GATITOS introduced by Jones et al. [519] does not include Sinhala among their 170 languages set. However, they use Sinhala data from other sources and include several observations for Sinhala in this task. They observe that PanLex [520] and GATITOS improve results in the En-Si direction for smaller models (Transformer Big, 475M) but weaken the results in the Si-En direction. For bigger models (Transformer 1.6B), they do not observe an improvement or degradation of En-Si or Si-En direction.

3.27.3 Sinhala-English Non-NMT

A series of work has been done by a group towards English to Sinhala translation as mentioned in some of the above subsections. This work includes; building a morphological analyzer [91], lexicon databases [94], a transliteration system [95], an evaluation model [100], a computational model of grammar [23], and a multi-agent solution [106]. After working on human-assisted machine translation [96], Hettige and Karunananda [99, 101] have attempted to establish a theoretical basics for English to Sinhala machine translation. A very simplistic web based translator was proposed [97, 98]. The same group have worked on a Sinhala ontology generator for the purpose of machine translation [105] and a phrase level translator [107] based on the previous work on a multi-agent system for translation [104]. Further, an application of the English to Sinhala translator on the use case of selected text for reading was implemented [102]. They later continued their work on multi-agent English to Sinhala translation with the AGR organizational model [108].

Another group independently attempted English-to-Sinhala machine translation [521] with a statistical approach [522]. Wijerathna et al. [523] and De Silva et al. [524] have proposed simple rule based translators. An example-based method applied on the English-Sinhala sentence aligned government domain corpus was proposed by Silva and Weerasinghe [525]. A translator based on a look-up system was proposed by Vidanaralage et al. [526]. In a preprint, Joseph et al. [527] proposes an evolutionary algorithm for Sinhala to English translation with a basis of Point-

wise Mutual Information (PMI) and claims that the code will be shared once the paper is accepted. However, they do not report any quantitative results to be compared and the reported qualitative results are also superficial. Pushpananda et al. [528] utilized statistical machine translation to translate between Sinhala and Tamil. Fernando et al. [529] tries to solve the Out of vocabulary (OOV) problem for Sinhala in the context of Sinhala-English-Tamil statistical machine translation. The approach proposed by Rajitha et al. [530] uses statistical machine translation and transliteration to align Sinhala and English documents.

3.27.4 Sinhala-English NMT

Fonseka et al. [531] used Byte Pair Encoding (BPE) for English to Sinhala neural machine translation. As another solution to the OOV problem, an analysis of subword techniques to improve English to Sinhala Neural Machine Translation (NMT) was conducted by Naranpanawa et al. [532]. A data augmentation method to expand bilingual lexicon terms based on case markers for the purpose of solving the OOV problem in the domain of NMT was proposed by Fernando et al. [533] which they later extended further [534]. Epaliyana et al. [535] proposed iterative filtering and data selection be used to improve Sinhala-English NMT. Perera et al. [536] used English Part-of-Speech (PoS) tags to improve English to Sinhala NMT. Lin et al. [537] used a model based on fairseq [538] to improve machine translation between English and Sinhala. The second half of the study by Arafath [539] dealt with translating Sinhala speech to other languages. Kugathasan and Sumathipala [540] proposed an NMT system for Sinhala-English Code-Mixed text using the standardized Sinhala Code-Mixed text they proposed earlier [412]. Later, they published a slightly different version of the work at a different venue but under the same title [413]. Nguyen et al. [541] introduced a new fine-tuning objective LAgSwAV (Language-Agnostic Constraint for SwAV loss), using which they obtained 5.4 BLEU for English-Sinhala. Later, this method was further discussed by Nguyen [542]. Attigala and Weerasinghe [543] conducted an analysis on the effectiveness of ChatGPT [544] in translating Sinhala songs to English. Utsa et al. [545] used transfer learning and back translation as well as focal loss of the Sinhala-English dataset from FLoRes [124] and reported better results than mBART [463]. The study by Ranathunga et al. [546] presents an evaluation of using parallel data from auxiliary domains to enhance EnSi NMT by fine-tuning or further pre-training the models. In their study on improving cross-lingual representation of multilingual language models for low-resource languages using linguistic entity masking, Fernando and Ranathunga [333] used SiTa and EnSi NMT on NLLB and CCAliigned as one of the evaluations. Tasnia and Al-Farsi [547], when introducing ContrastMT, their pivot language augmentation method that uses Hindi as the pivot, tested English to Sinhala as one of the translation pairs and reported a BLEU score of 24.61 on 48K pairs from Flores. Thillainathan et al. [548] proposed to use continual pre-training and intermediate task transfer learning for fine-tuning what they call multilingual sequence-to-sequence large language models (msLLMs). However, in actuality, what their experiments are on is mBART. The work by Zebaze et al. [549] introduces a new NMT paradigm

43. <https://bouquet.metademolab.com/>

that they call `CompTra` (compositional translation) which is based on LLMs. They report the results for Sinhala using the FLORES-200 [263] data set, where `CompTra` applied on NLLB [263] gives the best result. When introducing `ALOPE`, a translation quality optimisation study for LLMs, Sindhuja et al. [550] used Sinhala-to-English translation for their results. However, they achieved a better Spearman correlation score when they tested the same with `CometKiwi` [551]. Specialised Dictionary-supported LLMs are used to translate agricultural Packages of Practices (PoPs) into Sinhala in the work of Wilson et al. [552]. The work by Okadini et al. [553] tested Llama 3 and Llama 4 against NLLB-200 600M on the Sinhala-Tamil and Tamil-Sinhala translation task under multiple prompting strategies.

Fernando et al. [554] in their meta-study on low-resource language NMT, introduced debiasing heuristics for web mined parallel corpora that improve `EnSi` translation quality. They showed the improvements on `CCMatrix` and `CCAligned` datasets. The meta study by Velayuthan et al. [555] proposes an improved sequence level Knowledge Distillation method that improves encoder knowledge transfer. They used Sinhala-English NMT to discuss the α value ablation by using `CCAligned` [556], `OpenSubtitles` [501], and `SITA (Gov)` [529] corpora. Sinhala is one of the languages considered in the short survey by Perera and Sumanathilaka [557] on machine translation and transliteration for Indo-Aryan Languages. They mention a sub-set of works [445, 527, 530, 531, 558, 559] discussed in this survey for `EnSi` translation.

3.27.5 Singlish to Sinhala (Transliteration)

The `XTREME-UP` data set created by Ruder et al. [560] contains a Sinhala data set for the transliteration task. Perera and Sumanathilaka [561] introduced a `data set` to disambiguate Sinhala transliteration. The data set contains 600 sentences across 22 ambiguous Romanised Sinhala words. The `Dakshina` dataset by [562] contains Native Sinhala script data (200k train and 28k validation sentences) along with Romanised (Singlish) data (10k sentences) from Wikipedia. It also contains a Romanisation Lexicon (25k train and 5k test).

The early work by Goonetilleke et al. [563] attempted Sinhala transliteration through the Latin alphabet. However, their work does not use the word *transliteration* and instead focuses on the predictive aspect. Priyadarshani et al. [564] used statistical machine learning for transliteration of names between Sinhala, Tamil, and English. A rule-based system on trigrams was proposed by Liwera and Ranathunga [565] for Singlish to Sinhala transliteration of social media text. A Singlish to Sinhala converter which uses an LSTM was proposed by De Silva [566]. A rule-based approach for the same was proposed by de Silva and Ahangama [567]. The study by Nanayakkara et al. [568] introduced an English-to-Sinhala transliteration system.

Athukorala and Sumanathilaka [569] proposed a fuzzy logic-based Sinhala transliteration system named *Swa-Bhasha*. Sumanathilaka et al. [570, 571] then proposed a Trie [572] data structure-based algorithm for word suggestion for the *Swa-Bhasha* system. An extended analysis of the same work was presented in a later work [573]. A further

extension, *Swa-Bhasha 2.0*, by Dharmasiri and Sumanathilaka [574] used NMT to eliminate word selection ambiguity in the transliteration process. Subsequently, Sumanathilaka et al. [575] have released `😊 Swa-Bhasha Sentence (En-Si)`, `😊 Swa-Bhasha Sentence (Si-En)`, and `😊 Swa-Bhasha Word` datasets. *Swa-bhasha Resource Hub* [576] was then created by putting the previous work together for easy access, which also contains `😊 a trained BART model`. Continuing this work, Dias et al. [577] claim to introduce a dataset named `SinMix2Mono`; however, none of the supposed dataset links in the paper is clickable.

Amarasekara et al. [578] proposed a rule-based method supported by N-gram analysis and a corpus dataset to transliterate Singlish tweets to Sinhala. The study by Rajapaksha et al. [579] claims to have created a translation system, however, what they have created is a transliteration system between Singlish and Sinhala. While they also claim to have trained ASR and TTS systems, the paper does not identify them by name or citation. The same is true for the data sets they have used to train the various modules in their pipeline. The studies by Kumaravithana et al. [580] and Jayawardhana et al. [581] claim to have built Sinhala-English translators but in actuality are just invoking Google API for their translation tasks.

The work by Kirov et al. [582] used the *Dakshina* [562] dataset and compared the accuracy of Singlish to Sinhala transliteration with vanilla LSTM, vanilla Transformers, mT5 [154], ByT5 [213], and a non-neural finite-state transducer (FST) based on work by Bisani and Ney [583]. They discuss in detail how the zero-width joiner (ZWJ, U+200D) character cause issues in this task and Sinhala rendering as a whole. Their code is available on `GitHub`. Khiu et al. [584] fine-tuned mBART on Sinhala Government Corpus and Bible corpus to predict the NMT performance dependence on domain similarity. De Mel et al. [585] conducted a comparative analysis for Sinhala transliteration between an extension of the rule-based approach proposed by Tennage et al. [586] and the transliteration as a translation model proposed by Deselaers et al. [587]. They reported across-the-board better performance with the latter. All their code as well as the test-sets are `publicly available`. Perera et al. [588] attempted the same task with BERT. Their `code` is also publicly available. Rajapakse and Weerasinghe [589] evaluated multiple LMs on the task of Sinhala transliteration on a dataset of their own, which contains 1500 sentences.

In their short survey discussing Indo-Aryan Languages machine translation and transliteration, Perera and Sumanathilaka [557] covers a number of works [526, 564, 565, 567–569, 571, 574, 575, 586] on Sinhala transliteration which we have already discussed above. The survey by Deshpande and Deshpande [590] covers the works by Perera et al. [588] and De Mel et al. [585] for Sinhala transliteration, which we have discussed above.

3.27.6 Singlish and English (Translation via Transliteration)

An LSTM-based sequence-to-sequence model was used by Sandaruwan et al. [559] for Singlish to English NMT task. Nalinka et al. [591] used simple transformers with positional embedding to translate Singlish into English. The study

by De Silva et al. [592] claims to have achieved Singlish to English translation by simple stemming.

3.27.7 Sinhala to Singlish (Reverse-Transliteration)

For their data set development, de Silva and Ranathunga [593] created a Sinhala to Singlish transliteration tool and have made it publicly available⁴⁴.

3.27.8 Between Sinhala and Non-English Languages

Most of the cross-Sinhala and Tamil work has been done in the domain of machine translation. A neural machine translation for Sinhala and Tamil languages was initiated by Tennage et al. [594, 595]. Then they further enhanced it with transliteration and byte pair encoding [586] and used synthetic training data to handle the rare word problem [596]. This project produced *Si-Ta* [445] a machine translation system of Sinhala and Tamil official documents. In the statistical machine translation front, Farhath et al. [597] worked on integrating bilingual lists. The attempts by Weerasinghe [598] and Sripirakas et al. [599] were also focused on statistical machine translation while Jeyakaran and Weerasinghe [600] attempted a kernel regression method. A yet another attempt was made by Pushpananda et al. [601] which they later extended with some quality improvements [602]. An attempt at real-time direct translation between Sinhala and Tamil was done by Rajpirathap et al. [603]. Dilshani et al. [604] have done a study on the linguistic divergence of Sinhala and Tamil languages with respect to machine translation. Moganarangan [605] claims to have built a named entity translator between Sinhala and Tamil for official government documents. But this work is locked behind an institutional repository wall and thus is not accessible by other researchers. Arukgoda et al. [606] studied the possibility of using deep learning techniques to improve Sinhala-Tamil translation which they further improved later [607]. Pramodya et al. [608] compared Transformers, Recurrent Neural Networks, and Statistical Machine Translation (SMT) in the context of Tamil to Sinhala machine translation. The work by Nissanka et al. [609] used monolingual word embedding to improve NMT between Sinhala and Tamil. Thillainathan et al. [610] uses pre-trained mBART [463] models for six directional translations between Sinhala, Tamil, and English. Yashothara and Uthayasanker [611] discussed the use of the Hierarchical Phrase-Based Model for Tamil to Sinhala and Sinhala to Tamil translations. Pramodya [612] presented a comparison of SMT [528] and NMT models for Sinhala-Tamil translation. For NMT they have used base transformer, turned transformer, and mT5 [154]. The work by Su et al. [613] which compared 8 Parameter-efficient fine-tuning (PEFT) methods on the Si-Ta language pair using the Government corpus [529] and NLLB [263] reports that Houlsby adapter [295], with a 33.34 SacreBLEU [614] score, to be the best PEFT method. However, they also note that the Pfeiffer adapter [297] runs the fastest at 52.59 hours with a reasonable score of 31.24. (Comparatively, the Houlsby adapter runs for 78.65 hours). The work by Pramodya et al. [615] conducted a comparative analysis of the available

models for Sinhala-Tamil translation and reaffirmed the observations of Duh et al. [616] that, in low-resource scenarios, SMT and NMT both may work similarly, however, NMT needs more careful tuning to fit performance. The short survey [557] on Indo-Aryan Languages machine translation and transliteration, discusses a few works [445, 531, 606, 608–610] under SiTa translation which have already been discussed in this survey.

There have been attempts to link Sinhala NLP with Japanese by Herath et al. [39, 109, 110], Thelijagoda et al. [111], Thelijagoda et al. [617], and Kanduboda [12]. The translation component of the work of Rathnayaka and Fernando [618] translates Brahmi inscriptions to Sinhala with an LSTM with Byte Pair Encoding (BPE). Jayasinghe [619] discusses the importance of translation between Sinhala and the liturgical language of Buddhism, Pali [620–622]. Shalini and Hettige [623] attempted to use a dictionary-based machine-translation method for this task. The study by Anuradha and Thelijagoda [624] uses machine translation on the unique application of converting Sinhala and English Braille documents, which they have run OCR on, into voice. Natural Language Interfaces to Databases (NLIDB) where Sinhala natural language queries of a user is translated to SQL has been proposed [625, 626].

3.28 Phonological Tools

On the case of phonological layer, a report on Sinhala phonetics and phonology was published by Wasala and Gamage [627]. Wickramasinghe et al. [628] discussed the practical issues in developing Sinhala Text-to-Speech and Speech Recognition systems. *The Massively Multilingual Speech* (MMS) data set created by Pratap et al. [629] has Sinhala data for the Spoken Language Identification (LID) task. The meta-study by Al-Fraihat et al. [630] compared the status of Sinhala speech recognition research against 17 other languages. A [text-to-speech data set](#) with 3300 Sinhala sentences with 7.5 hours of recordings was released by the *Path Nirvana Foundation*. Buddhika et al. [631] introduce a crowd sourcing tool that they named *Voicer* to collect speech data. They claim that the tool is open source and that they have created a Sinhala Speech corpus of 10 hours with 39 different sentences in the banking domain. Neither the code nor the data is publicly linked in the research paper. However, a subsequent work by Hellarawa and Thayasivam [632] uses this data set. Thus, it can be assumed that this data set may be available if the authors are contacted. The study by Warusawithana et al. [633] created a [refined version](#) of the OpenSLR-52 speech corpus for Sinhala⁴⁵ by Kjartansson et al. [634]⁴⁶.

3.28.1 Text-to-Speech (TTS)

Based on the earlier work by Weerasinghe et al. [636], Wasala et al. [637] have developed methods for Sinhala grapheme-to-phoneme conversion along with a set of rules for schwa epenthesis. This work was then extended by Nadungodage et al. [638]. Weerasinghe et al. [639] developed a Sinhala text-to-speech system. However, it is not

44. <https://nisansa.lk/?s=GUUANj>

45. <https://openslr.org/52>

46. A later further overview on the corpus is also available [635].

publicly accessible. They internally extended it to create a system capable of helping a mute person achieve synthesized real-time interactive voice communication in Sinhala [640]. A rule based approach for automatic segmentation of a small set of Sinhala text into syllables was proposed by Kumara et al. [641]. An *ew prosodic phrasing* method to help with Sinhala Text-to-Speech process was proposed by Bandara et al. [642, 643, 644]. Sodimana et al. [645] proposed a text normalization methodology for Sinhala text-to-speech systems. Further, Sodimana et al. [646] formalized a step-by-step process for building text-to-speech voices for Sinhala. Both Jayamanna [647] and Mishangi [648] have created Sinhala document readers for visually impaired persons to be used on Android devices. An OCR and Text-to-Speech system for Sinhala named Bhashitha was proposed by De Zoysa et al. [649]. The works by Lakmal et al. [650] and Senarathna et al. [651] adapted MaryTTS [652] to synthesize Sinhala speech. The study by Jayawardhana et al. [653] used *Deep Voice* [654] for Sinhala and English TTS. Gamage et al. [655] included a Sinhala text-to-speech module as one of the three modules present in their currency recognition system. The study by Madhusa et al. [656] claims to have created a mobile app with *Sinhala Text-to-Speech and OCR to read books for visually impaired students*. Praveen et al. [201] claim to have created a Sinhala TTS system with a custom neural network that archives 98% accuracy. The work by Senarath [657] discusses using existing TTS systems for Sinhala. They especially highlight VAENAR-TTS [658] as the best candidate for this task. Nayanathara et al. [659], on the other hand, propose to use the  VITS architecture [486] for the same task. Botheju and Rajapaksha [660] used *SpeechT5* [661] model with *X-vector* speaker embeddings [662] for Sinhala Personalized voice synthesis and accent adaptation. Anuradha and Thelijagoda [624] proposed a machine translation system to convert Sinhala and English Braille documents into voice. A separate group has done work on Sinhala text-to-speech systems independent to above [663].

3.28.2 Speech-to-Text (ASR)

On the converse, Nadungodage et al. [664] have done a series of work on Sinhala speech recognition with special notice given to Sinhala being a resource-poor language. This project divides its focus on: continuity [665], active learning [666], and speaker adaptation [667]. A Sinhala speech recognition for voice dialling, which is speaker independent, was proposed by Amarasingha and Gamini [668], and on the other end, a Sinhala speech recognition methodology for interactive voice response systems, which are accessed through mobile phones, was proposed by Manamperi et al. [669]. A Sinhala speech to Unicode text converter for the disaster relief domain was proposed by Prasangini and Nagahamulla [670]. Priyadarshani [671] proposes a method for speaker-dependent speech recognition based on their previous work on: dynamic time warping for recognising isolated Sinhala words [672], genetic algorithms [673], and a syllable segmentation method utilising acoustic envelopes [674]. The method proposed by Gunasekara and Meegama [675] utilises an HMM model for Sinhala speech-to-text. A Sinhala speech recogniser supporting bi-directional conversion between Unicode Sinhala and phonetic English was

proposed by Punchimudiyanse and Meegama [676]. The work by Karunanayake et al. [677] transfer-learns CNNs for transcribing free-form Sinhala and Tamil speech data sets for the purpose of classification. Dilshan [678] conducted a study for the specific use case of transcribing number sequences in continuous Sinhala speech. Gamage et al. [679] explored the use of combinatorial acoustic models such as Deep Neural Network - Hidden Markov Model (DNN-HMM) [680] and Subspace Gaussian Mixture Model (SGMM) [680] in Sinhala speech recognition. In the follow-up work, Gamage et al. [681] extended that work with an end-to-end Lattice-Free Maximum Mutual Information (e2e LF-MMI) model [682], which is claimed to be a viable solution for low-resource language speech recognition by Carmantini et al. [683]. However, it was shown that the new model slightly underperforms compared to the state-of-the-art result. Later, they conducted further development on their model in a follow-up work [684]. Karunathilaka et al. [685] explore Sinhala speech recognition using deep learning models such as pre-trained DNN, DNN, TDNN, TDNN+LSTM. The first half of the study by Arafath [539] dealt with recognising Sinhala speech using LSTMs. Gamage et al. [655] included a Sinhala speech recognition module as one of the three modules present in their currency recognition system. Time-delay neural network architectures (including multistream CNN architectures) were used for acoustic modelling of Sinhala Automatic Speech Recognition (ASR) by Warusawithana et al. [686]. They have used the *Kaldi speech recognition toolkit* [687] for training the ASR models. As part of their child [sic] cognitive ability assessment model, Kahawanugoda et al. [688], proposed a Sinhala speech recognition system. The study by Azir et al. [689] attempts to identify number sequences spoken in Sinhala. *TacoSi* introduced by Arachige and Weerasinghe [690], is based on *Tacotron* [691] and has been evaluated with 10 human evaluators to determine its text-to-speech quality. Nanayakkara [692], Nanayakkara and Weerasinghe [693] used  *DeepSpeech* by Mozilla for Sinhala speech recognition. Gunarathne et al. [694] used an earlier version of the CMUSphinx toolkit⁴⁷ to transcribe Sinhala speech to text. The later work by Akeshe and Meegama [695] also used CMUSphinx toolkit on features extracted from Mel-frequency cepstral coefficients (MFCC) [696, 697] to automatically generate Sinhala subtitles from Speech. The work by Wickramaarachchi et al. [698] also used MFCC features but focused on Sinhala speech intonation recognition for the purpose of detecting speech impediments in young children. Dissanayaka et al. [699] have introduced *Word Sri* an application that is voice-activated and capable of Sinhala grammar checking and plagiarism checking. *tha* [700] have introduced a  *Sinhala text-to-speech* dataset transcribed from 60 videos with a total of 602 minutes. As the baseline, they report the lowest Diarization Error Rate (DER) of 6.0 for Powerset Cross-Entropy Diarization [701]. While the data set link points to a GitHub repository, it does not contain the data set; it points to a project website⁴⁸ which then points to a download page⁴⁹ where you need

47. <https://cmusphinx.github.io/>

48. <https://sita-speakerdiarization.github.io/>

49. <https://rtuthaya.staff.uom.lk/resources/dataset/44>

to send a request to the authors to obtain the data set. Omnilingual ASR Team et al. [702] includes 🗣️ [Sinhala](#) among the 1600+ languages for which they have created a 🗣️ [multilingual ASR system](#). Thameera et al. [703] also fine-tuned Whisper for Sinhala ASR. They have used multiple augmentation techniques to increase the size of their dataset from 3078 clips (5.83 hours) to 12312 clips (23.32 hours). Harischandra et al. [704] tested Whisper, Wav2Vec2-XLSR, and Wav2Vec2-BERT for the Sinhala ASR task and reported that Wav2Vec2-BERT obtains the best results with a Word Error Rate (WER) of 1.79% and a Character Error Rate (CER) of 0.33%. Prabhashwara and Weerakoon [705] implemented an ASR system with Wav2Vec2-XLSR-300M and a 5-gram KenLM [706] for phonetic error correction for the specific task of transcribing Sinhala lectures in the political science domain. Javed et al. [707] tested the Sinhala dataset created by Kjartansson et al. [634] on their model and reported a WER of 18.6.

3.28.3 Speech-to-Speech

Layansan et al. [708] created a speech-to-speech translation system for Sinhala on the Android platform. The system developed by Rajapakshe et al. [481] is also speech-to-speech in the sense that, it is a chatbot for scheduling medical appointments and giving medical advice where the front end contains speech recognition and voice synthesizer components that interfaced with a chatbot component in the back end. The work by Athas and Pirapuraj [709] uses Google APIs (speech-to-text API, text-to-text-translator API, and text-to-speech synthesizer API) to achieve voice translation for Sinhala and Tamil. Dilshani and Chandrasena [710] have conducted a survey on Speech-to-Speech systems available for Sinhala and Tamil translation. In their speech-to-speech chatbot, Weerakoon et al. [484] uses Wav2Vec2-BERT [485] for ASR and 🗣️ [VITS](#) [486] model for TTS. The Text-to-Text chatbot component in the middle is handled by an LLM with RAG.

3.28.4 Speech-to-Intent

The work by Karunanayake et al. [711] used English phoneme-Based Automatic Speech Recognition (ASR) for intent identification in Sinhala and Tamil. Ignatius and Thayasivam [712] proposed a speaker-invariant speech-to-intent classification model with i-vector based speaker normalization, which was then evaluated on Sinhala, and Tamil speech intent data sets. The later work by Yadav et al. [713] used pre-trained embeddings for Sinhala speech intent classification. Hellarawa and Thayasivam [632] proposes a BiLSTM-based ASR system for intent classification which they have tested on the banking domain Sinhala speech dataset created by Buddhika et al. [631]. For this, they report an accuracy of 98.53%.

3.28.5 Speech classification

Luger et al. [714] introduced a data set named 🗣️ [unsupervised_peoples_speech](#) that contains 981 hours of Sinhala speech data. The Sinhala speech classification system proposed by Buddhika et al. [283] does so without converting the speech-to-text. However, they report that this approach only works for specific domains with

well-defined limited vocabularies. The work by Dinushika et al. [715] uses automatic speech recognition of Sinhala for speech command classification. Extending that, Kavmini et al. [716] presented a Sinhala speech command classification system which can be used for downstream applications. The voice assistant system created by Senarathne et al. [717] is capable of handling Sinhala voice commands. Kathriarachchi and Pulasinghe [718] developed a system to classify and predict Sinhala speech clips of children into age groups. The work by Welarathna et al. [719] used CNNs to classify emotions (sad, disgust, surprise, neutral, happy, calm, fear, and angry) of Sinhala speech by Autistic children. Sundarapperuma [720] created a speech emotion detection system for Sinhala.

3.28.6 Lip Synchronization

The study by Weerathunga et al. [721] worked on lip synchronization for Sinhala speech where videos of people speaking Sinhala were mapped to a visemes alphabet created by them. Further of this line of study, Wakkumbura et al. [722] came up with Phoneme-Viseme mapping for Sinhala speech that they intended to be used for future applications of robotics.

3.28.7 Music to Notation

The work by Dulmi and Hettige [723] has created a pipeline which converts audio recordings to western music notation, which is then converted to Sinhala musical notations using an API call to a Large Language Model with a custom-tailored prompt.

3.28.8 Sinhala ASR as a Source

A rather unique study by Ilyas and Jayatilleke [724] uses Sinhala ASR as the source for transfer learning to improve Dhivehi ASR. While, as shown in Fig 2, Sinhala is very closely related to Dhivehi and thus transfer between the two languages should be possible, the lack of resources for both these languages so far discouraged any attempt at transfer learning between the two. But the success of this study is promising proof that the status of Sinhala NLP is improving. All the code for this study is 🗣️ [publicly available](#).

3.29 Optical Character Recognition Applications

While it is not necessarily a component of the NLP stack shown in Fig 5, which follows the definition by Liddy [48], it is possible to swap out the bottom-most phonological layer of the stack in favour of an Optical Character Recognition (OCR) and text rendering layer. Gunathilaka et al. [725] introduced 🗣️ [SinOCR](#), consisting of 100,000 images which include 1,135 🗣️ [handwritten Sinhala texts](#) and texts printed in 200 🗣️ [different Sinhala fonts](#). They also released 🗣️ [SinFUND](#), a fully annotated dataset of 100 manually filled Sinhala forms.

3.29.1 Printed Text

The 🗣️ [XTREME-UP](#) data set created by Ruder et al. [560] contains a Sinhala data set for the OCR task. The data was obtained from book transcriptions. Dilhara and Jayatilleke

[726] introduced 🤖 [sinhala-ocr-lk-acts-1010](#) an annotated dataset of 1,010 page images and their transcriptions collected from Sri Lankan Legislative Acts published between 1981-1989 and 2000-2019.

An attempt for a Sinhala OCR system has been taken by Rajapakse et al. [727] before any other work has been done on the topic. Much later, a linear symmetry-based approach was proposed by Premaratne and Bigun [728, 729]. They then used hidden Markov model-based optimization on the recognized Sinhala script [730]. Similarly, Hewavitharana et al. [731] used hidden Markov models for off-line Sinhala character recognition. Herath et al. [113], Herath and Medagoda [114] developed a preprocessing engine based on a template matching for printed Sinhala OCR. Statistical approaches with histogram projections for Sinhala character recognition is proposed by Hewavitharana and Kodikara [732], by Ajward et al. [733], and by Madushanka et al. [734]. Karunanayaka et al. [735] also did off-line Sinhala character recognition with a use case for postal city name recognition. A separate group had attempted Sinhala OCR [736] mainly involving the nearest-neighbor method [737, 738]. A study by Ediriweera [739] uses dictionaries to correct errors in Sinhala OCR. An early attempt for Sinhala OCR by Dias et al. [740] has been extended to be online and made available to use via desktops [741] and hand-held devices [742] with the ability to recognize handwriting. A simple neural network based approach for Sinhala OCR was utilized by Rimas et al. [743]. A fuzzy-based model for identifying printed Sinhala characters was proposed by Gunarathna et al. [744]. Premachandra et al. [745] proposes a simple back-propagation artificial neural network with hand crafted features for Sinhala character recognition. Another neural network with specialized feature extraction for Sinhala character recognition was proposed by Jayamaha and Naleer [746]. On the matter of neural networks and feature extraction, a feature selection process for Sinhala OCR was proposed by Kumara and Ragel [747]. Jayawickrama et al. [748] worked on Sinhala printed characters with special focus on handling diacritic vowels. However, they opted to refer to diacritic vowels as *modifiers* in their work. Gunawardhana and Ranathunga [749] proposed a limited approach to recognize Sinhala letters on Facebook images. A CNN-based methodology to improve printed Sinhala character OCR was proposed by Liyanage [750]. Printed Character Recognition (PCR) was used by Vasantharajan and Thayasivam [512] to create a large-scale Tamil-Sinhala-English parallel corpus. A meta-study on the effects of text genre, image resolution, and algorithmic complexity needed for Sinhala OCR from books and newspapers was conducted by Anuradha et al. [751]. Anuradha et al. [752] used *Tesseract* 3⁵⁰ [753] for Sinhala OCR. The later study by [754] improved the accuracy of *Tesseract* OCR engine on Sinhala from 53.22% to 86.16% for the data set they tested on. Maduranga and Jayalal [755] used an Artificial Neural Network (ANN) based on Universe of Discourse and Self-Organisation Map methods to recognise multi-style printed Sinhala characters. The study by De Silva et al. [756], even though ostensibly presented as translation research, is just invoking Google Cloud service for translation. Their novelty

in Sinhala NLP lies in the facility provided to OCR the text from photos and documents. The work by Thamarasee and Surendra [757] used a histogram-oriented gradient descriptor (HOG descriptor) [758] and SVMs to recognise 400+ variations of Sinhala characters. After conducting a survey of existing methods [759], Hulathdoowage and Kumara [760] proposed a U-Net [761] based system for Sinhala printed document layout analysis. For this task they have created a data set using Sinhala textbooks. However, they do not provide a public link to this data set. Jayatilleke and de Silva [762] conducted a comparative analysis on zero-shot Sinhala OCR using a 🤖 [synthetic Sinhala OCR data set](#). They concluded that 🌞 [Surya](#) to be the best Sinhala OCR system in this configuration. This aligns with the previous short study conducted by Velayuthan and Ambegoda [763]. However, later in their subsequent work, Jayatilleke and de Silva [764], they showed that for real-world applications Google Document AI⁵¹ obtains better results for Sinhala.

An OCR and Text-to-Speech system for Sinhala named Bhashitha was proposed by De Zoysa et al. [649]. A study on Sinhala text extraction from social media images (memes) was conducted by Samarajeewa and Ranathunga [765]. They specifically handled the character-touching issue. The study by Walawage and Ranathunga [417] attempted to devise a feature set to separately identify Sinhala and English text on social media images (memes). The study by de Silva and Liyanage [766] used Convolutional Spiking Neural Networks to extract Sinhala text from YouTube thumbnails. Chanda et al. [767] proposed a Gaussian kernel SVM based method for word-wise Sinhala, Tamil, and English script identification. The work by Vasantharajan et al. [513] adapted the *Tesseract* engine to handle non-Unicode (legacy fonts) in pdf documents to create a Tamil-Sinhala-English parallel corpus.

3.29.2 Handwritten Text

Fernando et al. [768] claim to have created a database for handwriting recognition research in the Sinhala language and further claims that the data set is available at the National Science Foundation (NSF) of Sri Lanka. However, the paper provides no URLs, and we were not able to find the dataset on the NSF website either. Weraduwa et al. [769] claims to have created a dataset of 373 Sinhala handwriting images from 84 participants where 73 are dysgraphic and 300 are non-dysgraphic. Their objective is that this dataset may be used to detect dysgraphia in children.

The work by Karunanayaka et al. [770] is focused on noise reduction and skew correction of Sinhala handwritten words. A genetic algorithm-based approach for non-cursive Sinhala handwritten script recognition was proposed by Jayasekara and Udawatta [771]. Nilaweera et al. [772] compare projection and wavelet-based techniques for recognizing handwritten Sinhala script. Silva and Kariyawasam [773] worked on segmenting Sinhala handwritten characters with special focus on handling diacritic vowels. A comparative study of few available Sinhala handwriting recognition methods was done by Silva et al. [774]. Silva et al. [775] uses contour tracing for isolated characters in handwritten Sinhala text. A Sinhala handwriting

50. <https://tesseract-ocr.github.io/>

51. <https://cloud.google.com/document-ai/>

OCR system which utilizes zone-based feature extraction has been proposed by Dharmapala et al. [776]. The study by Walawage and Ranathunga [777] and its follow up study by Walawage [778] specifically focus on segmentation of overlapping and touching Sinhala handwritten characters. Silva and Jayasundere [779] focused on recognizing character modifiers in Sinhala handwriting. The similarly named studies by Mariyathas et al. [780] and Wasalthilake and Kartheeswaran [781], both utilize CNN to recognize Sinhala handwriting; as does the study by Weerasinghe [782]. Ifhaam and Jayalal [783] used genetic algorithms to recognize Sinhala handwritten postal addresses for postal sorting. A segmentation-based approach that utilizes an n-gram model to recognize and validate Sinhala words written on touch screens was proposed by Mahesh and Priyankara [784]. They used a CNN classifier and were able to classify 19 different Sinhala characters. The study by Rowel et al. [785] frames their work as an E-Learning platform for hearing-impaired children. However, their research does not contain any work done towards Sinhala sign language to be included in Section 3.30. What they do have is an OCR system that they claim to recognise letters and digits. Even there, we are only given an example of a recognised digit. Whether or not their system can recognise Sinhala letters is not explicitly discussed. As part of their child [sic] cognitive ability assessment model, Kahawanugoda et al. [688], proposed a Sinhala handwriting letter recognition system. The study by Withana and Rupasinghe [786] used Sinhala handwritten text classification to detect Dyslexia and Dysgraphia. The study by eka [787] has used CNN to recognise Sinhala handwritten text. Karunarathne et al. [788] used a Gabor-initialised CNN (GCNN) on a dataset of 6000 handwritten Sinhala character images. *Akura*, introduced by Mihiran et al. [789], is a mobile application that includes handwriting recognition aimed at Sinhala language acquisition. Gunathilaka and Ranathunga [790] proposed using a feature-driven model incorporating stroke thickness, size consistency, spacing uniformity, slant angle, and baseline alignment for Sinhala handwriting detection. Their system obtained a Pearson correlation of 0.509 against human ratings when evaluated on a Likert scale. The study by Athukorala and Ranathunga [791] suggested a flow tangent-guided endpoint pairing algorithm and Bézier curve-based [792] stroke restoration for Sinhala handwritten text restoration prior to recognition. Even though the work by Ramanayake and Weerakoon [793] applies OCR on Sinhala handwritten text, its ultimate objective is not recognising the text itself, but the gender of the author. For this, they use several classical ML models as well as GoogleNet and ResNet.

3.29.3 Ancient Text

Summarising optically recognised old Sinhala text for the purpose of archival search and preservation was explored by Rathnasena et al. [794]. The work of Peiris [795] also focused on OCR for ancient Sinhala inscriptions. Building upon the architecture proposed by Ruwanmini et al. [796], a neural network-based method for recognising ancient Sinhala inscriptions was proposed by Karunarathne et al. [797]. The study by Wickramarathna and Ranathunga [798] created a system to recognise Brahmi characters, correct errors, and generate Sinhala meanings. Heenkenda and

Fernando [799] used Inception-v3 [800], VGG-19 [801], and ResNet-50 [802] to classify Sinhala inscriptions to historical time periods. They later extended their work [803] by proposing to use  YOLOv5 to detect ancient Sinhala inscriptions. The work by Gunasekara et al. [804] claims to have created a mobile app that is capable of recognising and translating early Brahmi characters. The work by Surasinghe and Thanikasalam [805] used CNN models to predict the historical period of ancient Sinhala text. In their subsequent work [806], they extended this work by adding attention to the CNN models. The work by Rathnayaka and Fernando [618] also used YOLO for recognising Brahmi characters.

3.30 Sinhala Sign Language

Meyler [809] claim that there is no such thing as a *Sinhala Sign Language* or a *Tamil Sign Language* and there is only *Sri Lankan Sign Language*. Further, Rishan et al. [810] claim that SSL was derived from British Sign Language (BSL). However, the literature uses *Sri Lankan Sign Language* and *Sinhala Sign Language* interchangeably with the same acronym, *SSL*. In this section, we will discuss works that have used either term. But when relevant we will point out the works that claimed to work on *Sri Lankan Sign Language* as opposed to *Sinhala Sign Language*. As far as we can observe, there does not seem to be a technical or functional difference between the two. We show the Sinhala Sign Language Fingerspelling Alphabet as reported by Punchimudiyanse and Meegama [807] and Munasinghe et al. [808] in Fig 7.

In the domain of Sinhala sign language, Liyanaarachchi et al. [811] claims to have created a signing dataset for the Sinhala sign language however only the abstract of their work can be publicly accessible. DISSANAYAKE et al. [812] also claims to have created a database of gestures that are included in Sinhala sign language. Charuka et al. [813] has published a dynamic word-level Sinhala Sign Language video dataset of 50 classes. The dataset contains 1110 videos. Abhishek and Sumanathilaka [814] claims to introduce *SSL400*, a word-level Sinhala sign language dataset consisting of 384 commonly used words. However, given that their paper is behind a paywall, it is not possible to verify whether the dataset is publicly available or not.

A study for the limited use case of translating 15 Sinhala signs to text was conducted by Fernando and Wimalaratne [815]. The work by Samarasinghe et al. [816] used a CNN model to convert Sinhala sign language to Sinhala text. Along with a Sinhala sign language recognition system. Wijegoonaratna [817] has created a neural network-based approach for real-time Sinhala sign language gesture recognition. The approach by Hettiarachchi and Meegama [818], the approach by Dilakshan and Priyadarshana [819], as well as the approach by Peiris [820] have used CNN to recognise the fingerspelling alphabet of Sinhala sign language. The work by Sewwantha and Ginige [821] used Mask Region-Based CNN (Mask R-CNN) to identify some Sinhala signs. Perera and Jayalal [822] also used CNN to translate Sinhala sign Language to Sinhala text. However, they have also explicitly used Scale-Invariant Feature Transform (SIFT) [823] features. The work by dha [824] used a CNN to classify features of static Sinhala signs, while dynamic signs were modelled as a tree structure of static signs. Dahanayaka et al. [825]

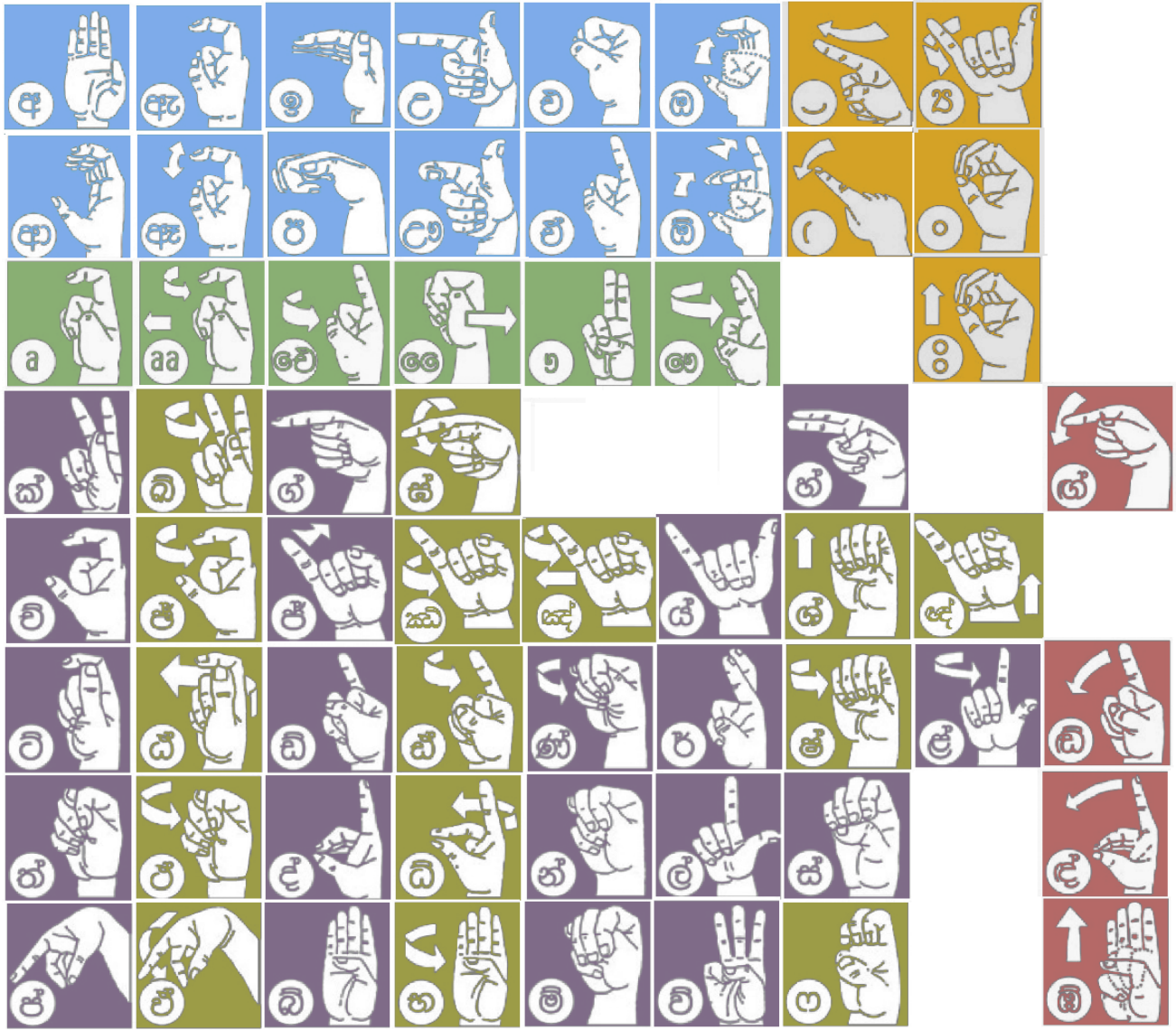


Fig. 7: Sinhala Sign Language Fingerspelling Alphabet as reported by Punchimudiyanse and Meegama [807] and Munasinghe et al. [808]. We have attempted to preserve the colour coding and alignment that we introduced in Fig 4 as much as possible.

also used CNN in their multi-model approach to recognise Sinhala sign language. Weerasooriya and Ambegoda [826] used Random Forest (RF), K-Nearest Neighbour (KNN), Logistic Regression (LR), and a rule-based classifier to recognise Sinhala fingerspelling. Kumarawadu and Izzath [827] used a *Leap Motion Controller*⁵² and Deep Neural Networks (which most probably is, but not explicitly stated, an MLP) to recognise Sinhala sign language. The study by Charuka et al. [813] introduces a novel skeleton-based sign language recognition method named *ProtoSign* built upon prototypical networks [828] to do few-shot learning for Sinhala sign language recognition. The study by Haputhanthri et al. [829] used *ResNet* followed by a transformer encoder with multi-head attention to recognise sentence-level Sinhala sign language. The study by Fernando et al. [830] introduced a mobile app capable of translating Sinhala Sign Language

to text. They further enhance the results of their system with a facial emotion detection system created with CNNs. The study by Gamage et al. [831] also combines Sinhala Sign language recognition and emotion recognition. Krishnananthan et al. [832] introduced *HANDTALK* web-based application designed to teach sign-language. As Such it has a component that translates the sign language gestures of the learner into text. In their study Gedaragoda et al. [833] used traditional machine learning approaches to recognize Sinhala Sign Language. A pose-based method for Sinhala Sign Language recognition was introduced by Indatissa [834]. The work by Perera [835] uses CNN for Sri Lanka Sign Language recognition. The work by Nadeesha and Wasalthilaka [836] used classical machine learning techniques to recognize 40 Sinhala signs from 100 pose videos. They reported a success rate of 86.7%. The work by Navodya et al. [837] reports accuracies between 87% and 92% for recognising Sinhala sign language. The work by Kavinda

52. <https://leap2.ultraLeap.com/leap-motion-controller-2/>

et al. [838] proposed a multi-modal system that converts both sign language and miming language to Sinhala text. Further, they have a component to convert Sinhala audio to miming language and Sinhala text as well. Rajapakshe et al. [839] used a hybrid CNN-LSTM model and reports 94% accuracy on the 20 Sinhala signs they tested on. Thennakon et al. [840] on the other hand reports 96% accuracy in recognising real-time Sinhala Sign Language signs. Heshanee and Tissera [841] attempted Sinhala sign language recognition with 3D CNN, LSTM, and Temporal Segment Networks (TSN). The study by Herath and Vidanagama [842] created a dataset of 23 SSL gestures representing major Sri Lankan cities and used CNN-LSTM and transformer-based models to recognise the said signs. [843] analysed commonly used Sinhala Sign Language detection models with SHAP [338] to report the efficacy of transfer learning in this matter. Rajamanthri [844] tested CNN, LSTM, and YOLOv8 for Sinhala sign language recognition. Silva et al. [845] claim to have built a system that, apart from recognising Sinhala sign language, is also capable of other relevant tasks such as providing sign language training, an environmental sound alert module, and a two-way communication module for visual and sign-based communication. Instead of using a vision-based approach, Madushanka et al. [846] utilises a wearable armband for this task. The work by Godage [847] uses *Myo* armbands for gesture-capturing in trying to recognise Sinhala signs at a sentence level. The work by Munasinghe et al. [808] used a sensor-based approach that requires a special gesture recognition glove.

Punchimudiyanse and Meegama [848] used a 3D avatar to display 200 known signs and 40 fingerprinting signs of SSL. Strides have been made in the domains of computer interpreting for written Sinhala [849] and animation of finger-spelt words and number signs [807]. Thrimahavithana et al. [850] used 3D avatars to show Sinhala sign language on screen. Their system was evaluated with a usability score, error rate in text-to-sign language conversion, and overall user satisfaction.

The work by Idushan et al. [851] claims to be able to convert Sinhala audio to SSL and also recognize SSL and interpret dynamic signs. The system proposed by Gunaratne et al. [852] attempts to build a full communication system where Sinhala text, Sinhala voice, Sinhala sign, and Sinhala Braille may be translated from one-another.

Wickramasuriya [853] used template matching to recognise Sri Lankan sign language. Dissanayake et al. [854] have introduced *Utalk*, a mobile app to interpret Sri Lankan Sign Language. Kumar et al. [855] introduced *EasyTalk*, a tool that translates Sri Lankan sign language to text as well as converts English text into Sri Lankan Sign Language. Both the approaches by Herath and Ishanka [856] and Perera and Kumarika [857] integrated *MediaPipe* [858] with Deep Neural Networks to interpret Sri Lankan sign language. Perera and Kumarika [857] only used LSTM while Herath and Ishanka [856] used LSTM, CNN, and CNN-LSTM. Rishan et al. [810] uses *Leap Motion Controller* (Similar to Kumarawadu and Izzath [827]) to translate Sri Lankan sign language to Sinhala text. Jayasekara et al. [859] uses *EfficientNet-B0* model [860] to detect static Sri Lankan Sign Language.

Priyankara et al. [861] have conducted a short sur-

vey on the available Sinhala sign language recognition systems. They look at 12 studies and conclude that image processing with CNN is the most used technique for Sign Language translation. The work by Chithrani et al. [862] compares a few Sinhala sign language recognition works [810, 815, 816, 818, 820, 824, 825, 854] against work conducted for other languages. The review by Abhishek and Sumanathilaka [863] on deep learning in SSL recognition also covers a subset [429, 808, 810, 812, 813, 815, 819–821, 824, 826, 827, 830, 831, 833, 836, 846, 847, 851–853, 855–857] of the works discussed in this section. The survey by Ahinsa et al. [864] discusses text-to-Sinhala sign language in the contexts of: finger-spelling and translation. It should be noted here that the *translation* they describe here is text-to-sign language translation and not translation between any two or more natural languages. Ahinsa et al. [865] in their survey of Sri Lankan sign language recognition and Sinhala text/speech to sign language translation cover a subset [807–810, 814, 815, 821, 829, 848, 851] of the works discussed in this section. Similarly, the survey by Maleesha et al. [866] on Sri Lankan Sign Language Interfaces for LLM integration covers a subset [811–815, 818, 821, 822, 829, 833, 836, 849, 863] of the works discussed in this section.

3.31 Sinhala Braille

Figure 8 shows the Sinhala Braille Alphabet. The work by de Silva and Liyanage [766] uses KNN, SVM, and a simple ANN system to recognize Sinhala Braille text. Vithanage [868] also claims to have created a conversion engine to easily convert the Braille text into the corresponding Sinhala text. The study by Madubashana [869], the study by Ariyaratna et al. [870] as well as the study by Weerasinghe [871] also focused on creating an automated Braille to Sinhala recognition system. Dampalessa and Ragel [872] used a dataset of 230 characters to train a model to transliterate Braille letters into Sinhala text. Anuradha and Thelijjagoda [624] uses OCR on Sinhala and English Braille documents on which they then run a machine translation system in order to convert them into voice. Dananjane et al. [873] used YOLOv8 for Sinhala Braille character detection and classification, which were then post-processed through an LLM to correct grammar and refine context. The fixed text was then sent to a TTS system to obtain Sinhala speech. Udana et al. [874] introduces a prototype tactile Braille display for Sinhala. They claim to be able to convert digital Sinhala text into Braille using an electromechanical process that also utilises a CNN. Gunaratne et al. [852] proposed a system that may translate between Sinhala text, Sinhala voice, Sinhala sign, and Sinhala Braille.

3.32 Large Language Models (LLMs)

Pramodya et al. [875] partially released, 🤗 *SinhalaMMLU*, a data set of more than 7000 Sinhala questions and answers covering secondary to collegiate education levels in Sri Lanka. Only around 1900 pairs are publicly available. The authors request that they be contacted to receive the rest of the data set. Further, it is released under CC BY-NC-ND 4.0 license; therefore, no derivatives of this data set can be published. They used this dataset to evaluate 26 LLMs and reported that Claude 3.5 achieved the highest average

අ a	ආ æ	ඈ i	ඉ u	එ e	ඔ o	ඌ		
ඊ ā	උ æ̃	ඉ ī	ඊ ū	එ ē	ඔ ō	ඌ m̃		
ඌ r̃	ඌ r̃̃	ඌ ai	ඌ au	ඌ l̃	ඌ l̃̃	ඌ h̃		
ක ka	ඛ kha	ග ga	ඝ gha	ඞ ṇa		හ ha	ඛ ṇga	
ච ca	ඡ cha	ජ ja	ඣ jha	ඤ ṇ̃	ය ya	ශ śa	ඤ jñā	
ට ṭa	ඨ ṭha	ඳ ḍa	ඳ ḍha	ණ ṇa	ර ra	ස sa	ළ ḷa	ඳ ṇḍa
ත ta	ථ ṭha	ද ḍa	ධ ḍha	න na	ල la	ස sa		ද ṇḍa
ප pa	ඵ pha	බ ba	භ bha	ම ma	ව va/wa	ෆ fa		ඹ m̃ba

Fig. 8: Sinhala Braille Alphabet. We have attempted to preserve the colour coding and alignment that we introduced in Fig 4 as much as possible. Given that Sinhala Braille is an extension of Bharati Braille and as such uses the same punctuation schema [867], we did not include it in this diagram.

accuracy of 67%. Singh et al. [876] introduced the 🧐 [Aya Data set](#), a human-curated multilingual instruction fine-tuning (IFT) data set for large language models (LLM). They also introduced 🧐 [Aya Collection](#), 44 instruction-style datasets that were created by transforming existing NLP datasets into pairs of *prompt* and *completion* and then using NMT on them to obtain data in multiple languages. Both of these data sets include Sinhala. Further, they also released 🧐 [Aya Evaluation suite](#) to measure multilingual open-ended generation quality.

The short survey by Unjum et al. [877] discusses the status of Sinhala in contemporary LLMs. They discuss the importance of cultural alignment of LLMs for Sinhala, but do not discuss GlobalPIQA [593, 878], which is the first dataset with Sinhala for LLM cultural alignment. Üstün et al. [127] introduced 🧐 [Aya](#), a massively multilingual generative language model trained on their *Aya data set* [876].

Amarakoon and Rajapakse [879] further fine-tuned the *Aya* model using Sri Lankan business and regulatory datasets to automate multilingual customer interactions and generate context-sensitive business insights in Sinhala and Tamil for Sri Lankan Small and Medium Enterprises (SMEs). [880] introduced 🧐 [Tiny Aya](#), a scaled-down version of the *Aya* model. Further, they have a model named 🧐 [Tiny Aya: Global](#) balanced for multilingual performance and specifically a model named 🧐 [Tiny Aya: Fire](#) specialised for South Asian languages. However, Sinhala is not listed in the HuggingFace description of the latter.

Aravinda et al. [334] introduced 🧐 [SinLlama](#), the first pre-trained Sinhala LLM-base model. They used continual pre-training with a 10-million-sentence Sinhala corpus on Llama-3-8B to obtain this model. They show the utility of this model with three text classification tasks. Further, for this task, they created a 🧐 [Sinhala tokeniser](#).

Abeynayake et al. [335] fine-tuned `SinLlama` for Sentiment analysis in Romanised Sinhala-English Code-Mixed social media text. Sandaruwan et al. [881] also claims to have trained a Sinhala LLM by continual pre-training on LLaMA 3.1. However, they do not build upon or compare with any other work in the Sinhala LLM domain other than their own prior work [882]. Further, neither their data nor their models are publicly available for any comparative study.

Jayakody and Dias [883] conducted a comparative analysis on LLMs for Sinhala. In it, they observed that the small (7B) models of Claude⁵³ and GPT-4o⁵⁴ handled Sinhala better than Llama 3 (8B)⁵⁵ and Mistral 7B⁵⁶. Further, they reported that they had to exclude Falcon 2⁵⁷ and Gemini⁵⁸ from the study due to poor or non-existent support for Sinhala. Hansaka and Sumanathilaka [884] also claims to have conducted a review on fine-tuning approaches for LLMs such as GPT-3, GPT-4, and LLaMA in the context of Sinhala. Chang et al. [885] introduced 🐟 *Goldfish*, a collection of monolingual language models which includes Sinhala trained on data from MADLAD [116], NLLB [263], and Glot500 [119]. The work by Ayesha et al. [886] used a fine-tuned Llama 3.1 (8B) model [887] to predict student performance by analyzing textual entries created by students in Sinhala and English. Tang et al. [888] evaluated the susceptibility of LLMs being tricked into answering illegal questions in low-resource languages when the same questions would have been successfully rejected had they been asked in a high-resourced language. They provide the framework for testing with the results on eight low-resourced languages, which include Sinhala. In their work to overcome data sparsity for LLMs, Lai et al. [889]⁵⁹ tested the Sinhala text generation and understanding results of Google Translate against LLaMA, BLOOM, and several other variations. Even though they point to a 🗄️ *repository* of the instruction data set they created, currently, the data set cannot be found at the given URL. Puranegedara et al. [891] proposed an intermediate-layer fusing method to LLMs so that they would perform better for low-resource languages such as Sinhala. They show their improvement with Sinhala News Results on the data set created by de Silva [40]. Senanayaka and Thanthriwatta [892] claim to have created a dataset named `SinMet-5K`, which is a multi-annotator Sinhala metaphor dataset aimed to be a benchmark to test the detection capabilities of LLMs. Devine et al. [254] introduced *Kakugo*, a pipeline designed to train general-purpose Small Language Models (SLMs)

Vayani et al. [422] analyzed the performance of a number of LLMs over 100 languages that included Sinhala. The basic dataset was created by machine translating English QA pairs from *LLaVA-Bench* (In-the-Wild) dataset [423]. This was further extended with culturally specific data points by searching for images under *country name*, *language name*, and *cultural category*. This dataset, which they named 🐟 *ALM-Bench*, is publicly available. They made a number

of observations in this aspect. First, they observed that for the Q&A task, the performance of GPT-4o [424] for Sinhala is better than Qwen2-VL [893]. Next, they report that including images in the question further improved Sinhala results of GPT-4o by 38.7%. In the error analysis of the same, they report that 36% of the errors are language errors (fluency), 31% of the errors are due to lack of knowledge, and 18% of the errors are due to a lack of cultural understanding. Shafique et al. [431] when introducing *ViMUL-Bench*, a multimodal video Q&A benchmark for LLMs that includes Sinhala, show that GPT4o has the best results for Sinhala.

3.33 LLM Alignment

As LLMs grow in usage, their bias towards *WEIRD* [894] (Western, Educated, Industrialised, Rich, and Democratic) cultures has been observed and attempts to align them (also) with other cultures have materialised. Chang et al. [878] created 🌍 *Global PIQA*⁶⁰, a multilingual extension to *PIQA* [895] in which they include physical commonsense reasoning triples covering 100+ languages, such that the examples are culturally aligned to the relevant languages. The objective of this dataset is to function as a benchmark to measure the cultural alignment of LLMs. Their analysis results, which include Sinhala, are also available on 🗄️ *GitHub*. Specifically, the Sinhala portion, created by de Silva and Ranathunga [593], can be accessed in multiple forms: 🐟 *Nonparallel*, 🐟 *Nonparallel-Romanised*, 🐟 *Parallel*, and 🐟 *Parallel-Romanised*. The 🐟 *CultureMarkers* dataset released by Sahu et al. [896] contains 9705 Sinhala entries, all sourced from 🐟 *Cultura-X* [897]. Muthugala et al. [898] introduced 🗄️ *LKValues*, the first survey-grounded resource suite for Sri Lankan value alignment. It includes *LKvaluesIT*, an instruction dataset to align LLMs to Sri Lankan values and *LKvaluesBench*, a benchmark dataset to test the aforementioned alignment. The datasets are in Sinhala, Tamil, and English.

3.34 Healthcare Applications

In discussing the management of certain mental conditions (eg: Autism, ADHD), this section covers systems targeted for teaching and evaluating individuals with such special needs. Information on systems handling Sinhala language acquisition and its evaluation in the general setting is discussed in Section 3.38 instead.

Using classical machine learning algorithms on TF-IDF features, Fazal and Farook [418] predicted depression in Sinhala-English code-mixed data from Twitter and Facebook. Hewapathirana and Sumanathilaka [899] introduced *EmoScan*, a method to detect depression in Romanized Sinhala (Singlish) tweets. The study by Herath and Wijayasiriwardhane [900] used classical machine learning models to predict suicidal ideation from Sinhala Facebook posts. Fazal and Farook [901] compared multiple algorithms to detect depression in Sinhala-English code-mixed social media posts and reported that *ExtraTreesClassifier* [902] obtains the best results. Vitharana et al. [903] also attempted

53. <https://docs.anthropic.com/en/docs/about-claude/models>

54. <https://openai.com/index/hello-gpt-4o/>

55. <https://llama.meta.com/docs/get-started/>

56. <https://docs.mistral.ai/>

57. <https://falconllm.tti.ae/>

58. <https://deepmind.google/technologies/gemini/>

59. Later elaborated as Lai [890].

60. <https://mrlbenchmarks.github.io/>

to detect depression, suicidal ideation, or irrelevance from Sinhala Facebook posts. They report the highest F1 of 0.88 with RoBERTa. The work by Sewwandi et al. [904] claims to have built a set of models to monitor depression, where the voice-based model achieves 86.69% test accuracy, social media depression detection achieves 97.15% accuracy, and typing behaviour analysis achieves 80% accuracy.

Welarathna et al. [719] used CNNs to classify emotions of Sinhala speech by Autistic children. They handle the following emotions: sad, disgust, surprise, neutral, happy, calm, fear, and angry. Jayawardena et al. [905] used a *NAO Robot* [906] to create an interactive system with Sinhala language content detection (Speech to Text), dialogue management, and voice synthesis (Text-to-Speech) for Autism Intervention in children. Continuing their work, Jayawardena et al. [907] used the RASA framework [473] to build a dialogue system that provides speech therapy to Sinhala-speaking Children with Autism Spectrum Disorder (ASD). Further, Manukalpa et al. [908] extended this work with cultural alignment. A Sinhala speech-based Autism intervention extension was then proposed by Bopage et al. [909].

The system by Kulasooriya et al. [910] used SVM and a simple ANN to detect psychological stress in Sinhala speech. Rathnayake et al. [911] used voice classification by biomarkers to detect Parkinson's disease with classical machine learning algorithms. The work by Alahakoon et al. [912] used Random Forest on Sinhala handwriting of children as a pre-screening for Attention-Deficit Hyperactivity Disorder (ADHD) diagnosis. [913] claim to have created an AI-IoT robotic system named *Huruwa* aimed at supporting Sinhala-speaking children with speech sound disorders.

Perera and Sumanathilaka [914] discusses the lack of annotated datasets and NLP resources to cater for Sinhala-speaking adults with dyslexia. *LexiLearn*, a gamified system intended to help Sinhala learners with Dyslexia was introduced by Weerasinghe et al. [915]. Dyslexia and dysgraphia were detected using Sinhala handwritten text classification by Withana and Rupasinghe [786]. A system named *Arunalu* that they claimed to use Voice recognition, Natural Language Processing, Machine Learning, and Deep Learning concepts to help individuals with dyslexia overcome problems of reading Sinhala was proposed by Sandathara et al. [916]. Lokubalasuriya et al. [917] proposed REaDDS, a tool for early detection and screening of dyslexia in Sinhala. Weraduwa et al. [918] used VGG16 to detect Dysgraphia among Sinhala-speaking primary school children. The work by Perera and Sumanathilaka [919] created a pipeline with: Whisper for speech-to-text, SinBERT for dyslexic error detection, mT5 and Mistral to generate corrected text, and finally GoogleTTS (gTTS) for text-to-speech. An Automated Essay Scoring (AES) system with integrated dyslexic writing-pattern classification named *Akura* was created by Fonseka et al. [920]. The *Nena Sawiya* system proposed by Thisera et al. [921] aims to detect and intervene in multiple Learning Disabilities (LD) such as: Receptive Language Disorder (RLD), Expressive Language Disorder (ELD), Visual Discrimination and Memory difficulties, and Visual Closure deficits. They employ speech-to-text as well as text classification with classical models such as Random Forest and regression models. *Shilpa* introduced by Ishara et al. [922] is a primary education mobile learning applica-

tion for students with visual, auditory, physical, or cognitive impairments. The system developed by Pathiraja et al. [923] uses multimodal emotional analysis with therapeutic story generation in Sinhala as well as a Sinhala voice-first mood prediction system using fine-tuned XLM-RoBERTa to help early detection of mental health issues in Children. [924] proposed a system they name IPITPH (Integrated Predictive Intelligence Tool for Pediatric Health). One of its features is voice-first caregiver interaction in Sinhala using a RAG-based multilingual conversational AI.

3.35 Legal and Parliamentary Domain

Perpetually maintained Sri Lanka Document Datasets by Senaratna [129] includes a number of data sets in the Legal and Parliamentary domain: [Hansards](#), [Legal Documents](#), [Cabinet Decisions](#), and [Police Press Releases](#). From this, Lasandi and Jayatilleke [925] derived  *SinhaLegal*, a curated corpus of Sinhala legal documents including 1,065 acts and 141 bills. They transformed the original PDF-only dataset by running OCR with data cleaning and structuring so that a dedicated machine-readable corpus is obtained. The OCR dataset  *sinhala-ocr-lk-acts-1010* created by Dilhara and Jayatilleke [726] uses Sri Lankan Legislative Acts published between 1981-1989 and 2000-2019 as their domain. This contains an annotated set of 1,010 page images and their transcriptions.

The work by [518] attempted to study the viability of using Google Translate for the legal domain English-Sinhala and Sinhala-English translation. Many concrete examples of legal phrases that need to be translated were extensively analysed by human experts in this work. Jayawardana and Vidanagama [926] conducted TF-IDF-based topic modelling for Sinhala parliamentary discussion categorisation into policy domains such as Education, Agriculture, and Health.

3.36 News and/or Social Media Recommendation

A trending topic detection model for Sinhala tweets using simple clustering and ranking algorithms was proposed by Jayasekara and Ahangama [927]. Sandamini et al. [928] proposed a post recommendation system, which supports Singlish, for social media. Tennakoon and Gamlath [929] proposes a hybrid system which uses skip-gram and collaborative Filtering on a Multi-Layer Perceptron for recommending categorised Sinhala news articles. Tennakoon et al. [930] then extends the system to also be able to identify grey sheep users while performing the aforementioned hybrid recommendation using LDA [281] and SVM. Following the above work, a news aggregator with news categorisation, comment filtering, and two types of recommendation systems was proposed by Malsha et al. [931]. Madhushika et al. [932] analysed Twitter trending topics to understand how Sinhala Twitter data affects news dissemination on mass media. They proposed calculating a *news value* to a tweet which can be utilised to sort tweets by their newsworthiness in order to give better recommendations.

3.37 Pali-Sinhala Applications

Pali is the liturgical language of Buddhism [620–622]. Given its cultural relation to the Sinhalese people, the Sinhala

script is one of the commonly used scripts for Pali. In addition to that, many Sinhala-Pali bridging NLP datasets and projects exist.

A collection of Sinhala-Sinhala, Sinhala-English, English-Sinhala, and Pali-Sinhala dictionaries is available on [GitHub](#) courtesy of [aruth.lk](#). Another Pali-Sinhala dictionary has been made [available](#) by the Path Nirvana project. The same project provides a large repository of Buddhist books with Sinhala and Pali text named as [books](#) and [buddhist-books](#). Further, they have some News data collected from [Silumina](#), [Mawbima](#), and [AdaDerana](#). An associated profile contains a large [Sinhala TTS data set](#) consisting of 6248 sentences with 13.8 hours of recordings over two speakers: one male and one female. Gurusinghe and Jayatilleke [933] introduced [SiPaKosa-Books](#) and [SiPaKosa-Sent](#), a large-scale Sinhala-Pali corpus combining classical Sinhala Buddhist texts from historical archives with complete canonical scriptures from web sources. Their [code](#) is also publicly available. The importance of translation between Sinhala and Pali was discussed by Jayasinghe [619]. Shalini and Hettige [623] attempted to use a dictionary-based machine-translation method for this task.

3.38 Language Acquisition and Evaluation

This section only discusses Sinhala language acquisition and its evaluation in the general setting. Information on systems targeted for teaching and evaluating individuals with special needs (eg: Autism, ADHD) is discussed in Section 3.34 instead.

Sandathara et al. [916] proposed a system which they named *Arunalu* that they claimed to use Voice recognition, Natural Language Processing, Machine Learning, and Deep Learning concepts to help individuals with dyslexia overcome problems of reading Sinhala. The learning bot *MiMi* proposed by Vithana et al. [934] assists children to learn to speak without stuttering. Nethmi et al. [935] introduced *Narrataa*, a mobile application targeted at developing the Sinhala vocabulary of children by using computer vision to caption images in Sinhala. The mobile application developed by Saranga et al. [936] helps individuals learn Sinhala letters with the help of images, colours, and shapes. The application also attempts to generate stories in Sinhala. The *Katha App* developed by Wijesooriya et al. [937] uses SVM and LSTM to identify phonological disorders from Sinhala speech. Walpola et al. [938] used Quadratic SVM and Subspace KNN to study Sinhala speed reading and regular reading-induced Functional Magnetic Resonance Imaging (fMRI). Fuzzy C-Means (FCM) clustering was used for data augmentation, and SHAP [338] was used for model interpretability.

Francis et al. [939] claims to use Sinhala OCR and PoS tagging in their effort to build the *SinhaLearn*, a system for improving the Sinhala Proficiency of grade 5 scholarship students. Praveen et al. [201] also claims to have created a web-based education system that helps grade 5 scholarship students who are learning the Sinhala language. They claim that they have created a Sinhala Spell checker with 82% accuracy and a Sinhala TTS system with 98% accuracy within

this system. Weerasinghe et al. [915] introduced *LexiLearn*, a gamified system intended to help Sinhala learners with dyslexia. The survey by Perera and Sumanathilaka [914] discusses the lack of annotated datasets and NLP resources to cater for Sinhala-speaking adults with dyslexia. An interactive platform capable of handwritten Sinhala character recognition and prediction was introduced by De Silva and Athukorala [940] for the purpose of teaching preschool children the Sinhala alphabet. Mihiran et al. [789] introduced *Akura*, a mobile application that includes handwriting recognition, gesture-based numeracy, emotion-aware lesson adaptation, and speech pronunciation feedback. Yet another system named *Akura* was created by Fonseka et al. [920]. However, their system is an Automated Essay Scoring (AES) system with integrated dyslexic writing-pattern classification. Lokuhewage et al. [941] have proposed, *Sinhala Learn*, an end-to-end educational assistant for Sinhala medium education with OCR on both printed and handwritten text, question answering and summarisation, voice recognition which includes Southern dialect normalisation. De Silva and Athukorala [942] introduced a digital platform to help students learn how to write Sinhala letters by using handwriting recognition. Jayasekara et al. [943] tested models such as BERT-base, RoBERTa-large, DeBERTa-v3 for Sinhala automated essay scoring (AES) using a Sri Lankan GCE O/L examination dataset.

3.39 Miscellaneous Applications

In this section, we discuss NLP tools and research which are either hard to categorize under the above sections or reasonably involve multiples of them. Fernando [944] proposed a method for inexact matching of Sinhala proper names. A study on determining canonical word order of colloquial Sinhala sentences using priority information was conducted by Kanduboda and Tamaoka [945] which they later extended [946–948]. A cross-language information retrieval system where Sinhala search queries are converted to English search queries by mapping Sinhala word embeddings to English word embeddings was proposed by Hisan et al. [558].

Rajitha et al. [530] has proposed a statistical machine translation and transliteration method to align Sinhala and English documents. Rajitha et al. [949] used the data set that they introduced in their previous work [266] to prove that task-specific supervised distance learning metrics outperform their unsupervised counterparts, for document alignment. Fernando et al. [506] have used pre-trained multilingual language models (PMLMs) to improve document and sentence alignment between Sinhala–English, Tamil–English, and Sinhala–Tamil language pairs.

Kumari and Hettiarachchi [950] proposed an algorithm for Sinhala topic modelling based on LDA [281] and RAKE [951]. Batawalaarachchi [952] proposed two methods using statistical features to select words to be included in Sinhala document titles. The study by Pallawala and Had-dela [953] conducted a comparative analysis of LDA, LSA, and NMF for Sinhala and concluded that NMF archives best results. Arambewela et al. [954] proposed a Sinhala writing assistant tool utilising CNNs. The study by Bandara et al. [955] has introduced a Sinhala virtual assistant with

rule-based decision-making for a pre-set collection of user queries and a general Transformer-based model for general user questions. Jayaweera et al. [956] used classical machine learning methods to propose dynamic stop word removal. They claim to have released a corpus of 100,000+ Sinhala documents in their paper. But they provide no information on where to obtain this corpus. Liyanage et al. [957] have created a small Sinhala treebank of 100 sentences. Minixhofer et al. [958] have used Sinhala from *OPUS100* [499] dataset to pre-train their multilingual punctuation-agnostic sentence segmentation model. Given that Sinhala is not one of their main target languages, they do not report sentence segmentation results for Sinhala in much detail. However, they do report that their method improves Sinhala sentence splitting F1 score to 86.0 above their SpaCy baseline of 75.8. A meta-study on how language data from social media are used in research was conducted by Hewapathirana [959].

The study by [222] performs best in this regard. used NER to cluster Sinhala news articles. Subasinghe and Samarawickrama [960] analysed the *Noto Sans Sinhala*⁶¹ font in the context of legibility on small-scale digital device screens. Zhao et al. [961, 962] discussed a Sinhala LLM-based system they utilized to conduct participatory research in Sri Lanka. While they discussed how the LLM system helped their goals and briefly touched on how they utilised a hybrid translation approach of NMT and rule-based methods to circumvent the lack of training data, the specifics of their LLM system are not given. Wijerathne et al. [963] introduced *ScheduleMe* an LLM multi-agent system to manage a Google calendar. They reported a success rate of 70% for Sinhala. It should be noted that among the languages tested by them, only Chinese achieved a worse result (65%) than Sinhala. de Silva [964] observed that Sinhala lacks a diachronic corpus. To alleviate this, Jayatilleke and de Silva [764] introduced  *SiDiac*, the first Sinhala diachronic corpus. It covers 58k tokens across 46 Sinhala literary works from the 5th to the 20th century CE under Religious, History, Poetry, Language, and Medical genres. They followed this up with  *SiDiac-v.2.0* [965], which covers 241k words across 185 literary works from the same time period and categorisation. In this extended version, they further introduced a number of advanced preprocessing steps that are not utilised in the earlier version. [966] evaluated the mathematical reasoning capabilities of LLMs in Sinhala compared to English and Tamil.

The work by Jayasiri and Peellawalage [967] attempted to identify Sinhala song genres by classifying their Mel-frequency Cepstral Coefficients (MFCCs) [968] spectrogram images using CNN models such as VGG-19, InceptionV3, EfficientNet, and DenseNet. Karunathilake et al. [969] used Support Vector Machine, k-Nearest Neighbours, Random Forest, and a Neural Network to classify a Sinhala song dataset of their own into Pop, Rock, Hip Hop, Classical, and Folk genres using the audio features in the songs. The earliest attempt at semantic analysis was done by Herath et al. [970] using their earlier work, which dealt with Sinhala morphological analysis [78].

4 PRIMARY SOURCES

Even though the main objective of this survey is to cover NLP tools and research, we noticed that much of these NLP tools and research depend on primary sources of Sinhala language such as printed books in the role of knowledge sources and ground truth. Therefore, for the benefit of other researchers who venture into Sinhala NLP, we decided to add a short introduction to the available primary sources of Sinhala language used by their peers. We note that the body of work by a single scholar, Disanayake [2, 65, 971–981], is quite prominent in the case of being used for NLP applications. For formal introduction of the language, the books by Disanayake [2] and Perera [3] are commonly used. In cases which deal with the Sinhala alphabet, the introduction by Indrasena [982] and by Disanayake [971, 972] have been used. An analysis of modern Sinhala linguistics has been done by Jayathilake [983] and by Pallatthara and Weihene [66]. The early study by Henadeerage [984] covers a number of topics on the Sinhala language such as grammatical relations, argument structure, phrase structure and focus constructions.

As we discussed in Section 3.3, a number of printed Sinhala dictionaries exist, Malalasekera [131] being the most prominent English-Sinhala dictionary among them. In addition to that seminal work, previous researchers of Sinhala NLP have utilized a number of other dictionaries of various configurations such as: English-Sinhala [133, 136, 137], Sinhala-Sinhala [132, 135], and English-Sinhala-Tamil [134].

A number of NLP applications have utilized first sources intended to teach children [985–989] or foreigners [32, 990, 991]⁶². This makes sense given that an introduction written for children would start with basic principles and thus be ideal for crafting rule based NLP systems and an introduction written for foreigners would have Sinhala language described in terms of English, making easy the process of rule based translation of English NLP tools to Sinhala.

For applications where a rule based approach for Sinhala spelling correction is utilized, the books by Disanayake [978, 979], by Koparahewa [992], and by Gair and Karunatilake [993] are used to provide a basis. A number of NLP applications which handle spoken Sinhala in the capacity of phonological layer (Section 3.28) applications or otherwise, make note of the fact that spoken Sinhala is considerably distinguishable from written Sinhala, as such, they refer primary sources which explicitly deal with spoken Sinhala [33, 65, 977, 981, 987, 990, 994].

Primary sources used in NLP application for Sinhala grammar are varied. A number of them provide overviews of the entirety of Sinhala grammar [24, 66, 69, 980, 995–1004]. There are specific primary sources focusing on verbs [976, 988, 1005], nouns [975, 989], prepositions [987], compounds [974], derivation [973], case system [1006], and sentence structure [70] of the Sinhala language. The book by Rajapaksha [1007] is commonly used in NLP applications as a guide for word tagging and punctuation mark handling. NLP studies that tackle the hard problem of handling questions expressed in Sinhala often refer to the book by Kariyakarawana [1008]. Kekulawala [1009] has

61. <https://fonts.google.com/noto/specimen/Noto+Sans+Sinhala>

62. Note that [991] is an extension of [32].

aptly discussed the much controversial topic of the situation of *future tense* of Sinhala.

5 CONCLUSION

At this point, a reader might think, there seems to be a significant number of implementations of NLP for Sinhala. Therefore, how can one justify listing Sinhala as a resource poor language? The important point which is missing in that assumption is that in the cases of almost all of the above listed implementations and findings, the only thing that is publicly available for a researcher is a set of research papers. The corpora, tools, algorithm, and anything else that were discovered through these research are either locked away as properties of individual research groups or worse lost to the time with crashed ancient servers, lost hard drives, and expired web hosts. This reason and probably academic/research rivalry has caused these separate research groups not to cite or build upon the works of each other. In many cases where similar work is done, it is a re-hashing on the same ideas adopted from resource-rich languages because of, the unavailability of (or the reluctance to), referring and building on work done by another group (Refer Fig. 14 in Appendix C). This has resulted in multiple groups building multiple foundations behind closed doors but no one ending up with a completed end-to-end NLP workflow. In their analysis of low-resourced languages, Ranathunga and de Silva [41] observed that only 11.43% of Sinhala NLP papers have released the relevant data sets. Further, according to them, code being released sits at 9.71% while tools being released sits at 5.71%. While the 11.43% figure of data release may induce a feeling of availability, Ranathunga and de Silva [41] further observes that it is only 1.14% has been released in public repositories. Research publications promising access to data and code only to be found lacking later is a common academic shortfall according to Gabelica et al. [60] however, given that Sinhala NLP is already having minimal work done as a whole compared to some other languages, we simply cannot afford to lose any of the generated data or code. In conclusion, what can be said is that, even though there are islands of implementations done for Sinhala NLP, they are of very small scale and/or are usually not readily accessible for further use and research by other researchers. Thus, so far, sadly, Sinhala stays a resource-poor language.

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APPENDIX A

EVOLUTIONARY ERAS OF THE SINHALA SCRIPT

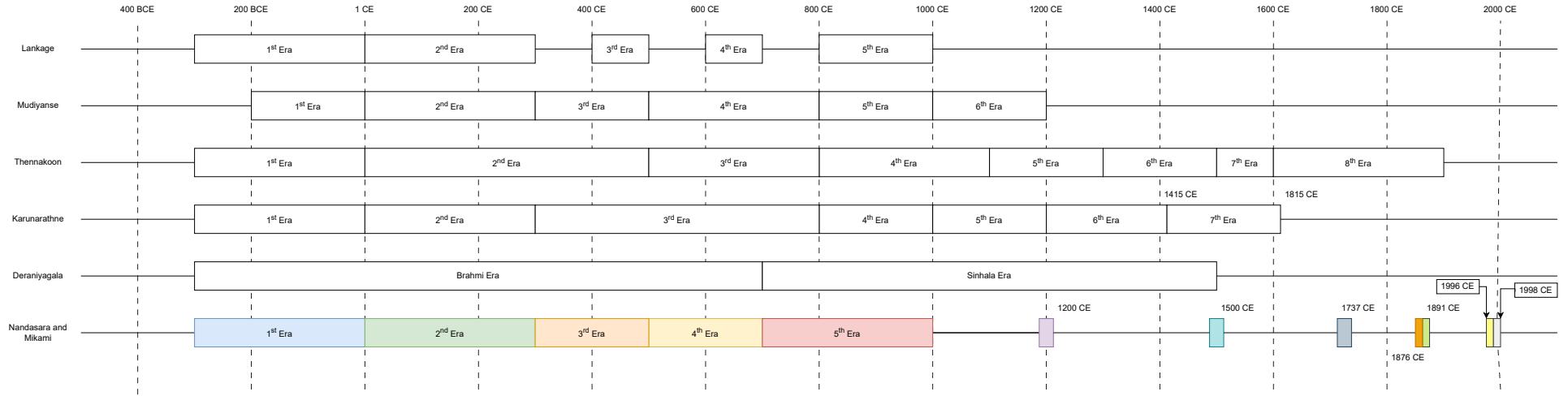


Fig. 9: Evolutionary eras of the Sinhala alphabet as proposed by Lankage [1010, 1011], Mudiyanse [1012], Thennakoon [1013], Karunaratne [1014], Deraniyagala [1015], and Nandasara and Mikami [90]

The evolutionary eras of the Sinhala script have raised much scholarly debate. We have shown some prominent categorizations in Fig 9. Other than Mudiyanse [1012], scholars generally agree with the fact that the oldest records start at around 300 BCE. Other than Deraniyagala [1015], all others agree that an era border should exist at 1AD. However, this is a peculiar observation. Lankage [1010, 1011], Mudiyanse [1012], Karunaratne [1014], and Nandasara and Mikami [90] place an era border at 300 CE while Lankage [1010, 1011], Mudiyanse [1012], Thennakoon [1013], and Nandasara and Mikami [90] place an era border at 500 CE. The next most common border placement is at 800 CE by Lankage [1010, 1011], Mudiyanse [1012], Thennakoon [1013], and Karunaratne [1014]. The last relatively common border placement is agreed between Lankage [1010, 1011], Mudiyanse [1012], Thennakoon [1013], and Nandasara and Mikami [90]. Given that the widest coverage is provided by Nandasara and Mikami [90], we have used their era definitions and examples in Fig 10 and Fig 11. The sources from which Nandasara and Mikami [90] have extracted the ancient script are given by the relevant superscripts as follows:

300 BCE - 1 CE

- 1 Periyankulama (207-197 BCE)
- 2 Mihintale (207-197 BCE)
- 3 Situlpawwa (161-137 BCE)
- 4 Korawakgala (77-63 BCE)
- 5 Ritigala Weweltanne (22-7 BCE)
- 6 Yatahalena Vihara (22-7 BCE)
- 7 Gallena Vihara (22-7 BCE)
- 8 Nuwaragala (22-7 BCE)
- 9 Ritigala Andiyakanna (22-7 BCE)
- 10 Boowattegala (22-7 BCE)
- 11 Rajagala (44-22 BCE)

1 CE - 300 CE

- 12 Anuradhapura (1-7 CE)
- 13 Situlpawwa (1-7 CE)
- 14 Maharatmale (7-18 CE)
- 15 Wallipuram (67-111 CE)
- 16 Viharagala (60-67 CE)
- 17 Pahala Kainattama (60-67 CE)

300 CE - 500 CE

- 18 Tonigala (301-328 CE)
- 19 Ruwanweliseya (337-365 CE)
- 20 Thissamaharama (406-428 CE)
- 21 Anuradhapura (437-452 CE)

500 CE - 700 CE

- 22 Kandanadu (517-518 CE)
- 23 Dhakshinathupa (639-650 CE)
- 24 Baron Paviliyan (639-650 CE)
- 25 Kuchchaweli (639-650 CE)
- 26 Murutawa (639-650 CE)

700 CE - 1000 CE

- 27 Thiriyaya (733-771 CE)
- 28 Viyauppotha (853-887 CE)
- 29 Dorabewila (915-923 CE)
- 30 Baddulla (946-954 CE)
- 31 Polonnaruwa (982-1029 CE)
- 32 Indikatusaya (982-1029 CE)

	IPA	Transliteration	300 BCE - 1 CE	1 CE - 300 CE	300 CE - 500 CE	500 CE - 700 CE	700 CE - 1000 CE	1200 CE	1500 CE	1737 CE	1876 CE	1891 CE	1996 CE	1998 CE
ං	~ *	m̐								ං	ං	අ(ං)	ං	ං
ඃ	h **	h̐										අ(ඃ)	ඃ	ඃ
අ	a,ə	a	අ ¹	අ ¹⁴	අ ²⁰	අ ²⁵	අ ³¹	අ	අ	අ	අ	අ	අ	අ
ආ	a:,a	ā	ආ	ආ			ආ ³⁰				ආ	ආ		ආ
ඇ	æ	æ					ඇ ²⁸			ඇ	ඇ	ඇ		ඇ
ඈ	æ:	ǣ					ඈ ²⁸				ඈ	ඈ		ඈ
ඉ	i	i	ඉ ¹	ඉ ¹⁵	ඉ ¹⁸	ඉ ²³	ඉ ³¹	ඉ	ඉ	ඉ	ඉ	ඉ	ඉ	ඉ
ඊ	i:	ī						ඊ			ඊ	ඊ/ඊ̄	ඊ	ඊ
උ	u	u	උ ²	උ ¹⁶	උ ²⁰	උ ²⁵	උ ³²	උ	උ	උ	උ	උ	උ	උ
ඌ	u:	ū		ඌ ¹²							ඌ	ඌ		ඌ
ඹ	ri,ru	r̥										ඹ	ඹ	ඹ
ඹා	ri:,ru:	r̄										ඹා		ඹා
ඬ	li	l̥										ඬ		ඬ
ඬා	li:	l̄										ඬා		ඬා
ඒ	e	e	ඒ ⁵	ඒ ¹⁷	ඒ ²⁰	ඒ ²⁴	ඒ ³¹	ඒ	ඒ	ඒ	ඒ	ඒ	ඒ	ඒ
ඒ	e:	ē									ඒ	ඒ		ඒ
ඓ	aj	ai									ඓ	ඓ		ඓ
ඔ	o	o	ඔ ⁶			ඔ ²²	ඔ ³¹	ඔ	ඔ	ඔ	ඔ	ඔ	ඔ	ඔ
ඔ	o:	ō									ඔ	ඔ		ඔ
ඕ	aw	au										ඕ		ඕ
ක	k	k	ක ²	ක ¹⁴	ක ²⁰	ක ²⁵	ක ³¹	ක	ක	ක	ක	ක	ක	ක
ඛ	k	kh	ඛ ⁴			ඛ ²⁵	ඛ ³²	ඛ	ඛ		ඛ	ඛ	ඛ	ඛ
ග	g	g	ග ²	ග ¹⁴	ග ²⁰	ග ²⁵	ග ³²	ග	ග	ග	ග	ග	ග	ග
ඝ	g	gh	ඝ ⁷	ඝ ¹³			ඝ ²⁷	ඝ	ඝ		ඝ	ඝ	ඝ	ඝ
ඞ	ŋ	ṇ						ඞ			ඞ	ඞ	ඞ	ඞ
ඟ	ṅ	ṅg								ඟ		ඟ	ඟ	ඟ
ච	t͡ʃ~t͡ʃ̄	c	ච ¹	ච ¹⁴	ච ²⁰	ච ²⁵	ච ³²	ච	ච	ච	ච	ච	ච	ච
ඡ	t͡ʃ~t͡ʃ̄	ch	ඡ ⁶				ඡ ³²	ඡ	ඡ		ඡ	ඡ	ඡ	ඡ
ජ	d͡ʒ~d͡ʒ̄	j	ජ ⁸	ජ ¹⁴	ජ ²⁰	ජ ²⁵	ජ ³¹	ජ	ජ	ජ	ජ	ජ	ජ	ජ
ඤ	d͡ʒ~d͡ʒ̄	jh	ඤ ²					ඤ	ඤ		ඤ	ඤ	ඤ	ඤ
ඳ	ɳ	ṅ̐						ඳ	ඳ	ඳ	ඳ	ඳ	ඳ	ඳ

Fig. 10: The evolution of the Sinhala script under the era classification proposed by Nandasara and Mikami [90] - Part 1.
Note: The IPA for (*) anusvāraya [1016] and (**) visargaya [1017] are approximations as there is no consensus on their representation. 1200 AD and 1500 AD from inscriptions and pillars [90]. 1737 from the first printed Sinhala book collected by Nandasara [1018]. 1876 from “Alfabeto des gesammten Erdkreises” (*Alphabets of the entire world*) [1019] reported as CINGALESISCH [1018]. 1891 from the alphabet shown by Gunasekara [32]. 1996 from the character set *Sarasavi* developed by S T Nandasara [1018]. Nandasara and Mikami [90] report that the gray cells are not included due to being possible to be produced using consonant modifiers. 1998 from the *Iskoola Pota* UNICODE font by Microsoft [1020].

	IPA	Transliteration	300 BCE - 1 CE	1 CE - 300 CE	300 CE - 500 CE	500 CE - 700 CE	700 CE - 1000 CE	1200 CE	1500 CE	1737 CE	1876 CE	1891 CE	1996 CE	1998 CE
ඳ	d̪ɔ̃ɳ	jñ								ඳ		ඳ	ඳ	ඳ
ඳ	nd̪ɔ̃ɳ	ñj										ඳ		ඳ
ඳ	t̪	t̪	ඳ ²	ඳ ¹⁴	ඳ ²⁰	ඳ ²⁵	ඳ ³²	ඳ	ඳ	ඳ	ඳ	ඳ	ඳ	ඳ
ඳ	t̪	th	ඳ ⁹			ඳ ²²		ඳ	ඳ		ඳ	ඳ	ඳ	ඳ
ඳ	d̪	d̪	ඳ ¹¹	ඳ ¹⁴	ඳ ¹⁸	ඳ ²⁴	ඳ ³⁰	ඳ	ඳ	ඳ	ඳ	ඳ	ඳ	ඳ
ඳ	d̪	dh	ඳ ³		ඳ ¹⁹	ඳ ²⁴		ඳ		ඳ	ඳ	ඳ	ඳ	ඳ
ඳ	n	ṇ	ඳ ²	ඳ ¹⁴	ඳ ¹⁸	ඳ ²⁵	ඳ	ඳ	ඳ	ඳ	ඳ	ඳ	ඳ	ඳ
ඳ	ṇd̪	ñd̪										ඳ	ඳ	ඳ
ඳ	t̪	t̪	ඳ ²	ඳ ¹⁴	ඳ ¹⁸	ඳ ²⁵	ඳ ³¹	ඳ	ඳ	ඳ	ඳ	ඳ	ඳ	ඳ
ඳ	t̪	th	ඳ ¹⁰		ඳ ¹⁸	ඳ ²⁵	ඳ ³²	ඳ	ඳ	ඳ	ඳ	ඳ	ඳ	ඳ
ඳ	d̪	d̪	ඳ ²	ඳ ¹⁶	ඳ ²⁰	ඳ ²⁵	ඳ ³¹	ඳ	ඳ	ඳ	ඳ	ඳ	ඳ	ඳ
ඳ	d̪	dh	ඳ ²	ඳ ¹⁶		ඳ ²⁵	ඳ ³²	ඳ	ඳ	ඳ	ඳ	ඳ	ඳ	ඳ
ඳ	n	n	ඳ ²	ඳ ¹⁴	ඳ ²⁰	ඳ ²⁵	ඳ ²⁹	ඳ	ඳ	ඳ	ඳ	ඳ	ඳ	ඳ
ඳ	ṇd̪	ñd̪										ඳ	ඳ	ඳ
ඳ	p	p	ඳ ²	ඳ ¹⁶	ඳ ²⁰	ඳ ²⁵	ඳ ³¹	ඳ	ඳ	ඳ	ඳ	ඳ	ඳ	ඳ
ඳ	p	ph					ඳ ³¹	ඳ	ඳ		ඳ	ඳ	ඳ	ඳ
ඳ	b	b	ඳ ²	ඳ ¹⁴	ඳ ²⁰	ඳ ²⁵	ඳ ³¹	ඳ	ඳ	ඳ	ඳ	ඳ	ඳ	ඳ
ඳ	b	bh	ඳ ⁷			ඳ ²⁵		ඳ	ඳ		ඳ	ඳ	ඳ	ඳ
ඳ	m	m	ඳ ²	ඳ ¹⁴	ඳ ²⁰	ඳ ²⁵	ඳ ³¹	ඳ	ඳ	ඳ	ඳ	ඳ	ඳ	ඳ
ඳ	mb	m̃b								ඳ	ඳ	ඳ	ඳ	ඳ
ඳ	j	y	ඳ ¹	ඳ ¹⁴	ඳ ²⁰	ඳ ²⁵	ඳ ³¹	ඳ	ඳ	ඳ	ඳ	ඳ	ඳ	ඳ
ඳ	r	r	ඳ ²	ඳ ¹⁷	ඳ ²⁰	ඳ ²⁴	ඳ ³¹	ඳ	ඳ	ඳ	ඳ	ඳ	ඳ	ඳ
ඳ	l	l	ඳ ²	ඳ ¹⁶	ඳ ²⁰	ඳ ²⁵	ඳ ³²	ඳ	ඳ	ඳ	ඳ	ඳ	ඳ	ඳ
ඳ	v	v	ඳ ²	ඳ ¹⁶	ඳ ²⁰	ඳ ²⁵	ඳ ³²	ඳ	ඳ	ඳ	ඳ	ඳ	ඳ	ඳ
ඳ	ʃ	ś	ඳ ¹			ඳ ²⁵	ඳ ³²	ඳ	ඳ	ඳ	ඳ	ඳ	ඳ	ඳ
ඳ	ʃ	ṣ	ඳ ⁷			ඳ ²⁵	ඳ ³²	ඳ	ඳ	ඳ	ඳ	ඳ	ඳ	ඳ
ඳ	s	s	ඳ ²	ඳ ¹⁴	ඳ ²⁰	ඳ ²⁵	ඳ ³²	ඳ	ඳ	ඳ	ඳ	ඳ	ඳ	ඳ
ඳ	h̪	h̪	ඳ ¹	ඳ ¹⁶	ඳ ²⁰	ඳ ²⁵	ඳ ³²	ඳ	ඳ	ඳ	ඳ	ඳ	ඳ	ඳ
ඳ	l̪	l̪		ඳ ¹⁴	ඳ ²⁰	ඳ ²⁶	ඳ ³⁰	ඳ	ඳ		ඳ	ඳ	ඳ	ඳ
ඳ	f	f										ඳ	ඳ	ඳ

Fig. 11: The evolution of the Sinhala script under the era classification proposed by Nandasara and Mikami [90] - Part 2
Note: 1200 AD and 1500 AD from inscriptions and pillars [90]. 1737 from the first printed Sinhala book collected by Nandasara [1018]. 1876 from “Alfabeto des gesammten Erdkreises” (*Alphabets of the entire world*) [1019] reported as CINGALESISCH [1018]. 1891 from the alphabet shown by Gunasekara [32]. 1996 from the character set *Sarasavi* developed by S T Nandasara [1018]. Nandasara and Mikami [90] report that the gray cells are not included due to being possible to be produced using consonant modifiers. 1998 from the *Iskoola Pota* UNICODE font by Microsoft [1020].

DISTRIBUTION OF SINHALA LETTERS TOWARDS THE BEGINNING, MIDDLE, OR END OF WORDS

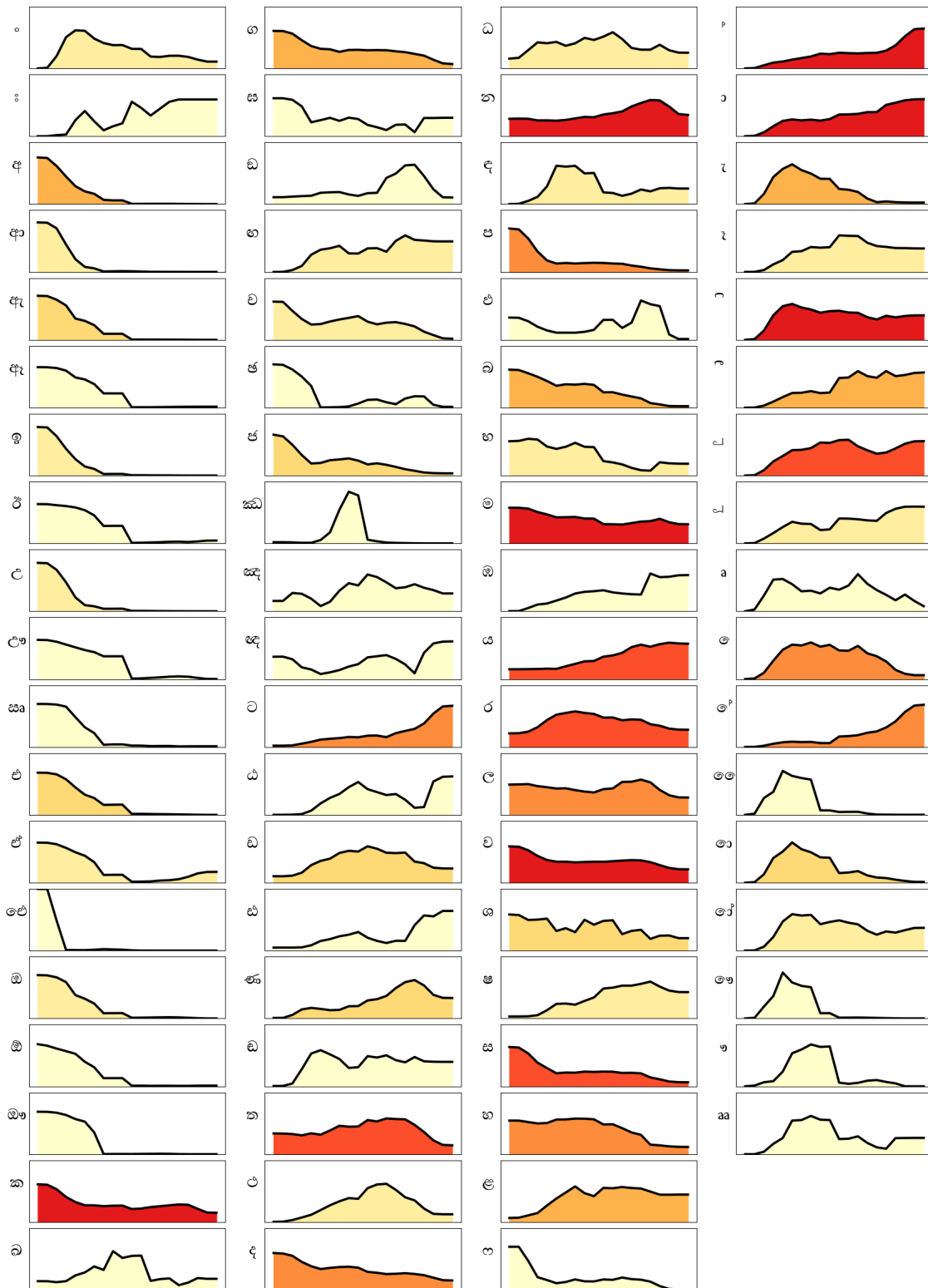


Fig. 12: An adaptation to Sinhala of the character position visualization proposed by ProffReader [27]. For this, we have used a portion of the *SinMin* corpus created by Upeksha et al. [28, 29]. Our sample included 3,990,838 unique words which appear more than once in the corpus. While the Spearman correlation used in this method dilutes the notability aspect such as pure Sinhala vowels only appearing as the first character, it brings to the front how diacritics, on the other hand, are generally biased towards the end.

APPENDIX C

AUTHOR META ANALYSIS

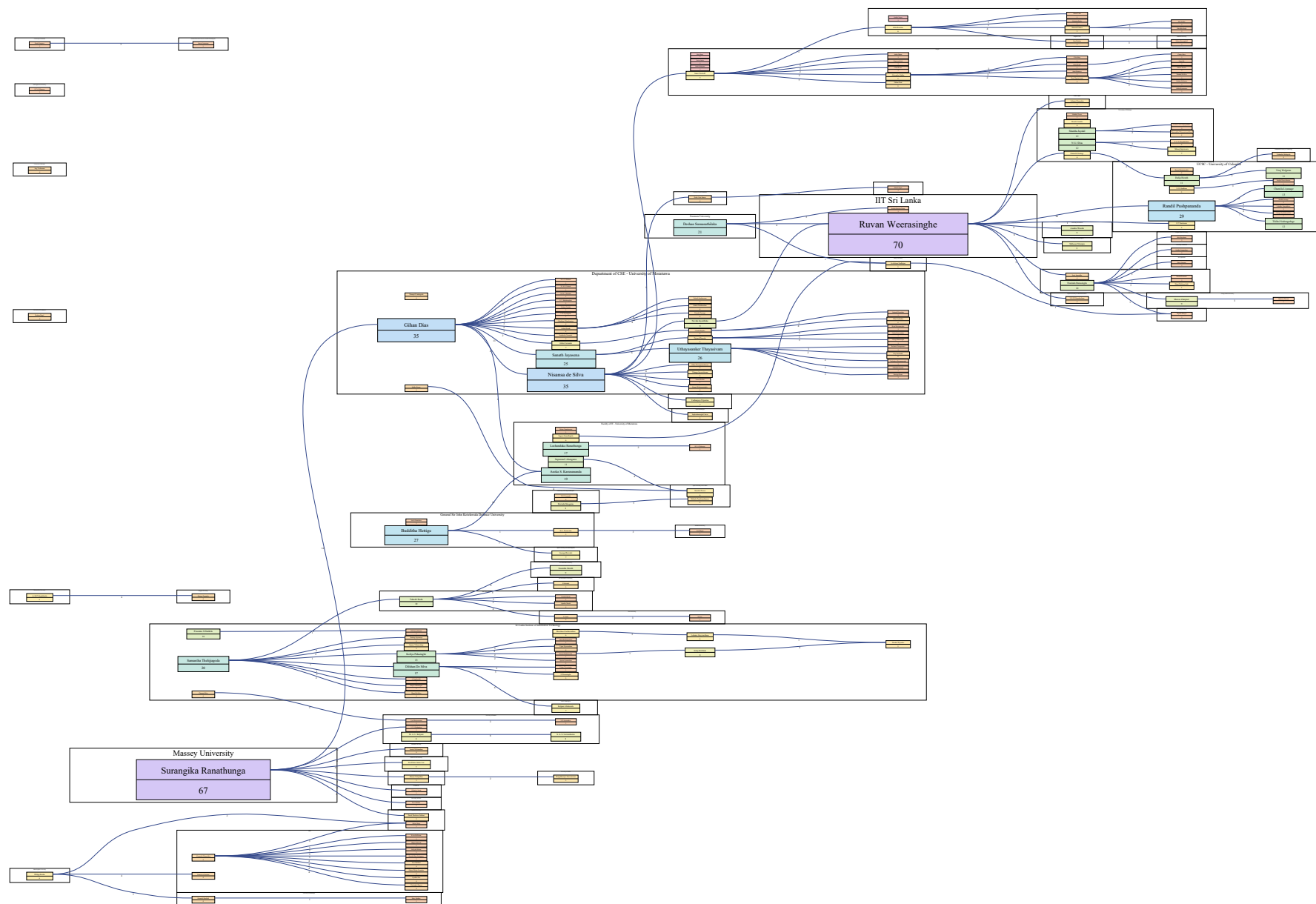


Fig. 13: Co-author graph of the most prolific researchers in the Sinhala NLP domain (Selected at the threshold of at least 3 publications)

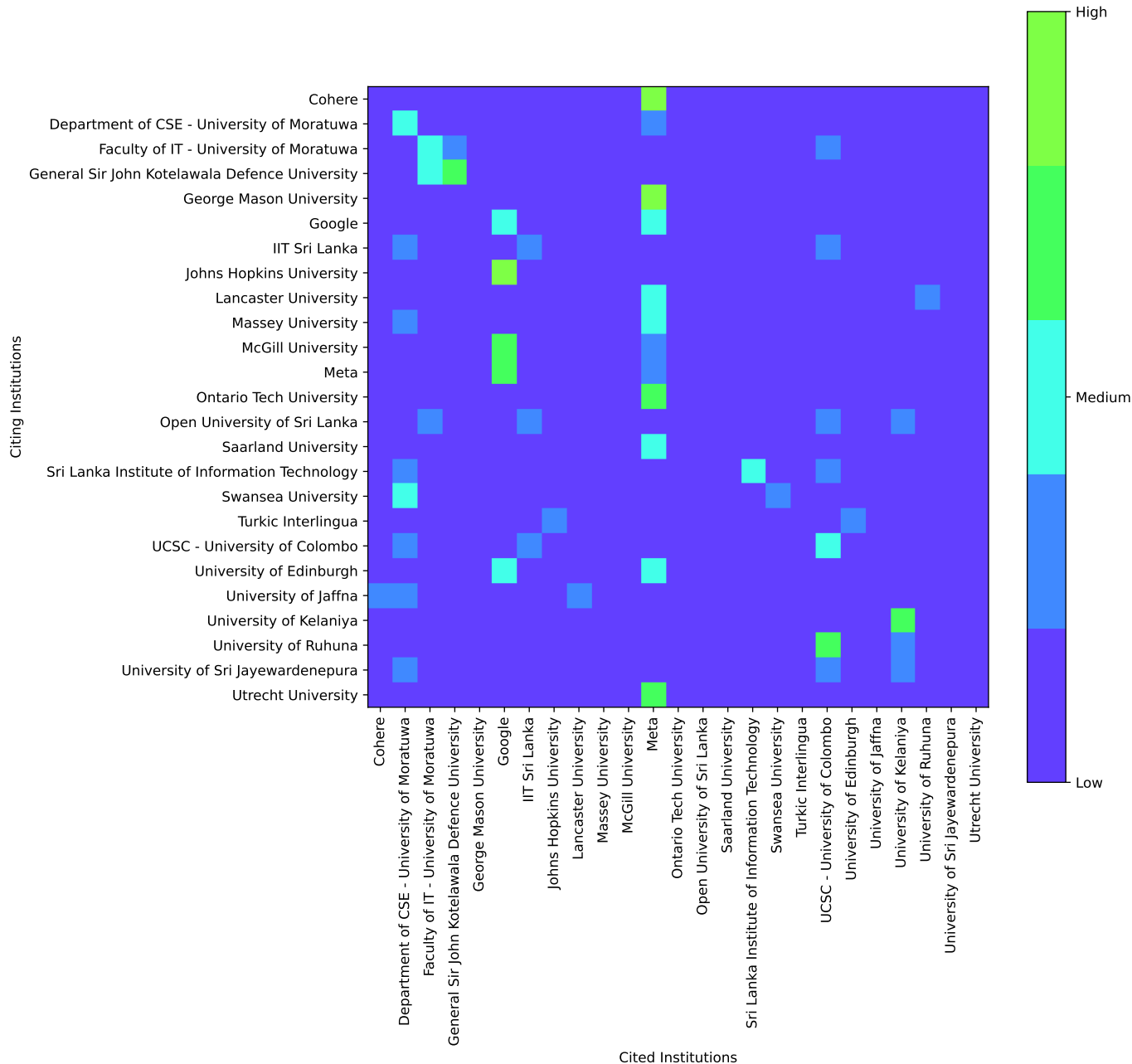


Fig. 14: The Probability of research from an institution citing that of other institutions. Note that the calculations are limited by 4 factors: (1) The availability of free to download pdf of the paper, (2) The aforementioned pdf containing references list (Some extended abstracts do not come with the references list), (3) The text extraction capabilities of *pdftotext*⁶³ (4) The accuracy of the research paper title look up. With all those limitations in mind, we still can make a few Interesting observations. The institute with the highest number of publications, *UCSC - University of Colombo* seems to be getting the most citations from most sources. However, they themselves seem to almost exclusively cite their own papers (0.7543). The only exception is the smaller number of works they cite (0.2114) from the *Department of CSE - University of Moratuwa* the institute with the second highest number of publications. This leaves them with a 0.0343 probability of citing anyone else. Comparatively, *Department of CSE - University of Moratuwa* seems to be more egalitarian in citing. They have a lower self-citation probability (0.5251) and a higher probability of citing *UCSC - University of Colombo* (0.3513). This results in a probability of citing others at 0.1236. Both the *Johns Hopkins University* and *University of Edinburgh* prefer to cite Google instead of work from Sri Lanka. *Faculty of IT - University of Moratuwa* prefers to cite *UCSC - University of Colombo* (0.2339) rather than *Department of CSE - University of Moratuwa* (0.0968) which is from the same parent institution.

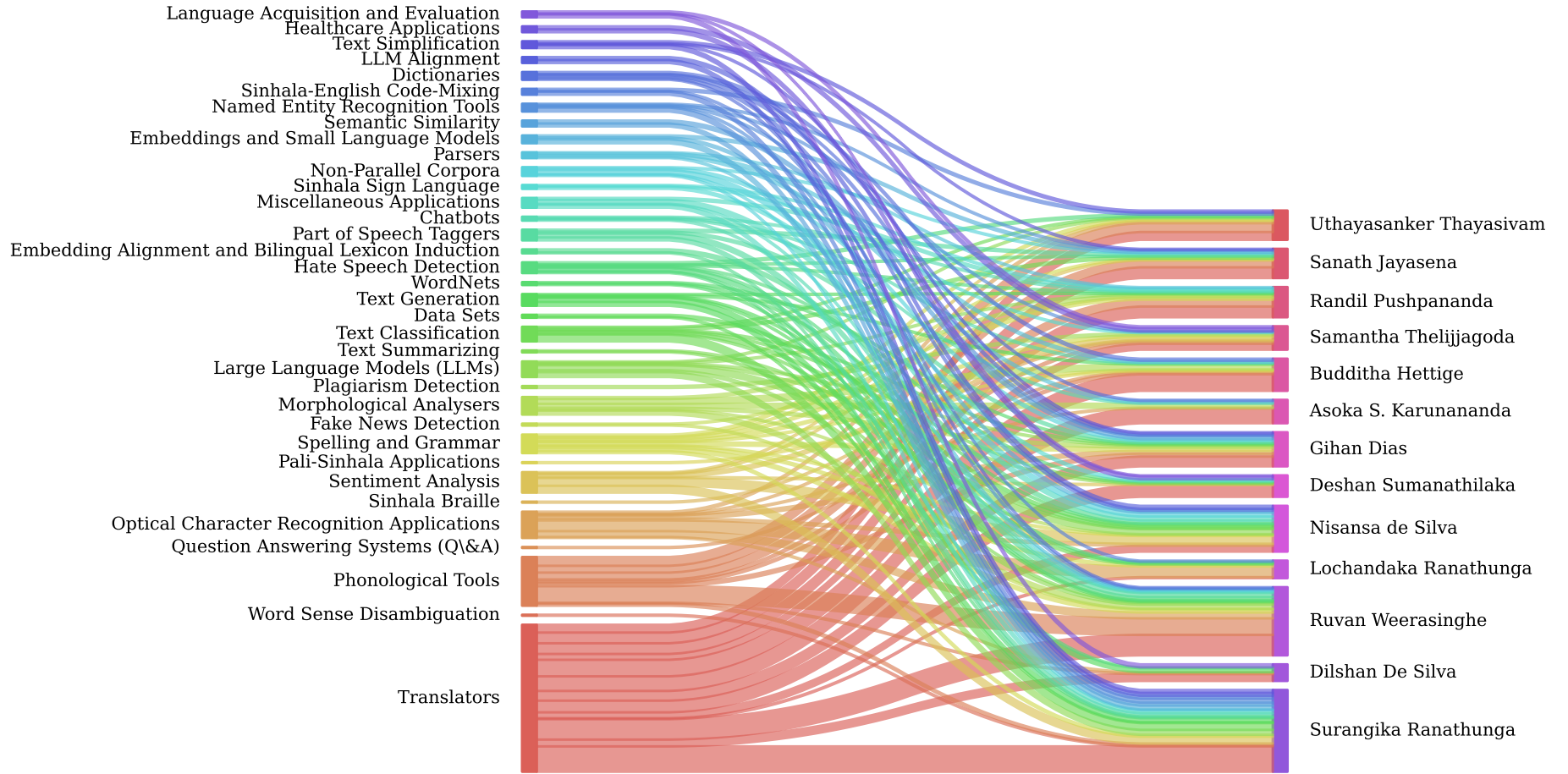


Fig. 15: Authors with at least 15 papers in the Sinhala NLP domain, mapped to their research interests, denoted by the subsection titles of the Section 3 of this paper.